Infant and young child feeding

Model chapter for textbooks for medical students and allied health professionals

Second edition

World Health Organization

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Glossary

Alveoli

Small sacs of cells in the mammary tissue that produce and secrete milk.

Areola

The pigmented area surrounding the nipple that contains Montgomery's glands which secrete lubricating and protective fluid.

Attachment

The positioning of the baby's mouth on the breast. Good attachment is essential for effective milk transfer and comfort. Also known as 'latch'.

Breast-milk substitute

Any milk (or milk-substitute, such as soy or oat 'milk') marketed as suitable for feeding to infants and young children up to the age of 3 years, including any product suitable as a partial or total replacement for breast milk in the diet of an infant or young child, other foods and beverages marketed or presented as suitable for feeding an infant less than six months of age.

Cleft lip and/or palate

An abnormal gap or split in the upper lip, upper mandible, palate or any combination of them that may extend into the nose.

Colostrum

The first milk produced in the mammary gland, which is rich in proteins and antibodies, and low in water and fat.

Commercial milk formula

A commercial product marketed as suitable as a partial or total replacement of breast milk in the diet of an infant or young child up to three years of age.

Complementary Feeding

The process of giving an infant food in addition to breast milk or milk formula, when these alone are no longer adequate to meet nutritional requirement. Complementary feeding starts at age 6 months and continues until 23 months of age.

Donor Human Milk

Human milk supplied by a donor human milk bank.

Engorgement

A state in which breasts become overfull of milk, leading to swelling, tightness, and sometimes pain.

Exclusive Breastfeeding

Feeding an infant only breastmilk, without any additional food or drink, not even water.

F-75

Powdered therapeutic milk that contains 75 kcal per 100mL when reconstituted properly, used in the treatment of infants and young children with severe wasting and/or nutritional oedema.

F-100

Powdered therapeutic milk that contains 100 kcal per 100mL when reconstituted properly used in the treatment of infants and young children with severe wasting and/or nutritional oedema.

Feedback Inhibitor of Lactation

A whey protein that regulates milk production by slowing it down when milk is not removed regularly.

Galactopoiesis

See 'Lactogenesis III'

Hyperlactation

A condition where the milk supply exceeds the infant's needs, potentially causing feeding difficulties and discomfort.

Hypoglycaemia

Low blood or plasma glucose concentration.

Involution

The final phase of lactation during which mammary cells undergo apoptosis and the breast returns to a simple ductal structure.

Intelligence Quotient

A validated measure of intelligence or cognitive ability.

International Code of Marketing of Breast-milk Substitutes

A set of recommendations for regulating the marketing of breast-milk substitutes, feeding bottles and teats with the aim of contributing to the provision of safe and adequate nutrition for infants, that was first adopted by the World Health Assembly in 1981 and includes subsequent relevant resolutions.

Kangaroo mother care

A method of caring for infants, especially preterm and low birth-weight neonates, being carried, usually by the mother, with skin-to-skin contact.

Lactation

The physiological process of producing and secreting milk from the mammary glands.

Lactogenesis I

The initial stage of milk production beginning in mid-pregnancy, involving the formation of colostrum.

Lactogenesis II

The onset of copious milk secretion, occurring typically 2–4 days after birth.

Lactogenesis III

The maintenance stage of lactation during which milk secretion is sustained by regular milk removal and nipple stimulation.

Latch

See 'attachment'.

Let-down Reflex

A reflex, triggered by oxytocin, that causes milk to be pushed from the alveoli through the ducts to the nipple.

Low birth-weight

Birth weight less than 2500g often, but not always, associated with premature birth.

Mammogenesis

The growth and development of the mammary glands during pregnancy in preparation for lactation.

Mature milk

The milk that follows colostrum and transitional milk, nutritionally complete for the infant's needs.

Milk Ejection Reflex

See 'Let-down reflex'

Milk transfer

The process by which milk is moved from the breast to the baby, often assessed by swallowing sounds and feeding patterns.

Mother's Own Milk

Milk expressed by an infant's birth-mother for feeding to a baby.

Montgomery's glands

Sebaceous glands in the areola that secrete protective oils and help guide the infant to the nipple.

Myoepithelial cells

Muscle-like cells that contract in response to oxytocin, pushing milk from the alveoli toward the nipple.

Neonate

An infant less than 28 days of age.

Oral rehydration solution

A mixture of water, glucose and electrolytes used to treat dehydration.

Oxytocin

A hormone that stimulates milk ejection during breastfeeding and promotes bonding between mother and infant.

Positioning

How the baby is held at the breast; proper positioning supports effective attachment and milk transfer.

Prelacteal feeds

Any food or fluid other than breast milk given before breastfeeding is initiated.

Preterm infant

An infant born before 37 weeks completed gestation.

Prolactin

A hormone that stimulates milk production, particularly in response to nipple stimulation.

Rooming-in

The practice of accommodating mothers and infants in the same room during inpatient postnatal care.

Responsive breastfeeding

Feeding in response to signs of hunger or the baby's cues rather, also known as "feeding on demand" or "baby-led feeding".

Responsive feeding

Encouraging a child to eat autonomously in response to their physiological and developmental needs, including by teaching them to recognise and respond appropriately to physical signs of hunger and satiety.

Skin-to-skin care

The practice of caring for an infant, wearing only a nappy (diaper), while held directly on the bare chest of a parent (usually the mother) or other caregiver. The infant may be held in place by the parent or with a long strip of fabric wrapped around both the parent and the infant.

Supplementary feeding

Feeding an infant less than six months old a therapeutic milk or breast-milk substitute in addition to breastfeeding when medically indicated to meet increased nutritional demands or to bridge a temporary gap between available breastmilk and the nutritional needs of an infant while the mother is supported to increase the volume of milk she is producing.

Sustainable Development Goals

A set of 17 goals that were adopted by the United Nations in 2015 as a universal call to action to end poverty, hunger, AIDS, discrimination against women, protect the planet, and ensure that all people enjoy peace and prosperity by 2030.

Transitional milk

Milk produced in the days following birth as the body transitions from colostrum to mature milk.

Weaning

The process of shifting an infant from exclusively breastfed or human-milk fed to not breastfed or human-milk fed at all. Weaning may be sudden or gradual and may begin with the introduction of complementary foods in addition to breastfeeding.

Abbreviations

ART	antiretroviral therapy
CL/CP	Cleft lip and/or palate
BFHI	Baby-friendly hospital initiative
FIL	feedback inhibitor of lactation
GNI	gross national income
HTLV-1	human T-lymphotropic virus type 1
IMCI	management of childhood illness
IUD	Intrauterine device
IQ	Intelligence quotient
KMC	kangaroo mother care
LAM	lactational amenorrhea method
LBW	low birth-weight
MOM	mother's own expressed breast milk
MUAC	mid upper arm circumference
NFS	WHO Department of Nutrition and Food Safety
SDGs	Sustainable Development Goals
sIgA	secretory immunoglobulin A
TB	tuberculosis
UNICEF	United Nations' Children's Fund
WHO	World Health Organization

Methodology

The process of revising Infant and young child feeding: model chapter for textbooks for medical students and allied health professionals, first published in 2009, began in 2021. Following examination of WHO and UNICEF publications on infant and young child feeding released since the first edition was finalized in 2009, WHO identified gaps, and redundancies in consultation with external partners.

WHO solicited experts in the fields of infant and young child feeding and medical education to form a Technical Advisory Group and assigned responsibility for the project to personnel with relevant qualifications and experience.

WHO commissioned research to investigate the learning needs of pre-service medical and allied health students and post-registration learners and coordinate content revision. Five focus group interviews were conducted with medical students and trainees recruited from the International Federation of Medical Students Association (international); Brown University Department of Family Medicine residency program (USA); and Academy of Breastfeeding Medicine (international). Insights gained from this research inform the structure and presentation of the second edition.

Julie Scott-Taylor performed the initial review and updated content. Revised drafts of each section were then presented to the WHO Technical Advisory Group (TAG) for multiple rounds of discussion and review. Sub-groups of subject matter experts incorporated the agreed changes and revisions under the guidance of Julie Scott Taylor and Nina Chad (WHO Department of Nutrition and Food Safety, Actions in Health Systems Unit). Julie Scott Taylor and Nina Chad to produce second drafts of each section. Second drafts were presented for review by all members of the TAG, who were asked to ensure changes were supported with evidence published in peer-reviewed publications or WHO guidance. WHO provided technical support to confirm consistency with WHO recommendations upon request from the TAG. Comments on the second drafts were discussed at TAG meetings held during the last quarter of 2021. These discussions were used to guide development of the final draft sections. The TAG met a total of six times to discuss process and content.

During the revision of this second edition, several new or updated WHO recommendations were published. These include the WHO Guideline on care of the preterm infant (2022); updated WHO guidelines on Complementary Feeding (2024); updated WHO guidelines on treatment and prevention of wasting and nutritional oedema (2023); and WHO guidance on scheduled child and adolescent well-care visits (2024).

WHO incorporated updated WHO recommendations, combined the sections, harmonised and restructured the content into the draft Infant and young child feeding: model chapter for textbooks for medical students and allied health professionals, second edition.

WHO recruited an experienced medical educator and a qualified lactation consultant to act as external peer reviewers, who could provide an objective and independent assessment of content accuracy, product credibility and feasibility of the draft Infant and young child feeding: model chapter for textbooks for medical students and allied health professionals, second edition.

All external experts and peer reviewers completed a declaration of interest disclosing potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of this guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meeting or addressed in the second edition.

SECTION 1

Introduction

Key messages

- Give mothers and families clear breastfeeding recommendations and timely, accurate information about complementary feeding.
- Facilitate and support initiation of breastfeeding within the first hour after birth.
- Recommend and support exclusive breastfeeding for six months (180 days).
- Educate families about introducing nutritionally adequate and safe complementary feeding from six months.
- Encourage and support breastfeeding up to two years of age or beyond.
- Inform families that breastfeeding prevents death from breast and ovarian cancers.
- Avoid exposing infants to breast-milk substitutes because it increases the risk of infectious morbidity and mortality in all countries – and the effect is dose-dependent.
- Be aware:
  - Breastfeeding reduces medical visits and prevents infant and young child mortality.
  - Improving breastfeeding practices is the responsibility of every health professional.

Section 1 outline

1.1 Purpose and scope

This second edition of the *Model Chapter for medical textbooks on infant and young child feeding* summarizes the essential knowledge that healthcare professionals and others need to protect, promote, and support optimal infant and young child feeding.

1.2 Recommendations

Clear recommendations and timely, accurate information can support patients to make informed decisions.

Clear recommendations and timely accurate information can support patients to make informed decisions.

WHO and the United Nations Children's Fund (UNICEF) recommend *(1)* that all infants:

1. **initiate breastfeeding within the first hour after birth**

   Breastfeeding begins as an instinctive behaviour. Immediately after birth, a healthy-term infant in skin-to-skin contact with their mother uses innate reflexes to find the nipple and attach. Healthcare workers can best support a first feed by not interfering with or rushing this process, which can take up to an hour *(2)*.

2. **breastfeed exclusively for six completed months (180 days)**

   Exclusive breastfeeding means feeding *only* breast milk. No other liquid or solid foods should be offered to infants less than six months old. Water is not necessary, even in hot weather, and should not be given to infants less than six months of age. Oral rehydration solution, drops or syrups consisting of vitamins, mineral supplements or medicines may be given if medically indicated without interfering with exclusive breastfeeding.

   Exclusive breastfeeding includes breastfeeding at the breast, feeding the mother's own expressed breast milk and feeding breast milk (MOM) expressed by another mother, including pasteurized donor human milk (DHM).

3. **consume nutritionally adequate and safe complementary foods from six months**

   Offer nutritionally safe and adequate complementary foods from six months (180 days), while breastfeeding continues for two years or beyond for as long as mothers and babies so desire.

   Nutritionally adequate, safe complementary foods should be offered in appropriate quantity and frequency to meet a young child's energy and nutrient needs from six months (see **Section 3).** Offering insufficient quantities of complementary food or offering foods that are nutritionally inadequate also increase the risk of acute malnutrition. Children 6–23 months of age should not consume unhealthy foods and beverages. Overweight and obesity in childhood increases the risk of non-communicable disease later in life *(3)*.

4. **continue breastfeeding alongside complementary foods for two years or beyond**

   Breastfeeding remains a critical source of nutrients for infants and young children after complementary feeding begins. Breast milk provides about one half of an infant's energy needs in the second half of the first year and up to one third of an infant's energy requirements during the second year of life *(3)*.

   There is no evidence that the quality of breast milk produced deteriorates over time – breast milk continues to provide high quality nutrients and protect children from infectious disease for as long as breastfeeding continues. Therefore, healthcare professionals should encourage and support mothers to continue breastfeeding for at least two years and thereafter, for as long as mother and baby wish to continue *(4)*.

1.3 Health outcomes

Infants who initiate breastfeeding two or more hours after birth are more likely to die in the first 28 days after birth and less likely to survive to three and to six months than infants who initiation breastfeeding within the first hour after birth. These infants are also less likely to be breastfeeding exclusively at one month and at three months. Infants who initiate breastfeeding more than two hours after birth are also more likely to stop breastfeeding altogether by one month and by three months *(5)*.

Exposure to any food or fluid other than breastmilk in the first six months of life – including water – increases the risk of gastrointestinal and respiratory morbidity and mortality.

Compared to exclusively breastfed infants, infants who are not exclusively breastfed during the first six months of life are at increased risk of all-cause and infection-related mortality in both high- and low-income settings *(6–9)*.

Exposure to any food or fluid other than breast milk – including water – in the first six months of life increases the risk of gastrointestinal and respiratory morbidity and mortality *(10)*.

Even just one dose of non-human milk compromises the infant's microbiome and increases susceptibility to infectious disease *(11)*. Exposure to breast-milk substitutes, including prelacteal feeds (such as ghee, honey, sugar water and tea) and formula milk, at any time in the first six months of life increases the risk of infection, especially if given before the baby's microbiome is established. Supplementation with formula milk, water or any other food or fluid deprives infants of the immunological protections in breast milk and alters their intestinal microflora. This renders infants more susceptible to infection and simultaneously exposes them to pathogenic organisms. As just one dose of formula milk is enough to disrupt the baby's microbiome, supplementary feeds, including any solid or liquid foods other than breast milk, should be avoided before six months of age unless medically necessary *(12, 13)*.

Children aged 6–23 months who are not breastfed are at increased risk of all-cause and infection-related mortality compared with children who continue breastfeeding to 23 months *(14)*. Some breastfeeding is better than none: the risks are highest for infants who are not breastfed at all and those who stop breastfeeding altogether.

Breastfeeding also provides protection from noncommunicable disease. Breastfeeding in infancy protects children and adolescents from obesity and diabetes *(15)*. Children who stop breastfeeding prematurely are at increased risk of morbidity and mortality from all causes *(16)*.

Children breastfed for less than a year achieve lower scores on intelligence tests that those who were breastfed for a year or more. Longer breastfeeding duration is associated with higher performance on intelligence tests among children and adolescents after controlling for maternal intelligence quotient (IQ). This results in improved academic performance, increased long-term earnings and productivity *(17)*.

Breastfeeding also protects mothers from adverse health outcomes. The risk of developing invasive breast cancer decreases by 6% for each year a woman breastfeeds. The longer a woman breastfeeds, the lower her risk of breast cancer becomes *(18)*.

1.4 Breast-milk substitutes

Early exposure to breast-milk substitutes, including commercial infant formula, carries health risks that are dose-responsive *(19–21)*. Even a single dose of breast-milk substitute can interfere with the immune protection and diminish the long-term health that is provided by breastfeeding *(22)*.

Exposure to breast-milk substitutes increases the risk of infectious morbidity and mortality in all countries – and the effect is dose-dependent.

Breast-milk substitutes, including commercial infant formula, can never be equivalent to breast milk. Unlike fresh breast milk, the composition of breast-milk substitute products is static and does not change to meet the evolving nutritional needs of growing infants and young children. The fats and proteins in breast-milk substitute products differ significantly from those found in human milk and the micronutrients in breast-milk substitutes are much less bioavailable than the micronutrients found in breast milk *(23,24)*.

Unlike fresh breast milk, breast-milk substitutes provide no protection from infectious disease and increase susceptibility to pathogens by raising gut pH and altering the microbiome *(23)*.

The development of the infant microbiome has an early-life critical window of susceptibility which begins with the maternal-to-foetal exchange of microbes in utero and continues through the first year of life and into early childhood.

Breast-milk substitutes are easily contaminated and therefore can itself become a vector for infectious disease. Powdered infant formula is not a sterile product. It carries an inherent risk of contamination with pathogenic organisms including cronobacter sakazakii and enterobactor salmonella, which can cause life threatening infections *(25)*. Safe preparation, storage and handling of powdered infant formula is discussed in **Section 4.**

Powdered infant formula should not be used to feed medically vulnerable infants, including small or premature infants and infants in the first 28 days after birth. Ready-to-feed infant formula does not carry the same risk of inherent contamination as powdered infant formula and may be given to infants who do not have access to sufficient breastmilk or DHM *(26,27)*.

The macro- and micro-nutrient compositions of animal milks are very different from that of human milk. Unmodified animal milks are not suitable for feeding infants less than six months of age. The renal solute load of these products is higher than that of breast milk and formula milk and it can cause kidney damage when used to feed infants less than six months of age *(28–30)*.

1.5 Infant and young child feeding practices

Optimal breastfeeding practices save lives and safeguard health throughout the life course.

Despite the enormous health and economic consequences of not breastfeeding, early termination of exclusive breastfeeding, and inadequate complementary feeding *(31,32)*, WHO and UNICEF guidelines for breastfeeding and complementary feeding are routinely not followed.

Data collected between 2017 and 2023 indicates that, globally, 46% of infants initiated breastfeeding within one hour of birth, 72% continued to breastfeed for at least one year and 46% were still breastfeeding at two years *(33).*

Globally, 46% of mothers initiate breastfeeding within one hour of birth, 71% continue to breastfeed for at least one year, and 45% are still breastfeeding at two years.

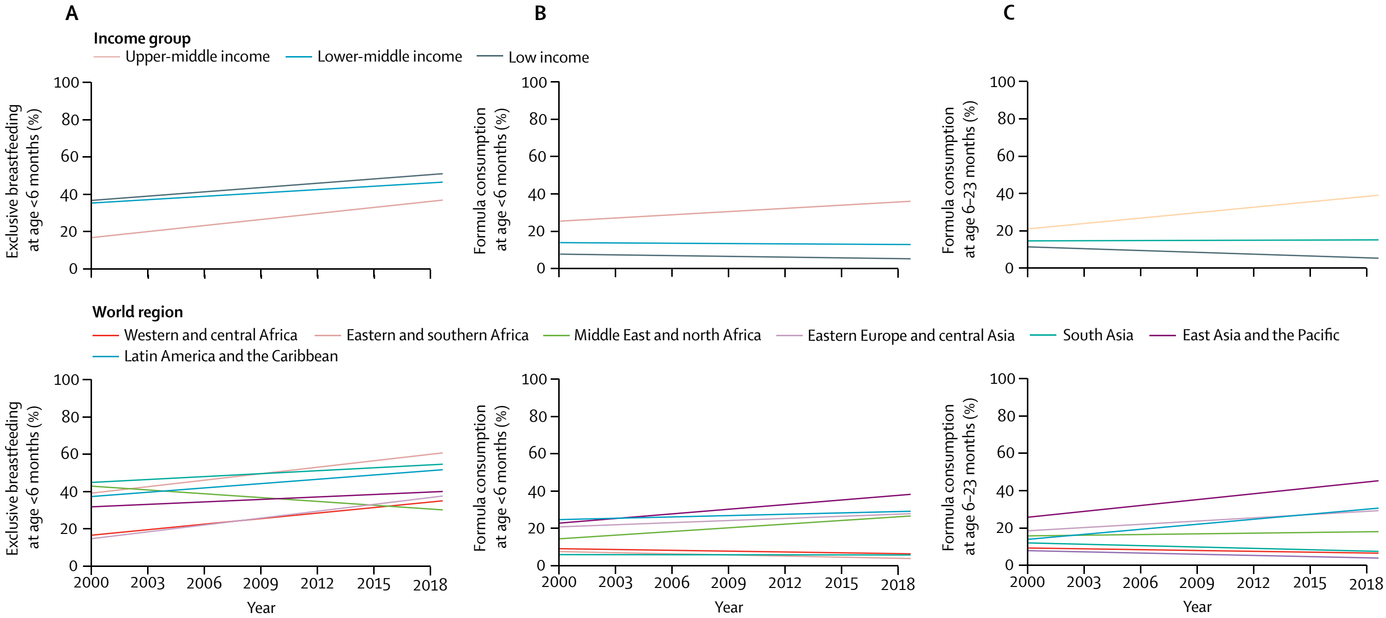
There have been sustained increases in the prevalence of exclusive breastfeeding since 2012. Of the 106 countries that reported on exclusive breastfeeding prevalence at least twice between 2017 and 2023, 75 recorded an increase. Of these, 23 countries recorded an increase >10% *(33).* **Fig. 1.1** demonstrates evolution of breastfeeding rates by country, age and region *(34).*

Globally 48% of all infants under six months of age were exclusively breastfed in 2024 *(33).* In high-income countries, fewer than 20% of children are breastfed for 12 months. In low- and middle-income countries, fewer than 40% of infants under six months are exclusively breastfed, and only two-thirds of children between six months and two years receive any breastmilk *(6)*.

A 2021 study of breastfeeding practices in 113 countries assessed time trends in the consumption of different types of milk (breast milk, formula, and animal milk) by children younger than two years from 2000 to 2019. Globally, exclusive and any breastfeeding at six months increased in various regions and income groups and infant formula consumption increased in upper-middle-income countries. Just under 90% of infants less than six months of age and around 80% of children one year of age were breastfed in 2019 (see **Fig. 1.1**) *(34)*.

There are large differences in breastfeeding rates and feeding practices across geographic regions, between and within countries. Complementary foods are often introduced too early or too late and may be nutritionally inadequate and unsafe. Practices vary across socioeconomic levels, exacerbating existing health inequalities.

Fig. 1.1 Evolution of the rates of exclusive breastfeeding and consumption of formula in low-income and middle-income countries, by age, country income group and world region, 2000–2019



(A) Exclusive breastfeeding in children younger than 6 months (83 countries, based on 344 data points), (B) consumption of formula in children younger than 6 months (83 countries, 328 data points), and (C) consumption of formula at age 6-23 months (83 countries based on 328 data points) *(34)*.

Source: Neves PAR, Vaz JS, Maia FS, Baker P, Gatica-Domínguez G, Piwoz E, Rollins N, Victora CG. Rates and time trends in the consumption of breastmilk, formula, and animal milk by children younger than 2 years from 2000 to 2019: analysis of 113 countries. Lancet Child Adolesc Health. 2021;5(9):619–30, Figure 2.

1.6 Public health and economic impacts of poor IYCF practices

Improving breastfeeding practices could prevent about 820 000 deaths a year, 87% of them infants under six months of age *(6)*. In 2014, it was estimated that promoting and supporting optimal breastfeeding in the first two years of life could avert almost 12% of deaths amongst children less than five years of age *(35)*. Breastfeeding also prevents almost 20 000 deaths from breast cancer each year and improving breastfeeding practices could prevent a further 20 000 deaths *(6)*.

Poor breastfeeding practices increase annual healthcare costs by an estimated US$ 312 million in the United States of America, US$ 48 million in the United Kingdom, US$ 6 million in Brazil and US$ 30.3 million in urban China. Nearly half of all diarrhoea episodes and one-third of all respiratory infections in infants and young children would be prevented by improving breastfeeding practices in low- and middle-income countries. There is growing evidence that breastfeeding prevents overweight/obesity and diabetes later in life *(6)*.

Globally, lower IQ associated with poor breastfeeding practices is estimated to cost US$ 300 billion annually, representing 0.49% of global gross national income (GNI). High-income countries lose more than US$ 230 billion annually (0.53% of GNI) due to low rates of breastfeeding. Low- and middle-income countries lose more than US$ 70 billion annually (0.39% of GNI) due to low rates of breastfeeding *(6)*.

Breastfeeding is critical to achieving all of the Sustainable Development Goals. In particular, breastfeeding safeguards nutrition (Goal 2), prevents child mortality and noncommunicable diseases

(Goal 3), and supports cognitive development and education (Goal 4). Breastfeeding also contributes to ending poverty, promoting economic growth and reducing inequalities *(36)*.

In 2003, the World Health Assembly and UNICEF adopted the Global Strategy for Infant and Young Child Feeding *(1)*. This strategy was developed to revitalize world attention to the impact that feeding practices have on the nutritional status, growth and development, health, and survival of infants and young children.

In 2018, the World Health Assembly affirmed that "the protection, promotion and support of breastfeeding contributes substantially to the achievement of the Sustainable Development Goals on nutrition and health and is a core element of quality health care" and recognized that "appropriate, evidence-based and timely support of infant and young child feeding in emergencies saves lives, protects child nutrition, health and development and benefits mothers and families". **Fig. 1.2** illustrates the SDGs relevance to breastfeeding. The World Health Assembly requested WHO "to continue to update and generate evidence-based recommendations" *(37)*.

Improving breastfeeding practices could prevent about 820 000 deaths a year, 87% of them infants under six months of age.

Fig. 1.2 Sustainable Development Goals' relevance to breastfeeding

Source: World Alliance for Breastfeeding. Used with permission.

1.7 Facilitating breastfeeding

> *Breastfeeding is a human rights issue for both the child and the mother. Children have the right to life, survival and development and to the highest attainable standard of health, of which breastfeeding must be considered an integral component, as well as safe and nutritious foods. Women have the right to accurate, unbiased information needed to make an informed choice about breastfeeding. They also have the right to good quality health services, including comprehensive sexual, reproductive and maternal health services. And they have the right to adequate maternity protection in the workplace and to a friendly environment and appropriate conditions in public spaces for breastfeeding which are crucial to ensure successful breastfeeding practices … States should take all necessary measures to protect, promote, and support breastfeeding, and end the inappropriate promotion of breast-milk substitutes and other foods intended for infants and young children up to the age of 3 years.*

Joint statement by the UN Special Rapporteurs on the Right to Food, Right to Health, the Working Group on Discrimination against Women in law and in practice, and the Committee on the Rights of the Child in support of increased efforts to promote, support and protect breast-feeding *(38)*

1.7.1 The International Code of Marketing of Breast-milk Substitutes

The aggressive marketing of breast-milk substitutes creates a major barrier to breastfeeding and influences nutritional decisions that impact children throughout their lives. In 1981, the International Code of Marketing of Breast-milk Substitutes (the Code) was adopted by the World Health Assembly. The Code is a set of recommendations outlining the minimum requirements for regulating the marketing of breast-milk substitutes (including formula milk represented as suitable for feeding a child up to three years of age), feeding bottles and teats, which aims to prohibit the aggressive and inappropriate marketing of breast-milk substitutes.

Aggressive marketing of breast-milk substitutes creates a major barrier to breastfeeding and influences nutritional decisions that impact children throughout their lives.

The Code seeks to prohibit all advertising and other forms of promotion of breast-milk substitutes, bottles, and teats, either to health-care systems or to the broader public. When the Code is successfully implemented, it protects families from the misinformation and commercial pressures that can dissuade women from breastfeeding. The World Health Assembly repeatedly calls on governments to give effect to the provisions in the Code through national, legally binding regulations.

A free elearning course for policy makers, legislators, health practitioners, UN agency staff, and civil society partners with an introduction to the aims, scope and contents of the Code is available from UNICEF's Agora platform *(39)*.

1.7.2 Investing in breastfeeding

Breastfeeding is one of the smartest investments a country can make to build its future prosperity. It offers children unparalleled health and brain-building benefits. It has the power to save the lives of women and children throughout the world, and the power to help national economies grow through lower health care costs and smarter workforces. Yet many societies are failing to adequately support women to breastfeed, and as a result, the majority of the world's children – along with a majority of the world's countries – are not able to reap the full benefits of breastfeeding.

Every dollar invested in breastfeeding generates a 35-fold return on investment.

Investing in improving breastfeeding practices is not only an investment in improving children's health and saving lives, but also an investment in human capital development that can benefit a country's economy (see **Table 1.1**). Every dollar invested in breastfeeding generates a 35-fold return on investment. Employers and health service providers benefit from investment in improving breastfeeding practices. Improved breastfeeding practices reduce annual healthcare expenditure and absenteeism *(40)*.

Table 1.1 Global breastfeeding investment case (2017)

	Future economic cost of mortality (USD $Billion)	Future economic cost of cognitive losses (USD $Billion)	Total combined economic cost as percent of GNI[a]
China	$6.13	$59.60	0.61
India	$7.10	$7.25	0.70
Indonesia	$2.32	$6.94	1.06
Mexico	$1.10	$7.13	0.67
Nigeria	$11.94	$9.10	4.10

Estimated annual economic costs attributable to not breastfeeding and poor breastfeeding practices at current levels compared to near universal coverage.

[a] GNI = gross national income

Source: Global Breastfeeding Collective. Used with permission. *(40)*

1.7.3 Facilitating policy actions

Improving breastfeeding practices is the responsibility of national governments and health care providers. A facilitating policy environment is required to support i) the initiation of breastfeeding within the first hour of giving birth, ii) breastfeeding exclusively for six months and iii) continuing to breastfeed up to two years and beyond.

The seven critical features of policy environments that facilitate breastfeeding are:

1. Increased funding to raise breastfeeding rates from birth through two years.
2. Full implementation of the International Code of Marketing of Breast-milk Substitutes, including related World Health Assembly resolutions, through strong legal measures that are enforced and independently monitored by organizations free from conflicts of interest.
3. Paid family leave and workplace breastfeeding protections, building on the International Labour Organization's maternity protection guidelines as a minimum requirement, including provisions for the informal sector.
4. National, system-wide implementation of the Ten Steps to Successful Breastfeeding in maternity facilities, including providing breast milk for sick and vulnerable newborns.
5. Providing skilled breastfeeding counselling to all pregnant women and new mothers as part of comprehensive breastfeeding policies and programmes in health facilities.
6. Strong links between health facilities and communities, and encourage community networks that protect, promote, and support breastfeeding.
7. Effective monitoring systems that track the progress of policies, programmes and funding towards achieving both national and global breastfeeding targets.

It is both ineffective and unethical to engage in health promotion activities aimed at persuading mothers to conform to recommended breastfeeding practices without also implementing the policies that make it possible.

Improving breastfeeding practices requires a whole-of-government approach that includes resource allocation, policy implementation and effective monitoring systems that are used to inform ongoing investment and activity.

It is both ineffective and unethical to engage in health promotion activities aimed at persuading mothers to conform to recommended breastfeeding practices without also implementing the policies that make it possible.

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SECTION 2

Anatomy and physiology of breastfeeding

Key messages

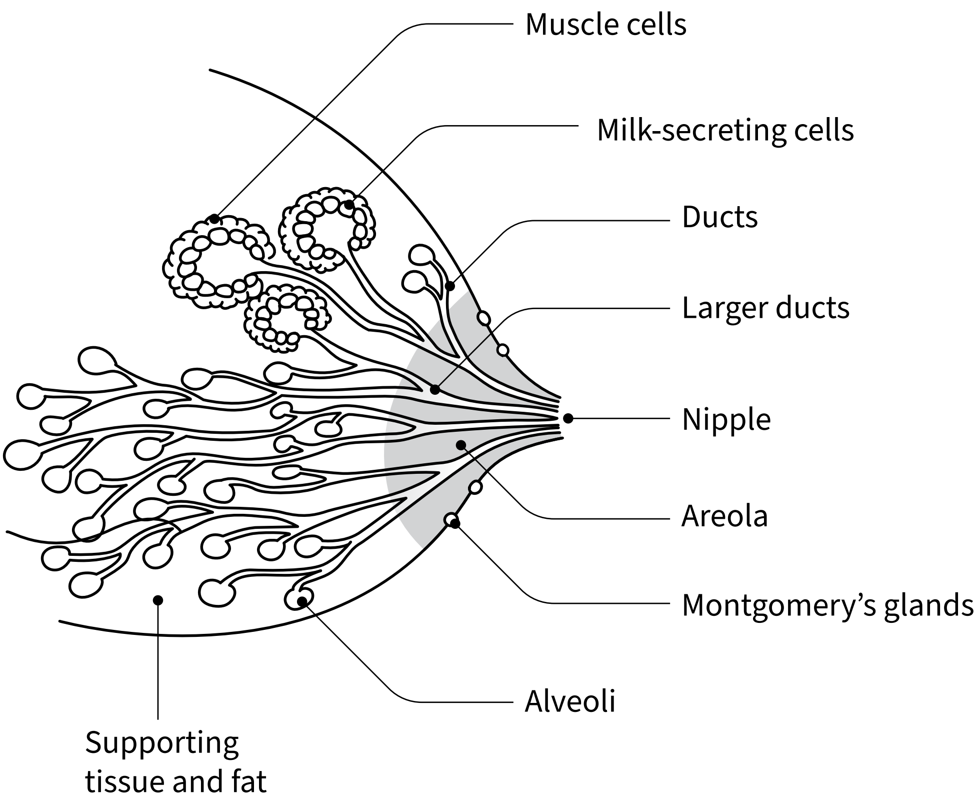
- Facilitate immediate and sustained skin-to-skin contact after birth to stimulate infant reflexes that initiate breastfeeding.
- Avoid separating infants from their mothers after birth.
- Teach mothers to recognise and respond to hunger cues and encourage responsive breastfeeding.
- Communicate key practices for establishing breastfeeding:
  - Removing milk from the breast and providing frequent nipple stimulation, which are critical to establishing and maintaining lactation.
  - If suckling at the breast is not possible, nipples must be stimulated and milk removed, manually or with a breast pump, very frequently to establish lactation.
  - Urine and stool output are more immediate indicators of breastmilk intake than weight gain or growth trajectory.

Section 2 outline

2.1. Anatomy of the breast

The anatomy of the breast is illustrated in **Fig. 2.1** below. The structure and function of mammary tissue, nipple and areola are significant during breastfeeding and lactation *(1)*.

Fig. 2.1 Anatomy of the breast



2.1.1 Mammary tissue

Mammary tissue includes the alveoli, small sacs of cells that secrete milk, and the ducts that carry the milk from the alveoli to nipple where it can be removed by the baby or expressed. Between feeds, milk collects in the lumen of the alveoli and ducts. The alveoli and ducts are surrounded by myoepithelial (muscle) cells. Under the influence of oxytocin, myoepithelial cells contract, squeezing the alveoli and dilating the ducts, forcing milk towards the nipple.

The breast is sufficiently developed to produce milk by the twentieth week of gestation.

2.1.2 Nipple and areola

The nipple is at the centre of the circular pigmented areola. The nipple contains sensory nerves and muscle fibres which contract and make it protrude when stimulated. There is an average of nine large milk ducts passing beneath the areola and through the nipple. During a feed, the ducts beneath the areola dilate and fill with milk.

Montgomery's glands, located in the skin of the areola, secrete an oily fluid that lubricates and protects the skin of the nipple and areola during suckling. The fluid is also responsible for the mother's individual scent which attracts her baby to the breast. These glands start functioning during pregnancy.

2.2 The physiology of lactation

Hormones produced during pregnancy stimulate the growth of mammary tissue and this process is known as mammogenesis. From about 16 to 20 weeks of pregnancy, colostrum is secreted in the breast. This stage of lactation is known as Lactogenesis I (or secretory differentiation).

Withdrawal of pregnancy hormones triggers the onset of copious milk production, known as Lactogenesis II (or secretory activation). This increase in milk volume usually begins during the first three or four days after birth. The milk produced during lactogenesis II differs from colostrum both in composition and volume. Known as "mature milk" it contains less sodium, more lactose and more water. The onset of lactogenesis II is commonly characterized by surplus milk production (tight, full or engorged breasts).

Lactogenesis I and II are driven by the endocrine system (progesterone withdrawal, prolactin reception) and fade completely within two weeks of a birth; lactation is then driven by an autocrine (local) response in which demand determines supply. This switch from endocrine to autocrine control is known as lactogenesis III (or galactopoiesis). During lactogenesis III, milk synthesis is primarily driven by removal of milk from the breast and stimulation of the nerve-endings in the nipple. Lactogenesis III is the maintenance stage of lactation and it may continue for many years.

The final stage of lactation occurs during weaning and is known as involution. As the volume of milk removed from the breast declines consistently, the cells responsible for milk synthesis are reabsorbed.

There are a number of hormones involved in initiating and maintaining lactation. Prolactin, oxytocin, and feedback inhibitor of lactation are the most important of these *(2)*.

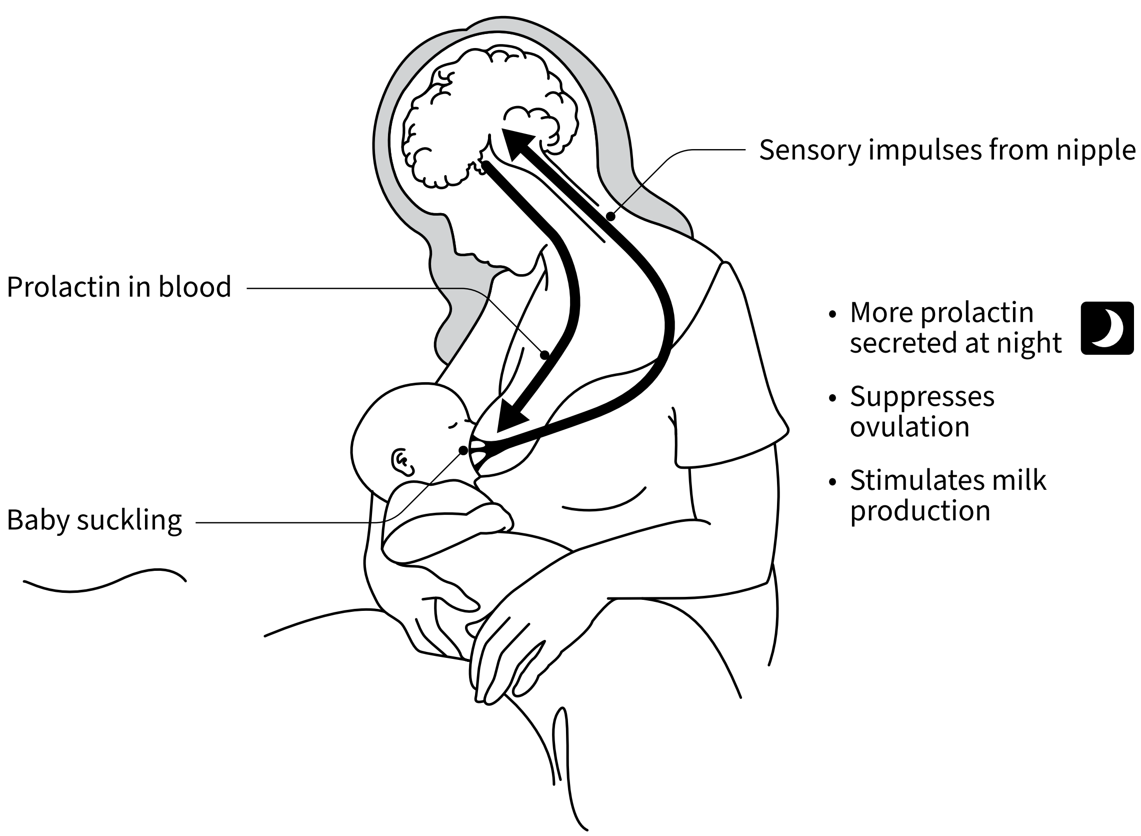
2.2.1 Prolactin

Prolactin is necessary for the secretion of milk by the cells of the alveoli. It stimulates both mammogenesis and lactogenesis. During pregnancy, prolactin is produced and blood concentration increases as pregnancy progresses. Oestrogen and progesterone inhibit milk synthesis and high blood concentrations of these hormones present during pregnancy prevent copious milk production. After expulsion of the placenta, progesterone concentrations in the mother's blood decline sharply, allowing milk production to increase rapidly *(2,4)*.

The more a baby suckles and stimulates the nipple, the more prolactin is produced, and the more milk is produced. This effect is particularly important while lactation is becoming established.

Prolactin is secreted in response to suckling during early breastfeeding (See **Fig. 2.2**) and frequent breastfeeding (or expressing) during this period is critical to establishing and maintaining breastfeeding. Blood concentration increases gradually and peaks around 30 minutes after the beginning of a breastfeed. During the first few weeks, the more a baby suckles and stimulates the nipple, the more prolactin is produced, and the more milk is produced. This effect is particularly important while lactation is becoming established. Prolactin secretion in response to breastfeeding (or expressing) is greater at night than during the day. Therefore, mothers should be encouraged to breastfeed or express their milk overnight, especially during the first few weeks after birth *(4).*

Fig. 2.2 Prolactin secretion

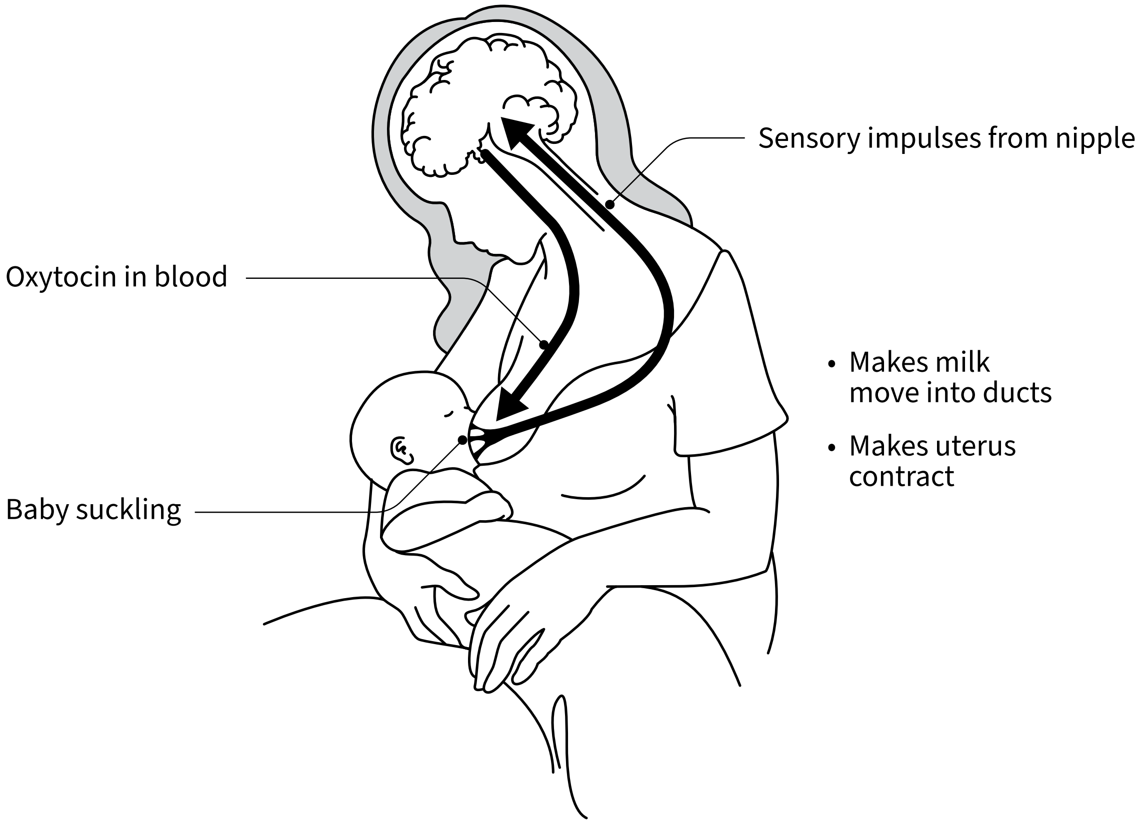


Although prolactin is still necessary for milk production during lactogenesis III, there is not a close relationship between the concentration of prolactin in the blood and the amount of milk produced *(3,4)*.

2.2.2 Oxytocin

Oxytocin's primary function is to make the milk that is already in the breast available to a baby during a breastfeed. Oxytocin is secreted by the posterior lobe of the pituitary gland in response to nipple stimulation and other stimuli, including when a mother sees, touches, smells, hears or thinks about her baby (see **Fig. 2.3**). Oxytocin causes the myoepithelial cells that surround the alveoli to contract, simultaneously compressing the alveoli and dilating the ducts that carry milk from the alveoli to the nipple. This process is known as the oxytocin reflex, the "letdown reflex" or the "milk ejection reflex" *(4)*.

Fig. 2.3 Oxytocin reflex



Anxiety, pain and extreme fatigue can cause temporary inhibition of the oxytocin reflex, delaying the movement of milk from the alveoli to the nipple. The oxytocin reflex can be restored very quickly by treating pain, ensuring the mother feels comfortable and safe while she breastfeeds, providing reassurance and helping her to manage anxiety. Separation of mothers and babies can also inhibit the oxytocin reflex and make establishing breastfeeding more difficult *(4)*.

The signs of an active oxytocin reflex are:

- a tingling sensation or feeling of fullness in the breast
- feeling thirsty or dry mouth
- milk leaking from the other breast during suckling
- change in the baby's suckling pattern from short, shallow sucks and infrequent swallowing to slow deep suck and swallow combinations

The most important and reliable sign of activation of the oxytocin reflex is that a change in the baby's suckling pattern occurs. If one or more of the other signs is present, this demonstrates activation of the oxytocin reflex. However, the absence of these other signs does not necessarily mean that the reflex is not active – the signs may not be obvious, and the mother may not be aware of them *(4).*

2.2.3 Feedback inhibitor of lactation

Milk production is also controlled by a factor that functions locally to inhibit milk synthesis. Feedback inhibitor of lactation (FIL) is a polypeptide present in breast milk. When milk is removed from the breast, milk synthesis resumes.

FIL enables the amount of milk produced to be determined by how much the baby takes. If the baby suckles effectively, the breasts produce the amount of breast milk the baby needs. This mechanism helps to protect the breast from the harmful effects of becoming too full and is particularly important for ongoing regulation after lactation is established.

Milk production is controlled by feedback inhibitor of lactation locally and independently within each breast. If a baby suckles only on one breast the other breast will stop making milk and may become smaller *(2,4)*.

2.3 How breastfeeding works

Breastfeeding begins as an instinctive behaviour. Immediately after birth, a healthy term infant in skin-to-skin contact with their mother uses innate reflexes to find the nipple and attach. Healthcare workers can best support a first feed by not interfering with or rushing this process, which can take up to an hour. With subsequent feeds, healthcare workers who have the appropriate knowledge and skills can then do more to help mothers and babies with breastfeeding *(5)*.

Babies are born with a number of innate reflexes that enable them to find their mother's nipple, attach and begin to suckle immediately after birth.

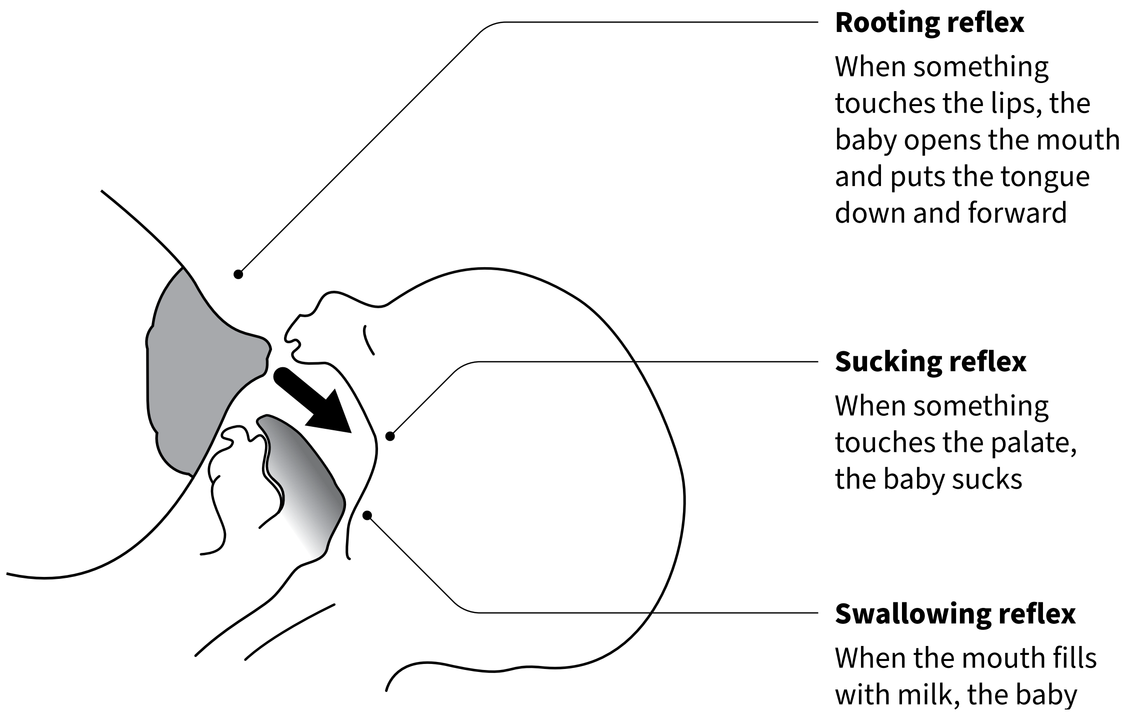
2.3.1 Innate reflexes

Babies are born with a number of innate reflexes that enable them to find their mother's nipple, attach and begin to suckle immediately after birth *(7,8)*. When a baby is placed prone on their mother's abdomen, they will pull their legs under their body and kick them out in a crawling motion. Babies use this reflex immediately after birth to crawl up to the breast and initiate breastfeeding. The crawling reflex may be present for up to two months *(2,4).*

Once they have located the breast, infants use a series of oral reflexes to breastfeed (see **Fig. 2.4**).

- **Rooting reflex:** turning towards a touch on the lips or cheek, open their mouths, and put their tongues down and forward.
- **Suckling reflex:** suckling in response to something touching their palate.
- **Swallowing reflex:** swallowing in response to their mouth filling with milk.

Fig. 2.4 Oral reflexes in the baby



2.3.2 Feeding cues

A mother should be taught to recognize early and late feeding cues and encouraged to respond to early feeding cues. Responding to early feeding cues makes breastfeeding easier. When early feeding cues are missed or ignored, babies may become distressed. A distressed baby may be fussy at the breast and need soothing before they can settle and attach to the breast to feed effectively (see **Fig. 2.5**) *(6,7)*.

Responding to early feeding cues makes breastfeeding easier.

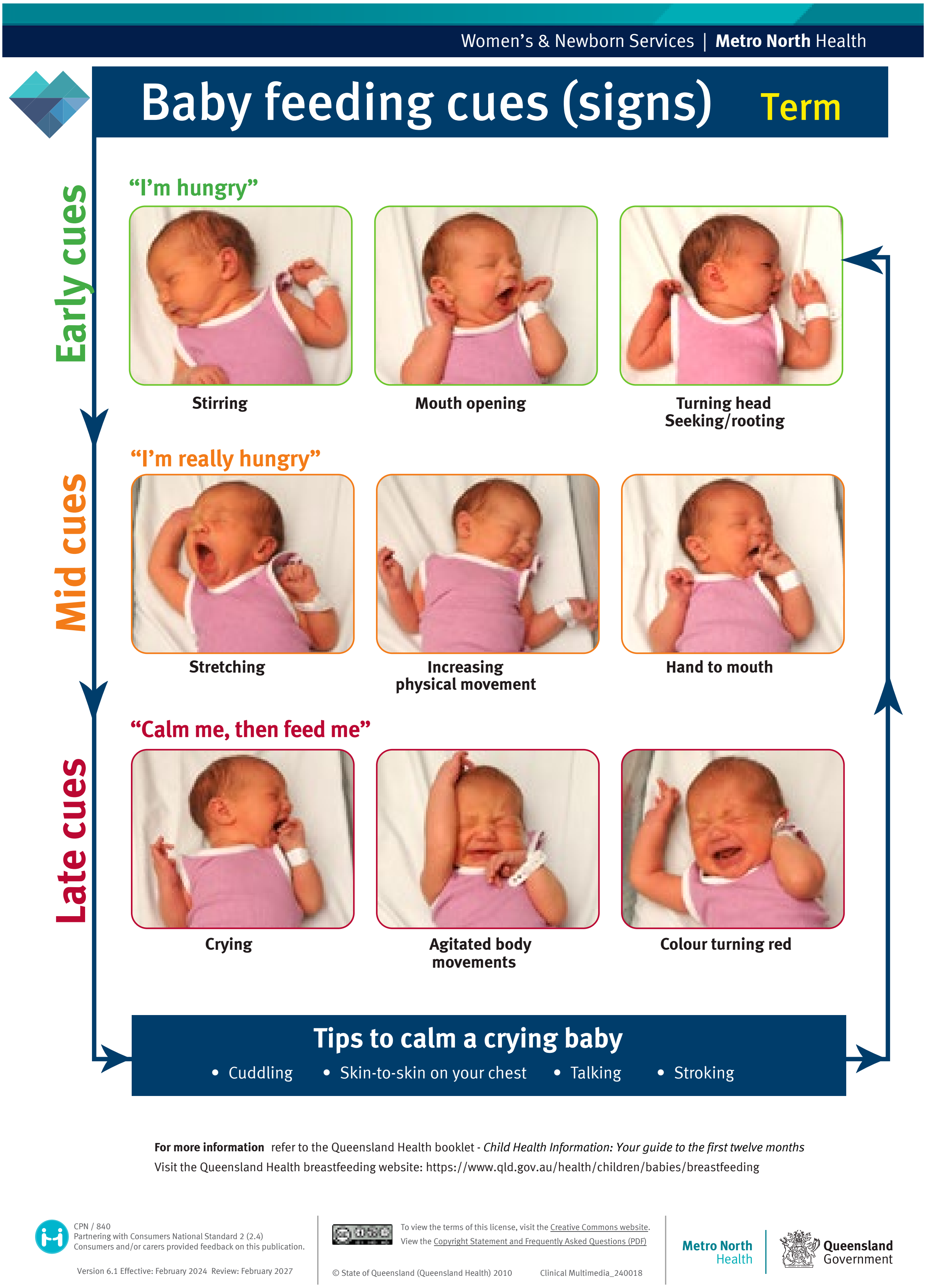
Early feeding cues include:

- opening the mouth and turning the head trying to find the breast (seeking or rooting)
- restlessness and waking
- vocalizing (grunting or squeaking)
- making hand to mouth movements
- making sucking movements
- sucking fingers
- head bobbing when held prone
- salivating.

Late feeding cues include:

- crying
- going back to sleep (withdrawing).

Fig. 2.5 Feeding cues in the newborn infant



Source: State of Queensland (Queensland Health); Metro North Health; Women's Newborn Services. Baby feeding cues (signs) term; version 6.1. Brisbane; Australia (2024). Licensed (https://creativecommons.org/licenses/by-nc-nd/4.0/).

2.3.3 Positioning and attachment

In the first few days after birth, many mothers need skilled support to ensure that their baby can attach and suckle effectively. Direct observation of a full breastfeeding session, including before and after the baby is attached, allows healthcare workers to support effective breastfeeding by observing whether a baby is properly attached and helping the adjust positioning and/or correct attachment *(7,8)*.

Providing immediate skin-to-skin care after birth and allowing the baby to initiate the first breastfeed establishes a positive pattern for subsequent feeds.

The first breastfeed usually occurs while the baby is receiving skin-to-skin care immediately after birth. An infant who is lying prone on the mother's abdomen or chest will begin to search for the breast. As the baby nears the breast, their head usually starts to bob up and down until their face makes contact with the nipple. Their jaw will open, their tongue will fall forward to cup the breast and they will take the nipple and much of the areola into their mouth. When the nipple touches the roof of the baby's mouth, the suckling reflex is triggered, and breastfeeding begins. Positioning the mother comfortably, in a semi-reclined position with her head well supported, usually with pillows, enables the mother to relax while watching the baby's first breastfeed and provide support under the baby's buttocks if required. Providing immediate skin-to-skin care after birth and allowing the baby to initiate the first breastfeed establishes a positive pattern for subsequent feeds *(9,10)*.

For later breastfeeds, a modified, usually more upright version of this position may help the mother and baby to breastfeed comfortably. However, mothers breastfeed in a range of positions, many of which are unique to an individual pair. As long as the mother is comfortable and the baby is properly attached, there is no need to intervene. Mothers can breastfeed sitting, standing, semi reclined or in a side-lying position. Whatever position she chooses, the mother needs to be relaxed, comfortable and well supported (see **Box 2.1**) *(8)*.

In the first few days after birth, many mothers need skilled support to ensure that their baby can attach and suckle effectively.

The baby can breastfeed in several different positions in relation to the mother. The most common position is known as the cradle hold. In this position the baby is held closely across the mother's abdomen and supported along her forearm, using the arm on the same side as the feeding breast. The baby's head is tilted back and resting near her elbow. This position can be used when seated or standing *(9)*.

Alternatively, the mother can use the far side arm to support the baby's body, leaving the near-side arm free to shape the breast or hold the baby's hand. In this position the mother's hand is placed behind her baby's neck and shoulders with the baby's rump resting near the bend of her elbow. This is known as the cross-cradle position *(9)*.

Box 2.1 How to help a mother to position her baby for attachment

- Greet the mother and ask how breastfeeding is going
- Assess a breastfeed, noticing baby's position and signs of proper or poor attachment
- Explain what might help and ask whether she would like you to show her
- Make sure that she is comfortable and relaxed
- Sit down yourself in a comfortable convenient position
- Explain how to hold her baby and show her the four key points (use a doll to demonstrate if necessary):
  - baby's head and body are in line
  - baby is held close to mother's body
  - baby's whole body is supported
  - baby approaches breast nose to nipple
- Show the mother how to support her breast only if needed:
  - with her fingers flat against her chest wall below her breast
  - with her first finger supporting the breast
  - with her thumb above
  - her fingers should not be too near the nipple
- Explain or show her how to help the baby to attach:
  - touch her baby's lips to her nipple especially the upper lip
  - wait until her baby's mouth is opening wide
  - move her baby quickly onto her breast, aiming the lower lip below the nipple
- Notice how she responds and ask her how her baby's suckling feels.
- Look for signs of good attachment. If the attachment is not good, try again.

See **Section 5.3** Communication and support skills

Source: World Health Organization, United Nations Children's Fund. Competency verification toolkit: ensuring the competency of direct care providers to implement the Baby-Friendly Hospital Initiative. Geneva: WHO; 2020.

Women who have a caesarean birth, very large breasts or are feeding twins often prefer placing their baby in an underarm or side lying position. The baby is positioned along the mother's near side forearm, with her hand supporting the neck and shoulders and their hips, legs and feed wrapped around the mother's body towards her back *(9)*.

The baby's body should face the mother and be held close to the mother's body.

Starting with the tip of the baby's nose level with the nipple encourages the baby to open their mouth wide.

When a mother is very tired or finds sitting uncomfortable, she may prefer to breastfeed in a side-lying position. In this position, the mother's back and head must be well supported. Pillows can be used for this. The baby is placed in a side-lying position, close to the mother's body and supported from behind with a pillow or the mother's arm *(9)*.

Regardless of the position a mother chooses for breastfeeding, the baby's body should face the mother and be held close to the mother's body. The baby's nose should be level with the nipple, neck slightly flexed, chin touching breast and body well supported and stable. There should be no gaps between the mother's and baby's bodies *(9).*

Starting with the tip of the baby's nose level with the nipple encourages the baby to open their mouth wide, bury the chin and flex the neck to reach over the nipple with the upper mandible. When the baby's mouth opens wide, rather than leaning forward to push the breast into the baby's mouth, the mother should pull the baby to her breast *(9)*.

The words, straight, close, supported and facing may be used to help remember these principles *(8)*:

STRAIGHT – ear, shoulder and hip should form a straight line

CLOSE – chin touching the breast, chest and tummy touching mummy

SUPPORTED – if the baby is not lying prone on the mother's body, the whole body is supported

FACING – baby's body facing the mother's body and not twisted at the neck or body, nose level with nipple.

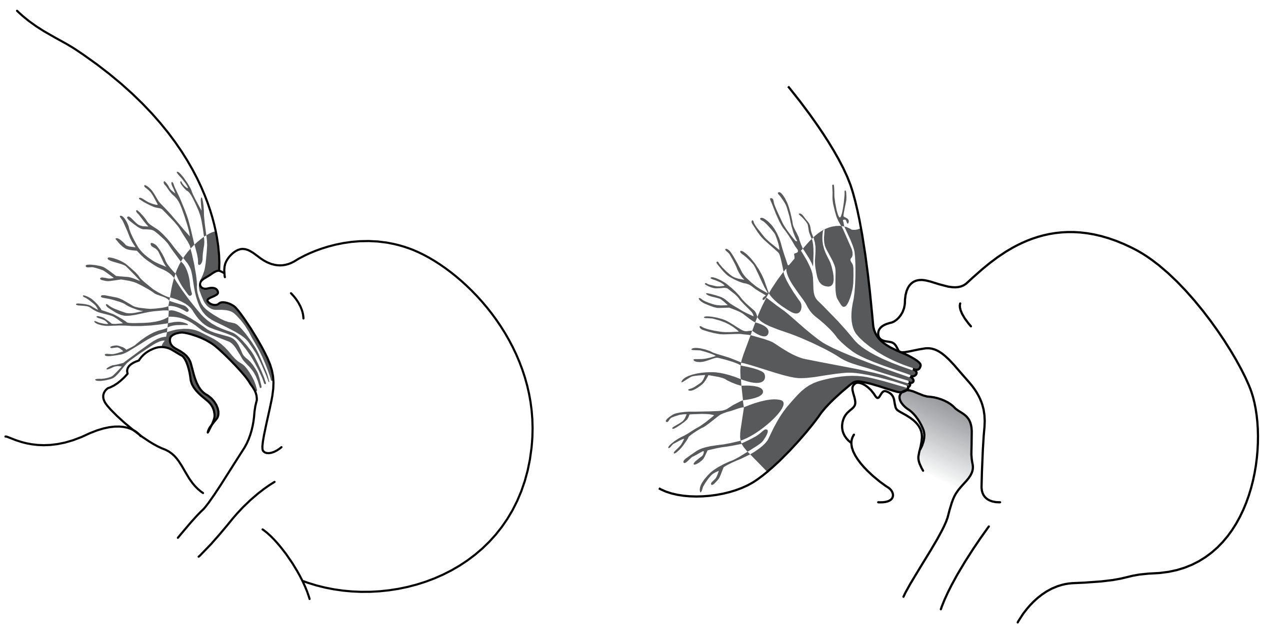
Grasping the back of a baby's head or pushing it towards the breast will cause the baby to pull their head away from the breast and prevent proper attachment. Instead, place a wrist behind the baby's shoulders and use the hand to support the baby's neck *(6,8,9)*.

Fig. 2.6 shows an internal view of a well attached baby (suckles from the breast, drawing the nipple beyond the hard palate) and of a poorly attached baby (sucks shallowly, only on the nipple).

Fig. 2.6 Internal view of a well attached baby compared to a poorly attached baby

Well attached baby: draws nipple beyond hard palate

Poorly attached baby: sucks only on the nipple



Notice that a well-attached baby:

- draws much of the areola and the tissues underneath it, including the larger ducts into the baby's mouth
- stretches the breast to form a long 'teat' but the nipple, itself, only forms about one third of the 'teat'
- extends their tongue over the lower gums and around the milk ducts (The baby's tongue is in fact cupped around the sides of the 'teat', but a drawing cannot show this)
- suckles from the breast and not from the nipple.

As the baby suckles, a wave passes along the tongue from front to back, pressing the teat against the hard palate, and pressing milk out of the sinuses into the baby's mouth from where he or she swallows it. The baby uses suction mainly to stretch out the breast tissue and to hold it in their mouth. The oxytocin reflex makes the breast milk flow along the ducts, and the action of the baby's tongue presses the milk from the ducts into the baby's mouth. When a baby is well attached, their mouth and tongue do not rub or traumatise the skin of the nipple and areola. Suckling is comfortable and often pleasurable for the mother. If breastfeeding is painful, the baby is not properly attached. Adjusting positioning and correcting attachment eliminate pain and allow damaged nipples to heal *(9).*

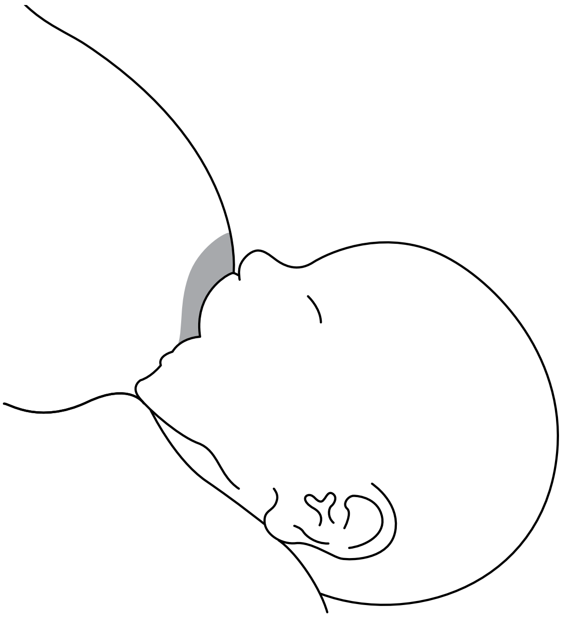
When a baby is not properly attached (see **Fig. 2.7**), they may not transfer milk effectively from the breast. Poor attachment can cause injury to the nipple including grazing and fissures *(4,9,11).*

If breastfeeding is painful, the baby is not properly attached.

Pain is the most important indication of improper attachment. A baby may suck harder to try and get the milk, and as a result, pull the nipple in and out of their mouth. The baby's tongue may rub against and damage the skin. The increased suction pressure may pinch and squash the nipple against the hard palate. Improper attachment is a very common cause of sore nipples and may result in fissures (or "cracks") which could lead to infections. Improper attachment may also interfere with milk transfer and may decrease milk output *(4,6,9,11).* Causes and management of poor attachment are addressed in **Section 7.**

Fig. 2.7 External view of a well attached baby compared to a poorly attached baby, with signs of proper attachment

A well attached baby



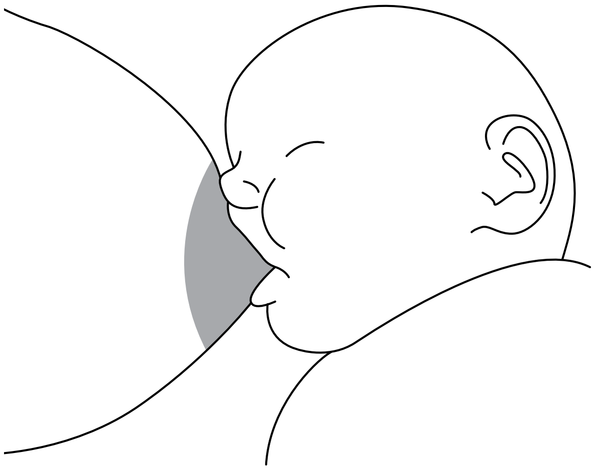
The four key signs of proper attachment are:

1. The baby's mouth is wide open.
2. The baby's lower lip is turned outwards.
3. The baby's chin is touching the mother's breast.
4. More areola is seen above the baby's top lip than below the bottom lip.

If **all** of these signs are present, the baby is well attached.

If even one of these signs is absent, the baby is not well attached.

A poorly attached baby



Signs of poor attachment:

1. nipple pain (this is **always** an indication of poor attachment)
2. chin not touching breast
3. little or no breast is drawn into the baby's mouth and they suck only on the nipple.
4. much of the areola is visible
5. dimpled cheeks
6. Infant makes smacking or clicking sounds when suckling
7. Infant suckles frantically and swallows infrequently (suck, suck, suck, swallow)
8. Infant fusses at the breast or pulls away and refuses to breastfeed
9. grazed or broken skin on the nipple
10. nipple is pinched after breastfeeding

If any one of these signs is present, then attachment needs to be improved.

2.3.4 Assessing milk transfer

When a baby is well attached, they remove and transfer milk effectively. You can often see and hear a baby swallowing milk when they suckle effectively. A baby who is not well attached (see above) is unlikely to transfer milk effectively *(4,5,6,8,11)*.

Signs of effective milk transfer

- After a period of short, urgent suckling and infrequent swallowing, the baby's suck pattern changes to a slow rhythmic pattern in which the chin can be observed pausing at its nadir as the baby swallows a bolus of milk. The swallow may be audible and the pattern repeat at a rate of about one suck per swallow per second.
- No clicking sounds.
- Periods of concerted suckling are punctuated by short pauses during which the baby rests for a few seconds.
- When a baby starts suckling again, they may suckle quickly a few times to stimulating milk flow, and then resume the slow, deep suck-swallow pattern.
- The baby's cheeks remain rounded during the feed and are not dimpled.
- The baby's hands relax and open during the feed.
- Towards the end of a feed, suckling usually slows down, with fewer deep suckles and longer pauses between. As the volume of milk taken reduces it becomes richer in fat.
- Babies usually release the breast spontaneously when they are sated and often fall into a deep sleep. A period of very quick, shallow, flutter-like sucking is common during the transition from active feeding to sleeping.
- The breast feels softer immediately after breastfeeding.
- If the baby does not remove breast milk effectively, the breasts may become engorged (breasts are painful and too full of milk). Because the baby does not get enough breast milk, they may be unsatisfied and cry. They may want to feed often or for a very long time at each feed, or they may get frustrated and have difficulty suckling. Eventually, if breast milk is not removed, the breasts may make less milk.
- If a baby cannot transfer breast milk effectively, they urinate less frequently, the urine becomes more concentrated, darker in colour and develops a stronger odour. If this situation persists, the baby will not gain weight as expected.

2.3.5 Responsive breastfeeding

Responsive breastfeeding *(8,11,12)* is when a mother responds to her baby's feeding cues, as well as her own desire to feed her baby. Other terms for responsive feeding are "demand feeding", "unrestricted feeding", "perceptive feeding", "ad-lib feeding" and "baby-led feeding." Responsive feeding provides comfort and reassurance for both the baby and the mother while nourishing the baby.

To ensure adequate milk production and flow for six months of exclusive breastfeeding, babies need to feed as often and for as long as they want, both day and night. Provided the baby is properly attached to the breast and breastfeeding effectively, it is important not to restrict the frequency or the duration of feeds. Infants who are feeding responsively, according to their appetite, obtain what they need for satisfactory growth.

Babies should suckle on the breast until they spontaneously release the nipple. After a short rest, babies can be offered the other side, which they may or may not want. Babies who have had enough milk at a feed are usually relaxed and quiet. They may fall asleep at the breast.

Encourage mothers to continue breastfeeding in response to her baby's hunger cues (see **Fig. 2.5**) and reassure them that breastfeeding alone provides all the nourishment (food and drink) babies need until they are six months old and ready to begin complementary feeding.

When babies are fed responsively, the frequency and duration of breastfeeds varies. During the first day or two, a baby may sleep for long periods and feed infrequently and then feed very frequently for a day or two. Frequent feeding and unsettled behaviour are common in the 24 hours before the onset of copious lactation and are thought to contribute to establishing breastfeeding. A more regular pattern with some longer intervals between breastfeeds will be established over the first few weeks of life. It is common for babies to breastfeed more frequently for a period of each day, usually in the late afternoon or evening. This pattern is called cluster feeding and can be, but is not always, associated with growth spurts.

Some babies need to breastfeed more often than others and some mothers need to breastfeed more often than others. Breasts vary in their capacity for synthesizing and storing milk and this is not closely related to breast size. Since milk removal triggers milk synthesis, responsive breastfeeding allows the baby to take sufficient breast milk over the course of a day and regulates milk production.

Breastfeeding never completely empties the breast and more milk is available from the first breast almost as soon as the baby has finished the second. If a baby seems to be still hungry after feeding from both breasts, the first breast can be offered again. This will increase daily milk production *(8).*

To ensure adequate milk production and flow for six months of exclusive breastfeeding, babies need to feed as often and for as long as they want, both day and night.

The duration of breastfeeds can vary in length from a few minutes to 30 minutes. During the first few weeks of life, some feeds can take even longer than 30 minutes. As babies grow, they can become very efficient at the breast and take a full breastfeed in fewer than five minutes *(8,11,12).*

If a baby stays on the breast for a very long time (more than a half to one hour for every feed) or if they want to feed very often (more often than every 1–1.5 hours), then a full breastfeeding session should be observed. Prolonged, frequent feeds can be a sign of ineffective suckling and inefficient transfer of milk to the baby. This is usually due to improper attachment, which may also lead to sore nipples. The baby's attachment needs to be checked and improved. Once the attachment is improved, transfer of milk usually becomes more efficient, and the feeds become shorter or less frequent. At the same time, the risk of nipple damage is reduced *(8,11).*

Encourage mothers to continue breastfeeding in response to her baby's hunger cues.

Babies take different amounts of milk at each feed. A typical 24-hour intake of milk varies between mother-infant dyads from 440–1220 g, averaging about 800 g per day throughout the first six months *(13)*.

Feeding volumes cannot be accurately measured during breastfeeding. Therefore, it is important for all healthcare workers to monitor measurable signs of adequate intake, such as urine output and stooling patterns as well as growth trajectory. In the first few days of life, output of urine and stool begins slowly and increases progressively. As a rough guideline, during the first week of life, the typical number of wet diapers/nappies and stools per day correlates with the infant's day of life. Meaning, an infant would have one wet diaper/nappy and stool once on the first day of life, two on the second day of life, up to six on day six *(4,9)*.

As described in detail in other sections, growth monitoring and observation of the baby's general condition can help to reassure healthcare workers and caregivers that a baby is taking the appropriate volume of milk needed for satisfactory growth.

2.4 Breast milk

Breast milk is uniquely adapted to feed human babies. There is no comparable commercial product because the content of breast milk is variable and changes throughout the course of lactation, including during each individual feed, in a 24-hour day, and over time as a baby grows.

The content of breastmilk is variable and changes throughout the course of lactation, including during each individual feed, in a 24-hour day, and over time as a baby grows.

2.4.1 Colostrum (or "first milk")

Colostrum is the "first milk" that is secreted, starting during the second trimester of pregnancy and continuing into the first few days after birth. This first milk is thicker than later milk and can vary in colour from transparent to yellow. It contains a larger percentage of protein, minerals, and fat-soluble vitamins (A, E and K) than later milk. It is produced in small amounts, appropriate for the size of an infant's stomach. It helps with passage of meconium through the gut *(14)*.

In addition to complete nutrition, colostrum provides important immune protection to infants against micro-organisms in the environment. It is rich in white blood cells and antibodies, especially secretory immunoglobulin A (sIgA). Colostrum also contains epidermal growth factor, which promotes the maturation of the intestinal epithelium, making it less permeable to infective microbes and allergens.

Colostrum helps with the establishment of a healthy microbiome. An infant's microbiome is the population of bacteria and other microbes in their gut and on their skin. It is formed largely from their mother's microbiome, which includes beneficial bacteria in her breast milk. The infant's microbiome matures between birth and early childhood. A healthy microbiome appears to be protective for life, thus reinforcing the importance of early infant feeding on long-term health and development. Routine supplementation is not recommended because it disrupts the microbiome *(15–18)*.

2.4.2 Transitional milk

The transition from colostrum to mature milk starts in the first few days after birth and takes as long as two weeks to complete. During this time, it is called transitional milk *(2)*.

Transitional and then mature milk are produced in larger quantities and have larger amounts of fat and lactose than colostrum. Increased milk production during the first week makes the breasts feel full. The milk is said to have "come in." On the third day, an infant is usually taking about 300–400 ml per 24 hours, and on the fifth day about 500–800 ml *(2)*.

2.4.3 Mature breast milk

Mature breast milk contains all the nutrients an infant needs for the first six months of life, including fat, carbohydrates, proteins, vitamins, minerals and water. It is easily digested and efficiently used. In addition to complete nutrition, breast milk also contains numerous components that have immune and gut-maturing properties *(4,9,6,11,19,20)*.

2.4.4 Nutritional components of breast milk

Fats

Breast milk fat contains long chain polyunsaturated fatty acids (docosahexaenoic acid or DHA, and arachidonic acid or ARA) that are needed for cell membranes and growth of the brain and nervous system. Breast milk contains about 3.5 g of fat per 100 ml, which provides about half the energy content of the milk.

The fat is secreted in small droplets, and the amount increases as a feed progresses. Towards the end of a feed, breast milk is higher in fat, hence the recommendation to let a baby finish feeding and release one breast before offering the other.

Bile-salt stimulated lipase is an enzyme in breast milk which becomes active in the small intestine, so the fats in breast milk are easily digested and metabolized. The fats in artificial milks are less completely digested *(2)*.

Carbohydrates

Lactose is a sugar found only in mammalian milk. It is the main carbohydrate in breast milk, a disaccharide made of glucose and galactose, and is an important source of energy. Breast milk contains about 7 g of lactose per 100 ml. Lactose is digested by lactase, an enzyme in the brush border cells of the intestinal mucosa. Glucose and galactose are absorbed into the blood and help with the absorption of certain minerals such as calcium and magnesium *(2)*.

Protein

Breast milk is species specific. Breast-milk protein differs in both quantity and quality from the milks of other mammals. Human milk contains a suitable balance of amino acids for a human baby. The concentration of protein in human breast milk is about 0.9 g per 100 ml, which is lower than in most animal milks *(2)*.

Human milk protein consists of whey, the soluble proteins which includes anti-infective factors, and casein, which forms curds, in a 60 to 40 ratio. Breast milk contains less casein than animal milk, and it has a different molecular structure. Casein in human milk forms soft curds that are easily digested; casein in cow's milk forms hard rubbery curds which are more difficult to digest.

Human infants need less protein because they grow more slowly than other mammals. Less protein is also protective of an infant's immature kidneys. A baby's immature kidneys have difficulty excreting the extra waste from the protein in animal milks *(2).*

Micronutrients

Breast milk normally contains sufficient micronutrients. Iron and zinc are present in breast milk in relatively low concentration, irrespective of maternal intake. However, their bioavailability and absorption are high. Therefore, breast milk provides all the iron and zinc that a healthy term infant needs for the first six months, provided that the infant is born with adequate iron stores *(2)*.

2.4.5 Immunological components of breast milk

Mature breast milk contains bioactive factors such as immunoglobulins and human milk oligosaccharides (HMOs) which provide protection against infection.

Anti-infective factors

Breast milk contains many factors that support the infant's immature immune system. For example immunoglobulins, principally sIgA and immunoglobulin G (IgG), adapt to the antigens a baby encounters in their environment and coat the intestinal mucosa to prevent pathogens from entering the cells; white blood cells can kill micro-organisms; antimicrobial peptides including alpha lactalbumin; lysozyme and lactoferrin can kill bacteria, viruses and fungi; cytokines influence the diversity of microorganisms that colonise the baby's gut and have an anti-inflammatory effect; hundreds of probiotic microflora, including lactobacillus and bifidobacteria, provide an essential foundation for development of a healthy immune system; and pro-and oligosaccharides prevent bacteria from attaching to mucosal surfaces *(21–23).*

The immunoglobulin sIgA contains antibodies formed in the mother's body against the bacteria in her gut, and against infections that she has encountered. These antibodies protect against bacteria which are particularly likely to be a risk in the baby's environment. Maternal supplementation with pre/probiotics may affect the immune composition of breast milk by increasing sIgA in colostrum/transitional milk *(21)*.

Oligosaccharides are the third largest component of human milk. They are complex sugars, and there are hundreds of different kinds in breast milk. They are not digested and have no direct nutritional value, but they help the intestinal epithelium to absorb nutrients. They promote the growth of beneficial bacteria in the microbiome *(21–23)*.

Oligosaccharides help to prevent infection by reducing the permeability of the epithelium to pathogens and allergens. They are similar to the binding molecules on epithelial cells to which microbes attach to enter and cause infection. The microbes attach to the oligosaccharides instead and then cannot attach to the cells. Oligosaccharides also reduce the risk of necrotizing enterocolitis (NEC), the most common and serious intestinal disease among premature babies *(24).*

2.4.6 Growth factors

Epidermal growth factor stimulates maturation and healing of the lining of the infant's intestine in a dose-dependent manner *(25, 26)*. The gut is then better able to digest and absorb nutrients, and is less easily infected or sensitized to foreign proteins.

Epidermal growth factor is found in colostrum (see **Section 2.4.1**). It is also found at increased levels in the milk expressed by mothers of premature infants and may be important for the development of extremely premature infants *(27)*. Lower gestational age of neonates is associated with higher risk of developing gastrointestinal disorders *(28)*.

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SECTION 3

Complementary feeding

Key messages

Educate and support families to:

- introduce complementary foods from six months (180 days) and continue breastfeeding up to two years or beyond;
- provide a diverse diet, including animal-source foods (such as meat, fish or eggs) and fruits and vegetables every day;
- avoid giving foods that are high in sugar, salt or trans-fats, or beverages sweetened with sugar or non-sugar alternatives;
- encourage infants and young children to eat autonomously and in response to their physiological and developmental needs (fed responsively); and
- choose either unsweetened, full-fat animal milk or yoghurt or infant formula for who are not breastfed or are fed milk other than breast milk formula milk.

Section 3 outline

3.1 Introduce complementary foods from six completed months of age

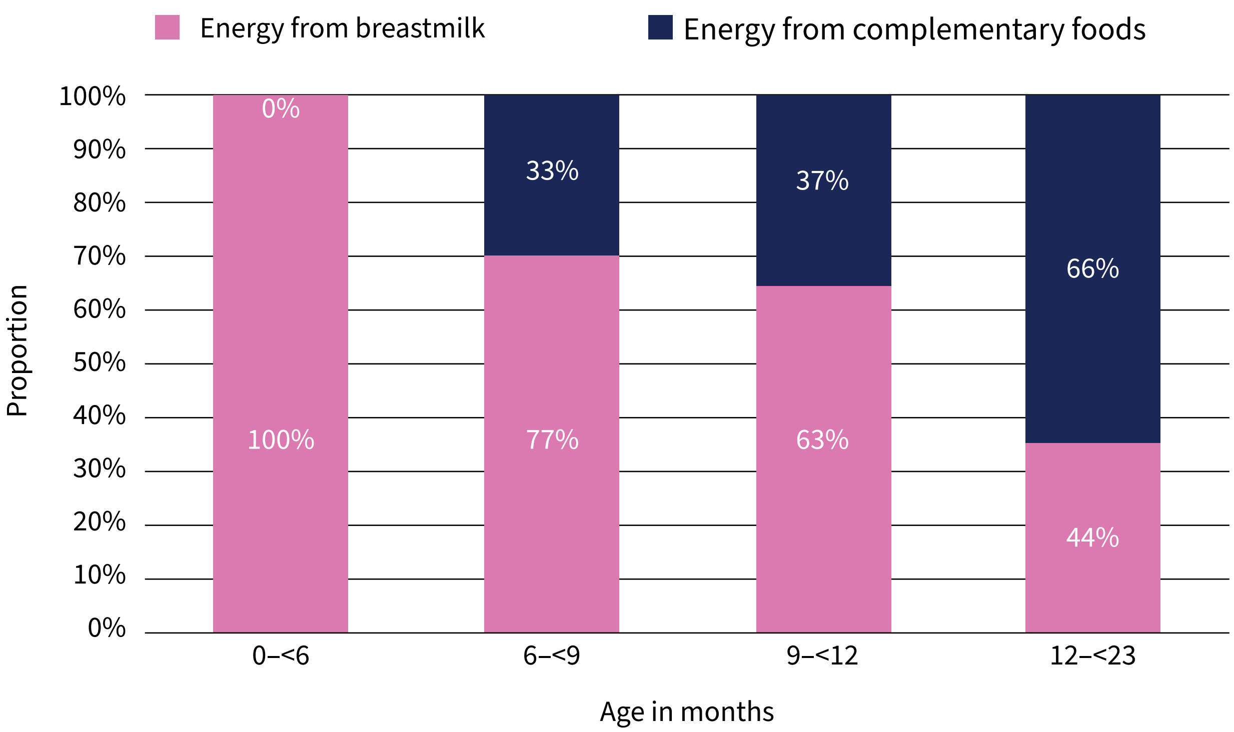
From six months infants need to diversify their diets to meet their nutrient needs *(1,2)*. At this age, they are generally developmentally ready to start eating solid or semi-solid foods. The ability to sit without support is an important indicator of readiness to start eating complementary foods as it is associated with other aspects of physiological development, including gastrointestinal, renal, and immunological system maturation. The reflex that causes very young infants to push foods out with their tongue recedes, enabling them to accept and hold semi-solid food in their mouths *(3,4)*.

Introducing complementary foods before six months risks displacing breast milk, which is a superior source of nutrients. Delaying introduction of complementary foods beyond the end of the sixth month risks depriving the infant of essential energy and nutrients needed to complement breast milk or milk from another source *(2)*.

3.2 Continue breastfeeding up to two years of age or beyond

Breast milk makes a significant contribution to children's energy intake as they transition to a diversified diet and continues to do so throughout the second year or beyond (see **Fig. 3.1**) *(2)*. Breastfeeding continues to provide effective immunological and nutritional support for as long as it continues *(1,2)*.

Fig. 3.1 Energy supplied by breastmilk and complementary foods

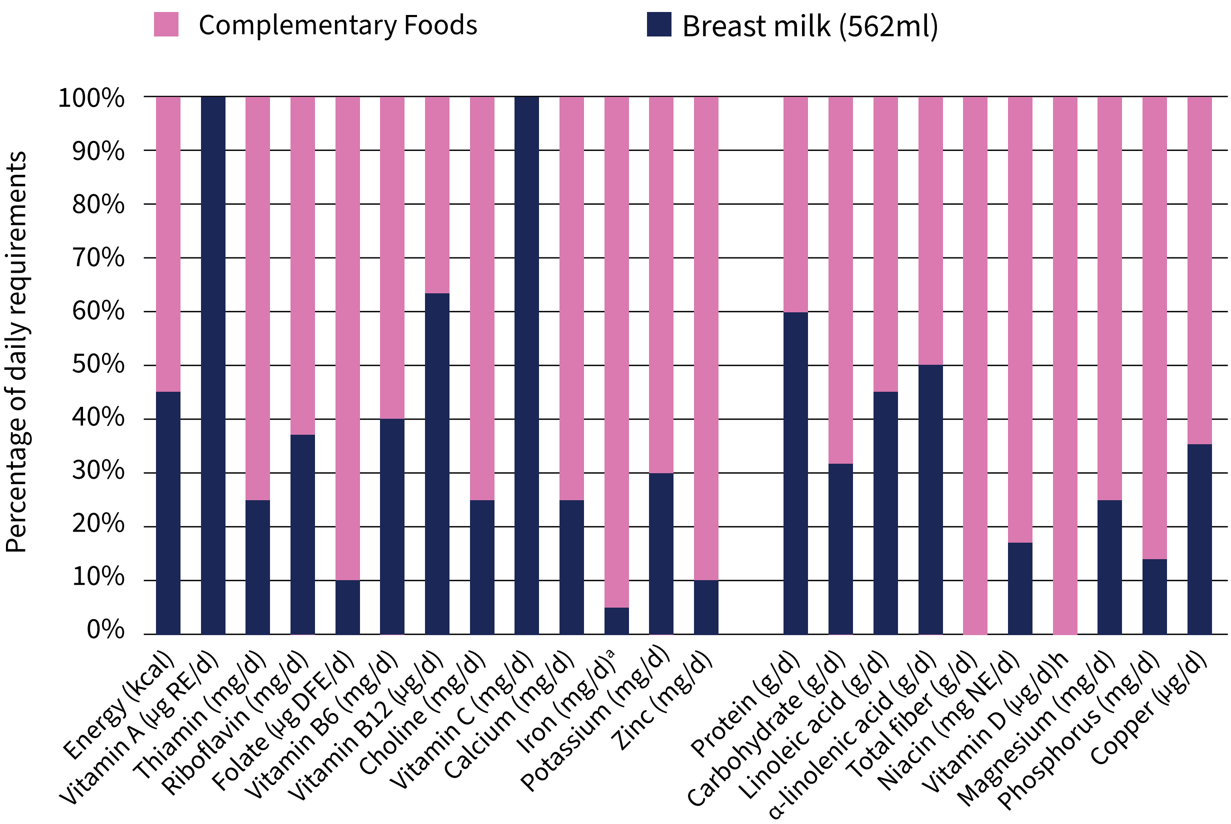


Source: WHO guideline for complementary feeding of infants and young children 6–23 months of age. Geneva: WHO; 2023.

Breast milk continues to contribute to meeting macro- and micro-nutrient needs as complementary foods are introduced up to two years of age and beyond. These nutrients are critical for brain development and function. **Fig. 3.2** illustrates the relative contributions made by breast milk and complementary foods to a healthy diet during the second year of life. The nutritional qualities of breast milk are particularly important in resource-poor settings and where starchy staples predominate *(1,2).*

During the second year of life, breast milk continues to provide a large array of non-nutritive substances that offer immune protection to the child during the second year of life. These include immunoglobulins, lysozyme, lactoferrin, oligosaccharides, white blood cells, anti-microbial peptides, cytokines, and probiotic microflora *(5–9).*

Fig. 3.2 Nutrition provided by breast milk / complementary foods 12–23 months



[a] 0.5mg/L with 50% absorbtion rate

Source: WHO guideline for complementary feeding of infants and young children 6–23 months of age. Geneva: WHO; 2023.

3.3 Practice responsive feeding

Not only what a child eats, but also how a child is fed is important. Responsive feeding has been defined as "feeding practices that encourage the child to eat autonomously and in response to physiological and developmental needs, which may encourage self-regulation in eating and support cognitive, emotional and social development" *(10).* **Table 3.1** compares the features of responsive feeding to non-responsive feeding practices.

Responsive feeding has been shown to promote healthy growth and development and to encourage children's self-regulation, which is important to prevent both under- and overfeeding. Responsive feeding is characterized by positive interactions between the caregiver and the child. There are four stages of responsive feeding:

i) the child signals hunger or satiety through actions and expressions

ii) the caregiver recognizes the cues

iii) the caregiver's response is prompt, nurturing, and developmentally appropriate

iv) the child experiences the caregiver's response.

Table 3.1 What is responsive feeding?

Responsive feeding	Non-responsive feeding
Encourages the child to eat, but does not force the child, being attentive to their hunger and satiety cues	Dominates the feeding situation through controlling and pressuring behaviours (e.g. Forcing a child to finish all the food on the plate even though the child has shown signs of being full of satiated)
Feeds slowly and patiently, encouraging the child to progressively self-feed. Recognizes that messiness is part of learning to self-feed	Feeds child directly, even when the child is able to self-feed, and does not give attention to prompts of being ready or full
Encourages family mealtimes to model healthy eating practices	Fails to direct child behaviours that interfere with the establishment of health food preferences and eating routines (e.g. frequent snacking on junk foods, consumption of sugar-sweetened beverages in place of water)
Minimizes distractions during meals as the child may easily lose interest in feeding	Ignores the child or is distracted during mealtimes (e.g. is preoccupied with a television or cellphone during mealtimes)
Understands that feeding times are periods of learning and a time to bond. Talks to the child during feeding, making eye-to-eye contact	Does not utilize mealtimes as an opportunity to interact with the child (e.g. does not talk to the child or make eye-to-eye contact)
If the child refuses certain foods, experiments with different food combinations, tastes textures and methods of encouragement.	Does not provide the child with enough opportunities to re-try foods once refused.

Source: World Health Organization, United Nations Children's Fund. Nurturing young children through responsive feeding: thematic brief. Geneva: WHO; 2023.

Responsive feeding helps children develop self-regulation over food intake and facilitates their transition to eating independently. Responsive feeding is a core element of nurturing care. Responsive feeding practices are described in **Fig. 3.3** *(10).*

Fig. 3.3 Practices related to responsive feeding

Practices related to responsive feeding

Responsive feeding in cases of feeding difficulties

responsive feeding, as part of broader services, is especially important for children with feeding difficulties or disabilities. Caregiver avoids non-responsive, force-feeding practices which may be common among caregivers of these children, especially if they have limited access to support services and assistive devices.

Daily routines/structure

caregiver establishes well-structured daily routines for eating, sleeping, playing, and bathing.

Avoid distractions during feeding

caregivers do not use screens or cell phones (e.g., texting) during meals to prevent the child from distractions interfering with feeding.

Eat as a family

infant or child can actively observe caregivers and other family members prepare and consume healthy foods and beverages in a clean and pleasant environment, and they bond as they eat together.

Caregiver-child interactions

caregiver interacts with the child using loving and stimulating language and actions, including eye-to-eye contact.

Soothing

caregiver does not use food to calm the child. Instead, the caregiver responds to thechild's needs, perhaps by rocking, changing the diaper, or playing.

Hunger and satiety cues

caregiver identifies and responds in an emotionally supportive and predictable way to hunger and satiety cues, recognizing changing cues as the child develops, and differentiating hunger from other issues that may cause an infant or young child to fuss or cry.

Feeding styles

caregiver is patient and does not pressure the child to eat or finish food or limit food intake (e.g., to prevent weight gain in child). Caregiver increasingly enables the child to self-feed but provides company and guidance, even when the child can eat independently.

Introduction of complementary foods

caregiver introduces complementary foods in a timely way, appropriate to the child's developmental readiness.

Flavour preferences

caregiver offers a diverse diet with repeated exposures to healthy foods and beverages and avoids offering ultra-processed or high-salt foods and sugar-sweetened beverages.

Food consistency

caregiver offers foods of appropriate consistency, evolving as the child grows and develops.

Portion sizes

caregiver offers foods and beverages in amounts and with eating utensils (e.g., cup, spoon) appropriate to the developmental stage and nutritional needs of the child.

Meal frequency

caregiver progressively increases the child's meals with age. The child should be fed 2–3 meals per day between 6–8 months of age and 3–4 meals per day between 9–23 months of age, with 1–2 additional snacks as needed.

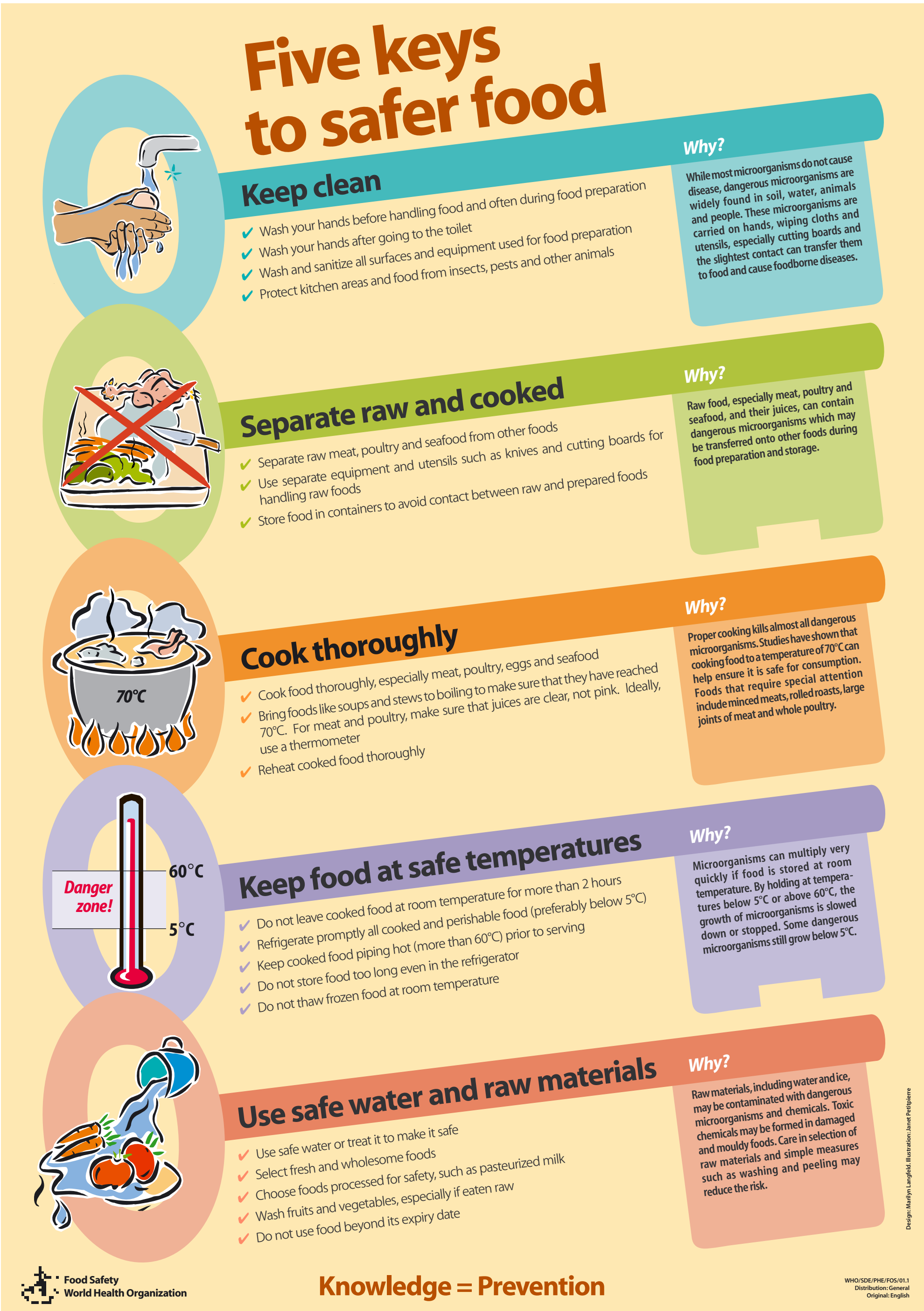
Source: World Health Organization, United Nations Children's Fund. Nurturing young children through responsive feeding: thematic brief. Geneva: WHO; 2023.

3.4 Practice good hygiene and proper food handling

Poor hygiene and improper food handling can make infants and young children very ill. Water and complementary foods may be contaminated by bacteria, viruses and parasites or toxins such as chemical substances and environmental pollutants. Contamination can cause diarrhoeal disease.

Parents and caregivers can prevent foodborne illness by following the Five Keys to Safer Food *(11)* described **Fig. 3.4** *(12)*.

Fig. 3.4 Five keys to safer food

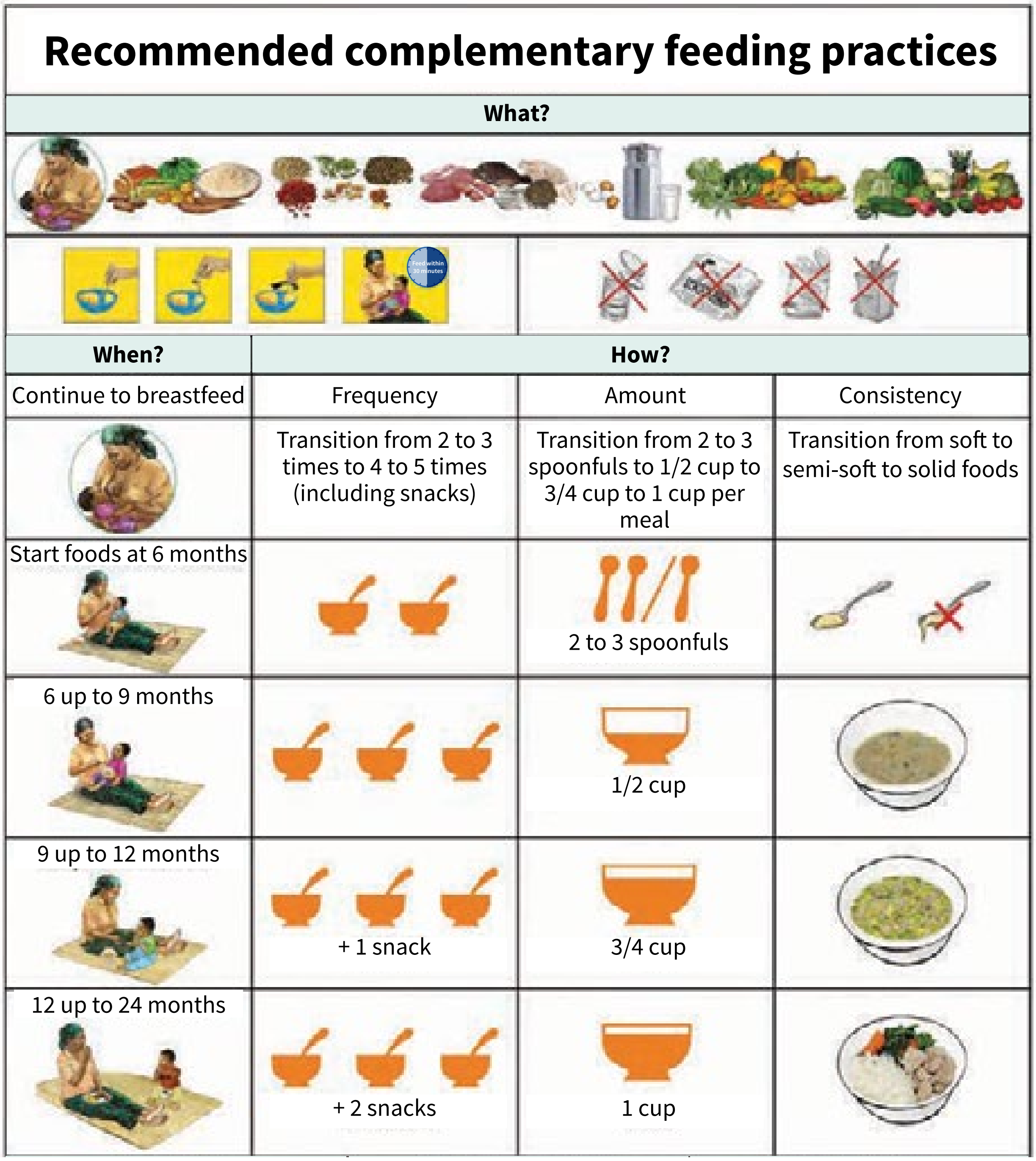


Source: Five keys to safer food manual. Geneva: World Health Organization; 2003.

3.5 Offer dietary diversity

Infants and young children need to consume a variety of foods to ensure their nutritional needs are met and to support healthy growth and development. A diet lacking in diversity increases the risk of nutrient deficiencies, many of which cannot be satisfied through nutrient supplements or fortified food products because they contain only a subset of the essential nutrients and bioactive substances found in food *(2)*. **Fig. 3.5** offers practical guidance on complementary feeding for infants 6 – 23 months *(13).*

Fig. 3.5 Practical guidance on the quality, frequency, and amount of food to offer children 6–23 months of age



Source: United Nations Children's Fund (UNICEF). Used with permission.

Foods with higher nutrient density should be key components of children's energy intake – fruits and vegetables, nuts, pulses and seeds, and animal-source foods have higher nutrient density than cereal grains. Starchy staple foods should be minimized. They commonly comprise a large component of complementary feeding diets, particularly in low resource settings, and do not provide proteins of the same quality as those found in animal source foods and are not good sources of critical nutrients such as iron, zinc and Vitamin B12. Many also include anti-nutrients that reduce nutrient absorption *(2)*.

WHO and UNICEF have defined eight key food groups for children *(2).*

To meet their nutritional needs, children need to consume foods from at least five of the following eight groups every day:

- breast milk
- flesh foods (meat, fish, poultry, and liver/organ meats)
- dairy (milk, yogurts, cheese)
- eggs
- legumes and nuts
- vitamin A-rich fruits and vegetables
- other fruits and vegetables
- grains, roots, and tubers *(19)*.

Beginning at 6 months, an infant can eat pureed, mashed or semi-solid foods. By 8 months most infants can also eat finger foods. By 12 months, most children can eat the same types of foods as consumed by the rest of the family. All members of the family need nutrient-rich foods.

As a child gets older and needs a larger total quantity of food each day, the food needs to be divided into a larger number of meals.

The number of meals that an infant or young child needs in a day depends on:

- **how much energy the child needs.** The more food a child needs each day, the more meals are needed to ensure that he or she gets enough.
- **the amount that a child can eat at one meal.** This depends on the capacity or size of the child's stomach, which is usually 30 ml per kg of the child's body weight. A child who weighs 8 kg will have a stomach capacity of 240 ml, about one large cupful, and cannot be expected to eat more than that at one meal.
- **the energy density of the food offered.** Healthy complementary foods contain at least 0.8 kcal per gram. If the energy density of food is lower than this, a larger volume of food is needed. This may need to be divided into more meals.

A breastfed infant 6–8 months of age usually needs 2–3 meals a day. A breastfed infant 9–23 months usually needs 3–4 meals a day *(14)*. Nutritious snacks may be offered between meals if children show signs, they are hungry. Snacks are often self-fed finger foods that are convenient and easy to prepare. Increases in meal frequency and size should be made in response to the child's appetite rather than their age.

If a child is not offered nutritious meals sufficiently frequently, they may not be able to meet their energy and nutritional needs. If a child is encouraged to continue eating after they have indicated they have had enough, they may breastfeed less or stop breastfeeding altogether. Displacement of breast milk may reduce the quality and amount of the child's total nutrient intake, especially in the first year of life.

3.6 Dairy products

Dairy products, including liquid animal milks, are part of a diverse diet and can contribute to nutritional adequacy. They are particularly important for children who are not breastfed and when other animal source foods are not available or affordable *(2)*.

For non-breastfed or partially breastfed (mixed fed) infants, commercial infant formula is not necessary after six completed months (180 days). Non-breastfed infants and children less than two years of age may be given full-fat animal milk (boiled for infants less than 12 months) or commercial formula. Pasteurized animal milk reconstituted evaporated (but not condensed) milk, fermented milk, or unsweetened yogurt are appropriate milk sources. Infants and young children should not be given flavoured or sweetened milks or yoghurts. Follow-up formula, including growing-up milk or toddler formula, is not necessary for children 12–23 months of age *(2)*.

3.7 Avoid unhealthy foods

Consuming foods and beverages that are high in salt, sugar or trans-fats is detrimental to children's health. These foods often displace healthier foods, and may be associated with nutrient deficiencies, excessive weight gain, and adverse cardiometabolic outcomes.

Children should not be fed foods that are high in sugar, salt and trans-fats, beverages sweetened with sugar or any food or drink that contains non-sugar sweeteners. Fast foods, convenience foods, and highly processed snack foods or "extra foods" are usually high in salt, fat and/or sugar. Children should not eat these foods.

Neither nutrient supplements nor fortified foods should be a substitute for a diverse diet consisting of healthy and minimally processed foods.

Consumption of undiluted fruit juice should also be limited because it is high in sugar and can displace more nutritious foods *(2)*.

3.8 Use nutrient supplements and fortified foods in certain contexts

Consumption of a diverse diet of locally available nutrient-rich complementary foods should always be the first choice for satisfying a young child's nutritional needs. However, in settings where such foods are not regularly available or affordable, nutrient supplements and fortified food products may help fill nutrient gaps.

WHO guidelines for micronutrient supplementation provide recommendations about the contexts when such supplements are recommended *(15)*. Micronutrient supplements should never be distributed as stand-alone interventions, rather they should always be accompanied by messaging and complementary support to reinforce optimal infant and young child feeding practices, including nutritionally adequate complementary feeding *(2)*.

Multiple micronutrient powders can provide selected vitamins and minerals without displacing other foods in the diet. For populations already consuming commercial cereal grain-based complementary foods or blended flours, fortification of these cereals can improve micronutrient intake. However, consumption of these foods should not be encouraged. Rather, parents should be encouraged to offer a diet rich in nutritious foods.

Small-quantity lipid-based nutrient supplements (SQ-LNS) may be useful in food insecure populations facing significant nutritional deficiencies. Neither nutrient supplements nor fortified foods should be a substitute for a diverse diet consisting of healthy and minimally processed foods.

3.9 Continue breastfeeding during illness

When children are ill, breast milk is an important source of nutrition. Several studies have shown that, whereas appetite for other foods decreases during illness, energy intake from breastfeeding is not affected *(2)*. Do not withhold breastfeeding during illness.

Breastfeeding more frequently meets the increased demand for fluids during illness and provides immunological support to expedite recovery. Frequent breastfeeding should be encouraged.

Children who are not breastfed who have diarrhoea can be given extra water and oral rehydration solution if they are older than six months.

Do not withhold breastfeeding during illness – breastfeeding more frequently meets the increased demand for fluids during illness and provides immunological support to expedite recovery.

A child recovering from illness or malnutrition may require extra assistance with feeding. Smaller, more frequent meals may encourage them to take enough food.

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SECTION 4

Supporting breastfeeding in perinatal care

Key messages

- The Ten Steps to Successful Breastfeeding are the benchmark for quality postnatal care.
- Mothers who have had intrapartum analgesia or anaesthesia may need more intensive monitoring and support.
- Mothers who have had caesarean births can initiate and establish breastfeeding with support.
- Infants with complications need more intensive support to initiate breastfeeding.
- Breastfeeding support is critical for low-birth-weight infants.
- Supplementary feeding can be avoided through simple interventions.
- Donor human milk is no substitute for effective breastfeeding support.

Section 4 outline

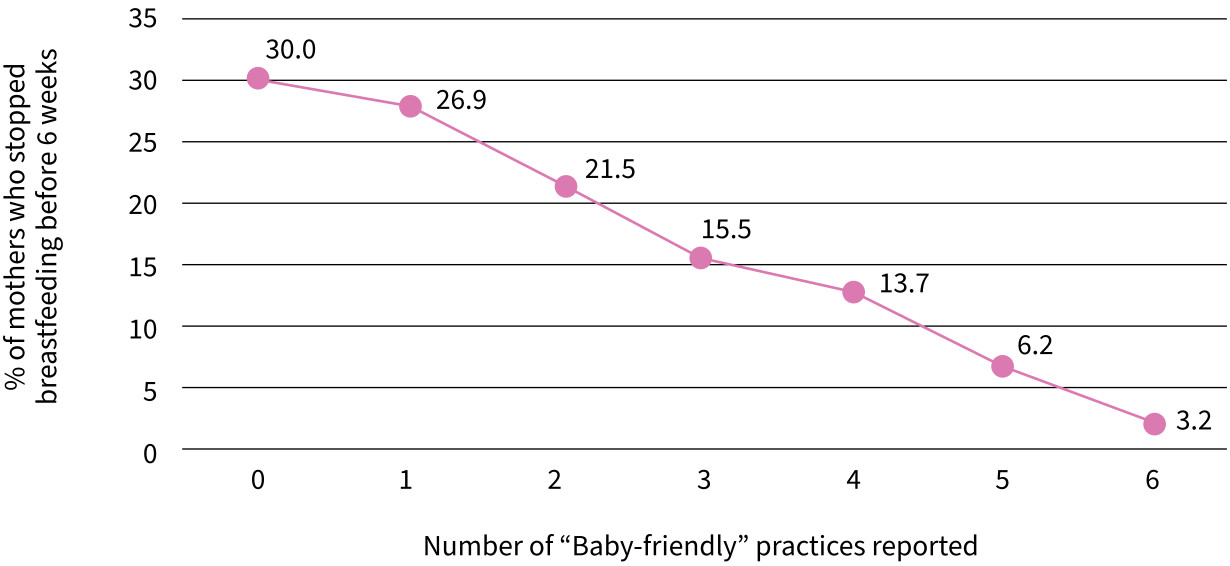
4.1 The Ten Steps to Successful Breastfeeding

Globally, three out of four babies are born in hospitals and other maternity facilities. In these clinical environments, routine antenatal and postnatal care practices impact breastfeeding and health outcomes for infants and mothers. Mothers need appropriate care and skilled help to initiate and establish breastfeeding, to anticipate and avoid difficulties, and to manage problems appropriately when they arise *(1–6)*.

In 1989, WHO and UNICEF spelled out Ten Steps to Successful Breastfeeding (the Ten Steps) to outline optimal clinical care for new mothers and their infants in the perinatal period. Since 1991, the Baby-friendly Hospital Initiative (BFHI) has helped to motivate facilities providing maternity and newborn services worldwide to better support breastfeeding based on the Ten Steps. Following an extensive literature review, the Ten Steps underwent a major revision in 2018. Along with the revision of the Ten Steps, the BFHI itself was reconceptualized as a broader public health initiative to facilitate widespread adoption of the practices *(1)*. Implementing the Ten Steps to Successful Breastfeeding significantly improves breastfeeding outcomes *(1)*.

The Ten Steps are grouped into 1) critical management procedures, which provide an enabling environment for sustainable implementation within the facility, and 2) key clinical practices that describe the gold standard care for mothers and infants (See **Box 4.1**). These clinical practices are evidence-based interventions that support mothers to establish breastfeeding. There is substantial evidence that implementing the Ten Steps significantly improves breastfeeding rates *(1)*. **Fig. 4.1** below shows the inverse relationship between increased exposure to the Ten Steps and breastfeeding cessation before six weeks amongst mothers who initiated breastfeeding and intended to breastfeed for more than two months *(7).*

Fig. 4.1 Exposure to Ten Steps and Breastfeeding Cessation <6wks



Source: DiGirolamo AM, Fein SB. Effect of maternity-care practices on breastfeeding. Pediatrics. 2008 Oct;122(Suppl 2):S43-49.

WHO has called upon countries to integrate the Ten Steps as routine practices across all facilities that provide maternity and newborn services *(1)*. Successful implementation requires integration of the Ten Steps into routine antenatal and postnatal care standards, routine monitoring for quality assurance, and national leadership (See **Box 4.2**).

Box 4.1 Ten steps to successful breastfeeding (revised 2018)

Critical management procedures	
Step 1	a. Comply fully with the *International Code of Marketing of Breast-milk substitutes* and relevant World Health Assembly resolutions. b. Have a written infant feeding policy that is a routinely communicated to staff and parents. c. Establish ongoing monitoring and data-management systems.
Step 2	Ensure that staff have sufficient knowledge, competence and skills to support breastfeeding.
Key clinical practices	
Step 3	Discuss the importance and management of breastfeeding with pregnant women and their families.
Step 4	Facilitate immediate and uninterrupted skin-to-skin contact and support mothers to initiate breastfeeding as soon as possible after birth.
Step 5	Support mothers to initiate and maintain breastfeeding and manage common difficulties.
Step 6	Do not provide breastfed newborns any food or fluids other than breast milk, unless medically indicated.
Step 7	Enable mothers and their infants to remain together and to practice rooming-in 24 hours a day.
Step 8	Support mothers to recognize and respond to their infants' cues for feeding.
Step 9	Counsel mothers on the use and risks of feeding bottles, teats and pacifiers.
Step 10	Coordinate discharge so that parents and their infants have timely access to ongoing support and care.

Source: World Health Organization, United Nations Children's Fund (UNICEF). Implementation guidance: protecting, promoting and supporting breastfeeding in facilities providing maternity and newborn services: the revised baby-friendly hospital initiative. Geneva: World Health Organization; 2018.

Box 4.2 Summary of updated directions for implementation of the Baby Friendly Hospital Initiative

1. Appropriate care to protect, promote and support breastfeeding is the responsibility of every facility providing maternity and newborn services. This includes private facilities, as well as public ones, and large as well as small facilities.
2. Countries need to establish national standards for the protection, promotion and support for breastfeeding in all facilities providing maternity and newborn services.
3. The Baby-friendly Hospital Initiative (BFHI) must be integrated with other initiatives for maternal and newborn health, health-care improvement, health-systems strengthening and quality assurance.
4. To ensure that health-care providers have the competence to implement the BFHI, this topic needs to be integrated into pre-service training curricula. In addition, in-service training needs to be provided when competencies are not yet met.
5. Public recognition of facilities that implement the Ten Steps and comply with the global criteria is one way to incentivize quality improvement. Several other incentives exist, ranging from compliance with national facility standards to performance-based financing.
6. Regular internal monitoring is a crucial element of both quality improvement and ongoing quality assurance.
7. External assessment is a valuable tool for validating the quality of maternity and newborn services. External assessments should be sufficiently streamlined into existing mechanisms that can be implemented sustainably.

Source: World Health Organization, United Nations Children's Fund (UNICEF). Implementation guidance: protecting, promoting and supporting breastfeeding in facilities providing maternity and newborn services: the revised baby-friendly hospital initiative. Geneva: World Health Organization; 2018.

4.1.1 Critical management procedures

Steps one and two detail the critical management procedures that should be adopted and maintained by facilities that provide maternity and newborn services in order to ensure universal and sustained application of the key clinical practices.

There is substantial evidence that implementing the Ten Steps significantly improves breastfeeding rates.

Step 1a: Comply fully with the International Code of Marketing of Breast-milk Substitutes and its relevant World Health Assembly resolutions

Families and healthcare professionals are vulnerable to the unethical and aggressive marketing of breast-milk substitutes, feeding bottles and teats. The International Code of Marketing of Breast-milk Substitutes (the Code) recognizes that infant formula is necessary for infants who cannot be exclusively breastfed and aims to ensure that it is not promoted in ways that undermine or discourage breastfeeding for those infants who can be exclusively breastfed *(8,9)*.

Manufacturers and distributors of products within the scope of the Code have a financial interest in encouraging the use of products such as formula milks. Families and healthcare professionals must be protected against commercial and financial influences that could impact infant feeding decisions *(9,10).*

There should be no promotion of feeding bottles or teats in any part of facilities providing maternity and newborn services, or by any healthcare professional or staff member because these products fall within the scope of the Code. Maternity facilities must avoid all promotion of breast-milk substitutes and related products, bottles and teats, and should not accept free or low-cost supplies or give out samples of those products.

Step 1b: Have a written infant feeding policy that is routinely communicated to staff and parents

Clinical leaders and managers are expected to incorporate clinical practices articulated in the Ten Steps into facility policies. Those policies include guidance on how each of the key clinical practices should be implemented and supported, together with clear management procedures. A written infant feeding policy serves as:

- a standard set of expectations for managers, healthcare professionals, and staff
- a vehicle for ensuring that families receive consistent, evidence-based care.

The Academy of Breastfeeding Medicine has developed a model policy that can be adapted for use in perinatal care services *(11)*.

Step 1c: Establish ongoing monitoring and data-management systems

Maintaining compliance with the Ten Steps requires careful continuous monitoring of defined indicators at the levels of the maternity facility, the health system, and at the country level (*12*). Key to this quality control approach is that maternity facilities employ a routine internal monitoring system that can track the indicators to identify and address failures quickly. Effective monitoring of adherence to the Ten Steps should be maintained as part of a continuous and ongoing programme of quality assurance and improvement in every facility that provides maternity care.

Step 2: Ensure that staff have sufficient knowledge, competence and skills to support breastfeeding

Healthcare providers must be equipped with a discrete set of competencies for helping mothers to initiate and establish breastfeeding, and to overcome any difficulties. Effective training equips healthcare professionals with the knowledge, skills and attitudes required to communicate effectively, give consistent messages, and implement policy standards *(3).* **Table 4.1** contains a list of the competencies health care professionals need in order to support breastfeeding in maternity facilities *(4).*

Table 4.1 List of competencies necessary for implementing the Ten Steps to successful breastfeeding

DOMAIN 1: Critical management procedures to Support the Ten Steps	
01. Implement the *Code* in a health facility	
02. Explain a facility's infant feeding policies and monitoring systems	
DOMAIN 2: Foundational skills: communicating in a credible and effective way	
03. Use listening and learning skills whenever engaging in a conversation with a mother	
04. Use skills for building confidence and giving support whenever engaging in a conversation with a mother	
DOMAIN 3: Prenatal period	
05. Engage in antenatal conversation about breastfeeding	
DOMAIN 4: Birth and immediate postpartum	
06. Implement immediate and uninterrupted skin-to-skin	
07. Facilitate breastfeeding within the first hour, according to cues	
DOMAIN 5: Essential issues for a breastfeeding mother	
08. Discuss with a mother how breastfeeding works	
09. Assist mother getting her baby to latch	
10. Help a mother respond to feeding cues	
11. Help a mother manage milk expression	
DOMAIN 6: Helping mothers and babies with special needs	
12. Help a mother to breastfeed a low-birth-weight or sick baby	
13. Help a mother whose baby needs fluids other than breast milk	
14. Help a mother who is not feeding her baby directly at the breast	
15. Help a mother prevent or resolve difficulties with breastfeeding	
DOMAIN 7: Care at discharge	
16. Ensure seamless transition after discharge	

Source: World Health Organization, United Nations Children's Fund. Competency verification toolkit: ensuring the competency of direct care providers to implement the Baby-Friendly Hospital Initiative. Geneva: World Health Organization; 2020.

4.1.2 Key clinical practices

Steps 3 to 8 describe the key clinical practices that facilities that provide maternity and newborn services should provide to patients in their care.

Step 3: Discuss the importance and management of breastfeeding with pregnant women and their families.

Breastfeeding counselling should be provided to pregnant women and their families tailored to meet the individual needs of each woman and her family *(4,5,6,13)*. **Box 4.3** describes essential content of breastfeeding education that should be provided during breastfeeding counselling.

Box 4.3 Essential issues to discuss with a breastfeeding mother

- Importance of exclusive breastfeeding for the first 6 months.
- Mother-infant eye-to-eye and body contact while feeding.
- Feeding cues and signs of an adequate latch, swallowing, milk transfer and infant satisfaction and how to recognize all of them.
- Average feeding frequency (at least eight times per 24 hours) with some infants needing more frequent feedings.
- How to breastfeed in a comfortable position and without pain.
- Infants should be fed in response to feeding cues, offered both breasts per feeding and fed until they seem satisfied.
- How to ensure/enhance milk production and let down.
- Why and how to hand express colostrum/breastmilk.
- How to correctly use and care for her breast pump (for a mother who needs to pump).
- Effects of pacifiers/ artificial teats on breastfeeding and why to avoid them until lactation is established.
- Very few medications or mother's illnesses contraindicated during breastfeeding.
- Accurate information resources.
- Reasons for a breastfeeding mother to avoid tobacco, alcohol and other drugs.
- Safe sleeping instructions (how to make co-sleeping safer).
- Recognize signs of undernourishment or dehydration in the infant and warning signs for calling a health professional management of most common breastfeeding difficulties.

Source: United Nations Children's Fund (UNICEF), World Health Organization. Implementation Guidance on Counselling Women to Improve Breastfeeding Practices. New York: UNICEF; 2021.

Step 4: Facilitate immediate and uninterrupted skin-to-skin contact with mothers and support them to initiate breastfeeding as soon as possible after birth.

Immediate skin-to-skin contact after birth facilitates timely initiation of breastfeeding (within the first hour). Uninterrupted skin-to-skin contact between mothers and infants should be initiated as soon as possible after birth. Delaying skin-to-skin contact compromises psychological and psychosocial outcomes for mothers and babies *(14–16).* Infants should be placed prone on the mother's abdomen or chest immediately or as soon as possible after birth. Babies should remain in this position for at least one hour after birth and not removed until the baby has initiated breastfeeding, regardless of the method of their birth.

Skin-to-skin care can be initiated immediately following a caesarean section provided regional anaesthesia (epidural) is used. If general anaesthesia is used, skin-to-skin care and initiation of breastfeeding can begin as soon as the mother is sufficiently alert to hold her infant. If mothers are not able to initiate breastfeeding during the first hour after birth, they should be supported to provide skin-to-skin contact at least until breastfeeding is initiated *(17–19).*

Sustained skin-to-skin contact and kangaroo mother care (KMC) are particularly important for preterm and low-birth-weight infants (see **Section 4.5.3**). Preterm infants may be able to root, attach to the breast and suckle from as early as 27 weeks' gestation. Early and frequent milk expression is critical to stimulating milk production and secretion for preterm infants who are not yet able to suckle. Transition to direct and exclusive breastfeeding is facilitated by regular, prolonged skin-to-skin contact *(20–22).*

During immediate skin-to-skin contact, and for at least the first hour after birth, healthcare professionals should use sensible vigilance and safety precautions to observe and manage any signs of distress. Mothers who are sleepy or under the influence of anaesthesia or drugs will require closer observation and support. If a mother has received analgesics, general or regional anaesthesia, the baby may also be less alert.

Step 5: Support mothers to initiate and maintain breastfeeding and manage common difficulties.

Initiation of breastfeeding is typically a direct sequela of uninterrupted skin-to-skin contact. All mothers should be supported to initiate breastfeeding as soon as possible after birth and within the first hour. Widström and colleagues have described and validated an innate sequence of observable pre-feeding behaviours that occurs when human neonates are placed skin to skin on their mothers' abdomens and these are described in **Table 4.2** *(23).*

Table 4.2 Recognizing the nine instinctive stages of the human newborn at birth

Stages	Behaviours
1. Birth cry	Intense cry just after birth, transition to breathing air.
2. Relaxation stage	Infant rests. No activity of mouth, head, arms, legs or body.
3. Awakening stage	Infant begins to show signs of activity. Small thrusts of head: up, down, from side-to-side. Small movements of limbs and shoulders.
4. Active stage	Infant moves limbs and head, more determined movements. Rooting activity, 'pushing' with limbs without shifting body.
5. Resting stage[a]	Infant rests, with some activity, such as mouth activity, sucks on hand.
6. Crawling	Pushing which results in shifting body.
7. Familiarization	Infant has reached areola/nipple with mouth positioned to brush and lick areola/nipple.
8. Suckling stage	Infant has taken nipple in mouth and commences suckling.
9. Sleeping stage	Infant closes eyes and falls asleep.

[a] The resting stage may be interspersed with all the stages.

Source: Widström A-M, Brimdyr K, Svensson K, Caldwell K, Nissen E. A plausible pathway of imprinted behaviors: Skin-to-skin actions of the newborn immediately after birth follow the order of fetal development and intrauterine training of movements. Med Hypotheses. 2020;134:109342.

Early suckling is important for stimulating milk production and establishing the maternal milk supply. Suckling at the breast facilitates initiation and continuation of breastfeeding as well as the passage of antibodies. The removal of feedback inhibitor of lactation (FIL) with the first feed is crucial for increasing the volume of breast milk *(24)*. Initiation of breastfeeding prompts a hormonal surge in the mother, stimulating the progression of the lactation process.

While breastfeeding is a natural human behaviour, all mothers need individualized support and practical assistance to learn how to breastfeed and even experienced mothers may encounter new challenges when breastfeeding a new baby. Direct observation of a full breastfeeding session is necessary to ensure that the infant is able to attach to and suckle at the breast and that milk transfer is occurring *(3,13)*. More information about assessing breastfeeding and observing a breastfeed is given in **Section 5**. If a baby is not well attached, then hands-off assistance can be offered to improve the baby's position and attachment (see **Section 2.3.3**).

Step 6: Do not provide breastfed newborns any food or fluids other than breast milk, unless medically indicated.

Exclusive breastfeeding means that the baby receives only breast milk. Giving newborns any foods or fluids other than breast milk in the first few weeks after birth interrupts the physiology of lactation and interferes with breastfeeding.

Prelacteal feeds should not be given unless medically indicated. This includes foods traditionally given to infants in the hours or days after birth, including ghee (refined butter), dates, honey, tea, sugar water, sugar juice, raw cow/goat milk.

If a baby is not able to breastfeed, a mother can express her breast milk to establish and maintain her milk supply. For term infants, there is not sufficient evidence that one method of expression (hand expression, manual pump or electric pump) is more effective than another. Healthcare professionals can give practical help to make sure a mother is able to express her milk by hand. Global Health Media has created video resources that can be used to support teaching hand expression in a variety of languages *(25).*

Step 7: Enable mothers and their infants to remain together and to practice rooming-in 24 hours a day.

Rooming-in means that the mother and the infant are accommodated in the same room. The baby may be placed in a bedside cot or a three-sided cot attached to the mother's bed. When not being receiving skin-to-skin care, the baby remains within arms-reach of the mother.

Rooming-in enables mothers to establish breastfeeding and to practise responsive feeding. When the mother and infant are kept together, day and night, it is easier for the mother to learn to recognize feeding cues and respond to them. The close presence of the mother to her infant facilitates successful breastfeeding.

It is rarely necessary to separate mothers from babies. Separation should be a rare and exceptional occurrence that is only considered when a mother is too sick to care for her baby or if the baby's condition is metabolically unstable and requires intensive medical care that cannot be provided in the mother's room. Most tests and procedures can be done at the mother's bedside. If the baby must be treated in a procedure room the mother should be encouraged and supported to stay close to her infant while treatment is administered. Rooming in can be practiced in intensive care units to facilitate KMC.

Step 8: Support mothers to recognize and respond to their infants' cues for feeding.

Responsive feeding is also called "demand feeding", "baby-led", "ad-hoc" or "unrestricted" breastfeeding. It involves offering a breastfeed whenever the babies show cues for closeness, comfort or feeding, and usually before they start crying. Responsive feeding is part of a nurturing relationship. During a breastfeeding session, babies should remain at the breast for as long as they wish. Scheduled feeding, which involves a predetermined, time-restricted frequency and schedule of feeds, is not recommended.

Step 9: Counsel mothers on the use and risks of feeding bottles, teats and pacifiers.

Breastfed babies rarely need feeding bottles, teats and pacifiers. Parents need to be aware of the risks linked to their use, including missing the baby's cues and potential challenges with oral formation and hygiene. Bottles may interfere with breastfeeding if they are introduced before lactation is well established. As a baby suckles at a bottle, the baby suckles less at the breast, breaking the physiological demand and supply mechanism.

Step 10: Coordinate discharge so that parents and their infants have timely access to ongoing support and care.

Continuing breastfeeding support in the community is essential to sustain exclusive breastfeeding for the first six months and continuation of breastfeeding for two years or beyond.

Mothers may be discharged from the hospital before their milk comes in and without enough time for informative conversations with healthcare professionals during the immediate post-partum period. The list provided in **Box 4.4** can be used to develop individualised feeding plans for mothers before they are discharged.

Healthcare professionals should ensure the smoothest transition from their care to accountable skilled resources in their communities. Healthcare professionals routinely engage with mothers before they are discharged home, to revise and reinforce the basic knowledge and skills they need to care for their infants, breastfed or not.

Mothers require ongoing support from diverse sources in their own community: skilled healthcare professionals, home visits, breastfeeding or well-baby clinics, peer counsellors, mother-to-mother support groups. When leaving the maternity facility, all mothers should be given written information about trustable, Code-compliant resources that are easily accessible in their own communities *(26).*

Box 4.4 Elements to review/reinforce before discharge (individualized feeding plan)

First, assess:

- a feed
- the general health of mother and baby.

Then, choose a few points that are relevant to the specific mother's and baby's needs to review or reinforce:

- mother's understanding of her baby's unique feeding cues;
- baby's ability to achieve a comfortable latch;
- signs of milk transfer with infant swallowing;
- signs of adequate of adequate intake (stools and urine);
- mother's understanding of her baby's need to feed frequently, at least eight times in 24 hours or more;
- importance of eye-to-eye contact with baby while feeding;
- importance of letting the baby finish nursing on the first breast, then offer the other breast until the baby seems satisfied by releasing the breast;
- mother's position (how she holds baby) to assure comfortable, pain-free feeds;
- mother's understanding of ensuring/enhancing milk production and let-down;
- mother's understanding of hand-expressing colostrum/breast milk and why this is helpful;
- mother's awareness of risks of other fluids and importance of exclusive breastfeeding for 6 months;
- mother's awareness of risks and uses of pacifiers and teats;
- very few medications or illnesses are contraindicated during breastfeeding;
- accurate sources of information and how to get help if needed;
- information for continued breastfeeding and general health support in the community;
- adequate food and drink support her general health because special foods are not needed for breastfeeding; and
- mother's understanding of safe sleeping (breastfeeding and co-sleeping) arrangements.

And, as applicable:

- appropriate guidance specific to the mother-infant dyad
- mother's ability to correctly use and care for her breast pump
- mother's ability to correctly prepare and use infant formula.

Source: World Health Organization, United Nations Children's Fund. Competency verification toolkit: ensuring the competency of direct care providers to implement the Baby-Friendly Hospital Initiative. Geneva: WHO; 2020.

4.2 Intrapartum analgesia and anaesthesia

Studies examining the impact of epidural analgesia on breastfeeding have been limited and their results conflicting *(27,28)*. A recent observational study showed that fentanyl given during labour, either intravenously or by epidural, is associated with breastfeeding difficulties *(29)*. Healthcare professionals should be aware of these potential risks and provide adequate support to initiate breastfeeding for mothers who receive anaesthesia during labour and birth *(29)*. Detailed clinical protocols for analgesia and anaesthesia during labour and breastfeeding are available *(30)*.

4.3 Caesarean birth

Once breastfeeding is successfully established, rates of exclusive breastfeeding at six months are comparable for mothers who had caesarean births and those who had vaginal births *(31)*. Any woman who is having a planned, or scheduled, caesarean birth needs both anticipatory guidance in advance of the surgery and additional support during the postpartum period, especially the first 24 hours after birth.

Most mothers of babies born by caesarean section can establish breastfeeding during the first hour of life, unless there is a complication that requires them to be separated. If a mother has had spinal or epidural anaesthesia, a healthy term infant can be placed skin-to-skin on her chest and initiate breastfeeding during the first hour of life, similar to babies born vaginally. If a mother has had general anaesthesia, she can be encouraged to start skin-to-skin contact and initiate breastfeeding as soon as she is able to respond, ideally within the first hour and usually within two hours after the birth of her baby.

Encourage mothers to feed their babies on demand after a caesarean birth. For the first few days after a caesarean birth, most mothers will need help from healthcare workers and family members to position themselves and their babies comfortably for breastfeeding. Reclined, side-lying, or football-hold positions may be most comfortable.

If a mother is too ill to breastfeed, the options available are, in order of preference:

- MOM
- DHM
- commercial infant formula.

Formal consent should be required before giving formula milk supplements.

4.4 Infant conditions

Many common breastfeeding issues can be prevented. The management of any difficulties should include protection of the lactation process.

4.4.1 Neonatal jaundice

In infants, blood levels of total serum bilirubin can be elevated because of a combination of three factors: increased production of bilirubin, increased intestinal absorption of bilirubin, and hepatic immaturity that decreases uptake and conjugation of bilirubin. Effective feeding reduces the risk of jaundice in breastfed infants *(32,33)*.

Early jaundice (also known as physiological jaundice) appears between two and seven days of life. This condition is often physiological and clears after a few days. A thorough evaluation for feeding technique difficulties and signs of dehydration must be ruled out. Jaundice can make a baby sleepy so they suckle less and perpetuate a suboptimal feeding. Early initiation of breastfeeding and frequent breastfeeding reduces the severity of early jaundice. Colostrum is a natural laxative that helps prevent jaundice in the newborn. Physiological jaundice resolves more quickly when breast-milk intake is increased.

Observe a breastfeed to check the baby is feeding effectively (see **Section 5.7**). Encourage mothers to breastfeed frequently and offer extra expressed breast milk by cup or supplemental feeding at the breast if necessary (see **Section 4.6**). Water and glucose water are not recommended and may make a baby suckle less at the breast.

If severe jaundice is detected, either clinically through the colour of the skin or by elevated serum bilirubin levels, early phototherapy (light treatment) may be needed. Phototherapy can be interrupted for breastfeeding *(32)*. If the baby must be separated from the mother, she should be encouraged to express milk frequently and resume breastfeeding as soon as possible.

Prolonged jaundice starts after the seventh day of life and continues for as long as 12 weeks. (Prolonged jaundice used to be known as 'breast milk jaundice' but using this term undermines confidence in breastfeeding and should be avoided.) It is usually due to hormones or other substances in the mother's milk, is harmless and resolves by itself. If the jaundice is due to a more serious condition there are usually other signs, such as pale stools, dark urine, or an enlarged liver and spleen.

Babies with prolonged jaundice should be assessed to exclude serious condition such congenital abnormality, inborn errors of metabolism, or sepsis. Non-physiological causes of jaundice that must be considered include bruising, dehydration, rhesus incompatibility, and other causes of haemolysis *(32,33)*.

4.4.2 Neonatal hypoglycaemia

Transitional neonatal hypoglycaemia must be distinguished from persistent pathological hypoglycaemia. Delayed breastfeeding (greater than one hour after birth) is the most common risk factor for hypoglycaemia. Maternal risk factors include diabetes and pre-eclampsia. Neonatal risk factors include small for gestational age, prematurity, and perinatal stress *(32,33).*

A rapid fall in glucose levels in the first few hours of life is expected, self-limited and usually asymptomatic. The glucose level of neonates can fall as low as 20–25 mg/dl (1.11–1.39 mmol/L) over the first two hours of life. These levels rise after around three hours of life even without oral milk intake and then stabilize between 43–90 mg/dl (2.4–5 mmol/L) by 12 to 24 hours of age. If an infant's glucose levels are persistently low or are recurrent, further evaluation is needed *(32,33)*.

Signs and symptoms such as irritability, tremors, tachypnoea, sweating, pallor, hypothermia, poor suck or refusal to feed are used to guide which baby might need treatment. Apnea, hypotonia, seizure and coma can be seen if left untreated *(33)*.

Every baby must start breastfeeding during the first hour of life *(21)*. Skin-to-skin contact, early feeding and rooming in to support responsive feeding are practices which decrease the risk of hypoglycaemia. A baby who is at risk of hypoglycaemia requires close attention and immediate feeding, ideally at the breast or with expressed colostrum, which may be collected before or after birth. Some of these babies may need an alternative feed until they can start breastfeeding. Alternative feeds may be given by cup so that they do not interfere with the establishment of breastfeeding *(32,33)*.

Routine blood glucose screening is not necessary. Blood glucose levels should only be measured in infants at increased risk of neonatal hypoglycaemia (see **Table 4.3**) *(32,33)*.

Table 4.3 At-risk infants for whom routine monitoring of blood glucose is indicated

Maternal conditions
Pre-existing or gestational diabetes, or abnormal result of glucose tolerance test, especially if poorly controlled
Pre-eclampsia and pregnancy induced, or essential hypertension
Previous macrocosmic infants (as a proxy for undiagnosed diabetes in pregnancy)
Substance abuse
Treatment with beta-agonist tocolytics
Treatment with oral hypoglycaemic agents
Late antepartum or intrapartum administration of IV glucose
Neonatal conditions
Intrauterine growth restriction or marked wasting
Low birth weight (<2,500g)
Small for gestational age (birthweight <10th percentile)[a]
Babies with clinically evident wasting of fat and muscle bulk
LGA (>90th percentile for weight & macrosomic appearance)[b]
Discordant twin (weight 10% < larger twin)
Infants of poorly controlled diabetic mothers
Prematurity (<35 weeks or late preterm infants with clinical signs or extremely poor feeding)
Perinatal stress; severe acidosis or hypoxia–ischemia
Cold stress
Polycythaemia (venous haematocrit >70%) / hyperviscosity
Erythroblastosis fetalis
Beckwith–Wiedemann syndrome
Microphallus or midline defect (indicating an underlying endocrine condition)
Suspected infection
Respiratory distress
Known or suspected inborn errors of metabolism or endocrine disorders
Any infant admitted to the neonatal intensive care unit
Infants displaying signs associated with hypoglycaemia

Abbreviations: IV: intravenous; LGA: large for gestational age.

Notes:

[a] In the USA and WHO definition is <10th percentile and British <2nd percentile.

[b] In unscreened populations, LBA may represent undiagnosed/untreated maternal diabetes.

Source: Wight NE, Academy of Breastfeeding Medicine. ABM Clinical Protocol #1: Guidelines for Glucose Monitoring and Treatment of Hypoglycemia in Term and Late Preterm Neonates, Revised 2021. Breastfeed Med. 2021;16(5):353-65.

Frequent breastfeeding, along with dextrose gel, is the first-line oral treatment of infants with hypoglycaemia. Breastfeeding, along with IV glucose, may be necessary in infants with abnormal clinical signs and glucose levels <20–25 mg/dl (<1.1–1.4 mmol/L) *(33)*.

4.4.3 Congenital defects

Tongue-tie

When a baby's lingual frenulum is abnormally short, tongue mobility is restricted, preventing the tongue from being pushed forward over the gum and under the mother's nipple. Known as tongue-tie or ankyloglossia, this can make attachment difficult, which may cause challenges with breastfeeding such as sore nipples. The baby may not suckle effectively and may have a low intake of breast milk resulting in poor infant weight gain *(34,35)*.

Tongue-tie is a potential risk for breastfeeding difficulties. It can affect the attachment at the breast, leading to pain and sore nipples, long feeds, poor drainage of milk leading to mastitis, weight loss or poor weight gain. The decision to treat surgically requires expert clinical skills in the diagnosis and management of tongue-tie as well as lactation support *(34).*

Frenotomy is a simple, safe procedure that is usually performed in early infancy. Studies have demonstrated improvement in maternal nipple pain after frenotomy. However, there is not conclusive evidence of its impact on long-term breastfeeding success *(35).*

Cleft lip and/or palate (CL/CP)

Infants with CL/CP may have difficulty creating a seal around the breast. If the baby cannot achieve a secure seal, they will not be able to generate negative pressure in the oral cavity, which is required to remove milk from the breast. Infants with a small defect or cleft lip may have better success with breastfeeding than those with cleft palate or CL/CP *(36)*.

If only the lip is affected, the breast covers the cleft, and the baby may be able to suckle effectively. If there is enough palate for the tongue to press the nipple against, a baby with a cleft palate can suckle quite well *(36)*.

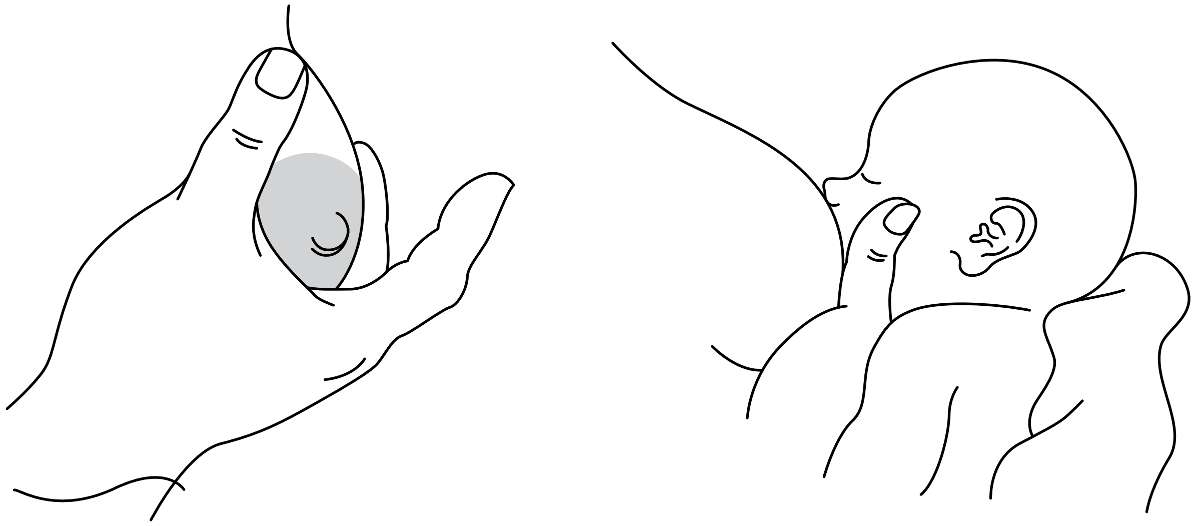
Mothers of babies with CL/CP can be helped to hold the baby in an upright sitting position at the breast with their legs on either side of the mother's thigh. This position makes swallowing easier and may help the baby to breastfeed, fully or partially. An orthopaedic device in the form of a very long teat can be put over the nipple as the baby suckles to facilitate breastfeeding. Alternatively, she can express her milk and feed it to the baby by cup until surgical help is available.

Muscle weakness or hypotonia

Babies with Down syndrome or cerebral palsy may have difficulty attaching to the breast and suckling because of the weakness. Special breastfeeding support is needed.

The mother should be shown how to help the baby to attach to the breast by using the dancer hand position. She supports the baby's chin and head to keep the mouth close on to the breast. These babies may feed slowly. It may be necessary for the mother to express her milk and give some feeds by cup or supplemental tube feeding. The mother will need extra counselling and support.

Fig. 4.2 Dancer hand position



Congenital heart or kidney problems

A baby with a congenital anomaly may fail to grow or have other associated symptoms even if there is no apparent difficulty with breastfeeding or breast-milk supply. These abnormalities are not always obvious and require careful examination of the baby. Optimal nutrition with breastfeeding and breast milk may help to minimize complications associated with these conditions.

4.5 Low birth weight

Implementation of the Ten Steps to Successful Breastfeeding in facilities caring for populations of small, sick and preterm infants can dramatically increase breastfeeding rates amongst infants with low birth weight *(20)*.

Term low-birth-weight infants (born after 37 or more weeks gestation) can usually breastfeed effectively. Proper attachment at the breast is important. When a low-birth-weight baby first suckles, they may pause quite often and for long periods during a feed. The baby may need to continue feeding for as long as an hour. Do not to take the baby off the breast during these pauses *(21)*.

Preterm infants should start breastfeeding as soon as they are clinically stable and should be allowed to suckle frequently and on demand. Mothers should be supported to recognize their baby's hunger cues and respond appropriately. To stimulate breastfeeding, these babies should be allowed to suckle or lick the breast on demand *(21)*.

If the baby is having difficulties attaching properly at the breast, encourage the mother to express her milk directly into her baby's mouth (See **Box 4.5**). If a baby has difficulty suckling effectively, tires quickly at the breast or does not gain adequate weight, either offer expressed milk by cup after the breastfeed or alternate breast and cup feeds. Offer MOM by cup or tube until they are suckling well and gaining weight. **Fig. 4.3** shows an infant being cup-fed. Cup feeds can be reduced gradually as the infant's strength and stamina increase *(20,21)*.

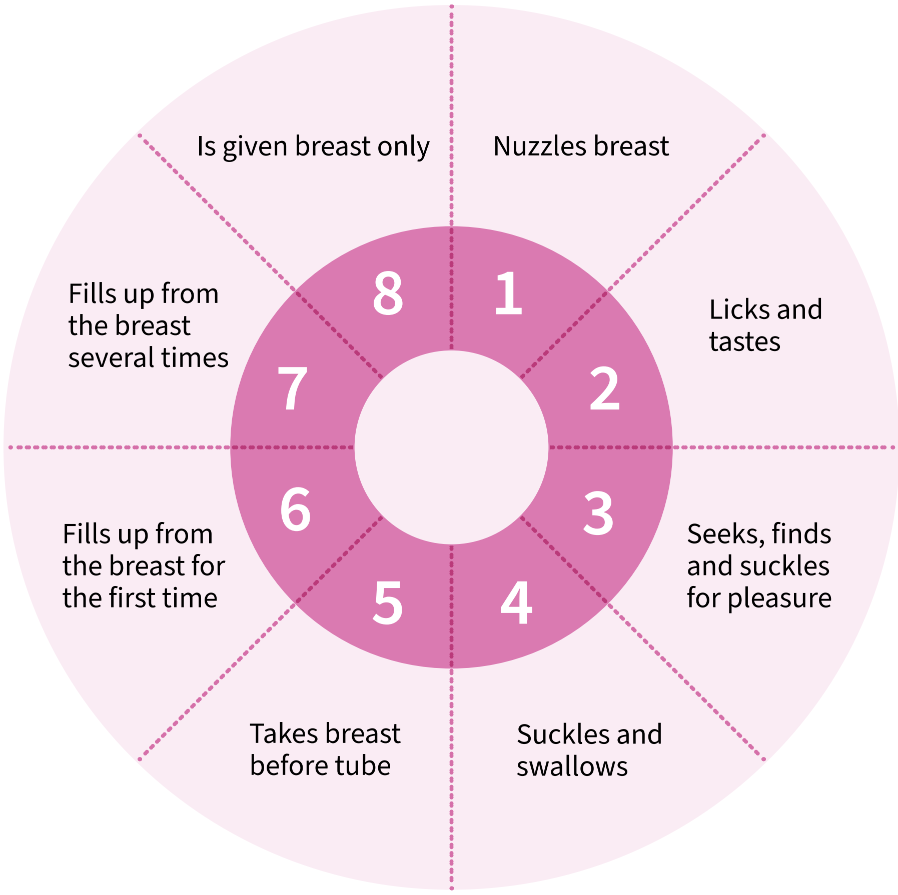
If MOM is not available, DHM should be given wherever possible. Infant formula should only be considered when neither MOM nor DHM are available. Preterm infant formula may be considered for very preterm (< 32 weeks' gestation) or very low birth weight infants. Powdered infant formula should not be used to feed infants in the first 28 days after birth *(20,37,38)*.

Fig. 4.3 Cup feeding a low-birth-weight baby

Multicomponent fortification of human milk is not routinely recommended for preterm or low-birth-weight infants but may be considered for very preterm (< 32 weeks' gestation) or very low birth weight (< 1.5 kg) infants who are fed MOM or DHM *(20).*

4.5.1 Feeding small, sick or premature infants

Fig. 4.4 Breast transition sequence



Source: Adapted from Nyqvist KH, Sjödén PO, Ewald U. The development of preterm infants' breastfeeding behavior. Early Hum Dev. 1999;55(3):247–64.

Initiating breastfeeding

Placing the baby skin-to-skin stimulates innate reflexes, initiating a sequence of behaviours that culminates in the initiation of breastfeeding. This sequence is described in **Fig. 4.4** *(39)*.

Infants who are not able to suckle effectively should be fed from a cup. Gastric tube feeding should only be considered for infants who are unable to feed safely from a cup. Avoid bottle feeding, as it may interfere with the baby learning to breastfeed. Bottle-feeding has been shown to negatively impact breastfeeding success for preterm infants *(44)*.

In health-care facilities, scheduled feeding every 2–3hrs may be considered rather than responsive feeding for preterm infants born before 34 weeks' gestation, until the infant is preparing for discharge. In preterm or low birth weight infants, including very preterm (< 32 weeks' gestation) or very low birth weight infants, who need to be fed by an alternative feeding method to feeding at the breast (e.g. gastric tube feeding or cup feeding), feed volumes can be increased by up to 30 ml/kg per day.

Box 4.5 How to express breast milk directly into a baby's mouth

Ask the mother to:

1. wash her hands
2. hold baby skin-to skin, positioned as for a breastfeed, with the baby's mouth close to her nipple
3. express some drops of milk onto her nipple
4. wait until baby is alert and opens the mouth widely
5. stimulate the baby if they appear sleepy
6. let the baby smell and lick the nipple and attempt to suck
7. let some breast milk fall into the baby's mouth
8. wait until the baby swallows before expressing more drops of breast milk.

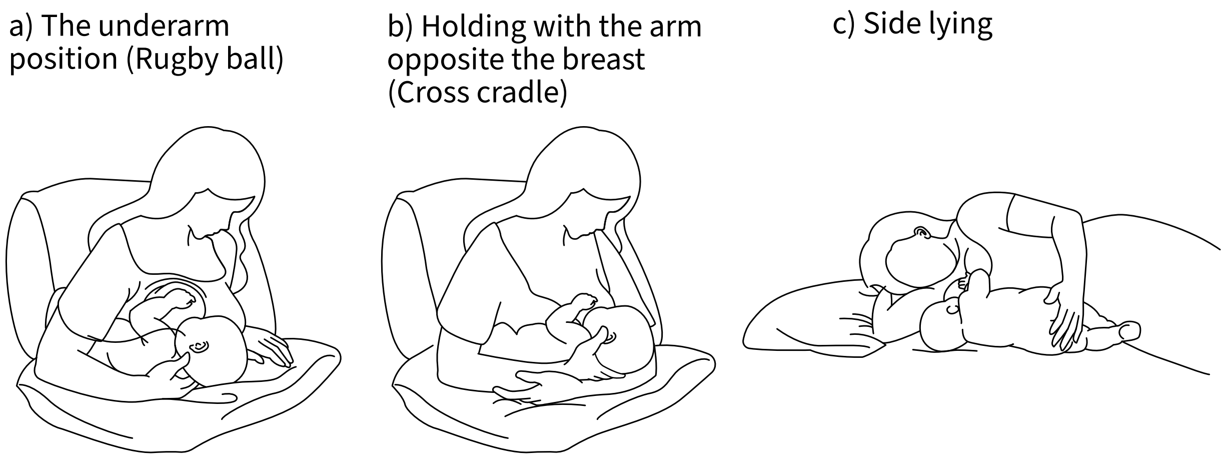
When the baby has had enough, they will close their mouth and will take no more milk.

Ask the mother to repeat this every 1 to 2 hours if her baby is very small, or every 2 to 3 hours if her baby is bigger.

Breastfeeding positions for low-birth-weight babies

There are certain positions that are especially useful for breastfeeding very small babies, including the underarm position, the cross-cradle position, and the side-lying position.

Fig. 4.5 Useful positions for holding a low-birth-weight baby for breastfeeding



Expressing breast milk

If an infant is unable to suckle at the breast, the mother will need skilled help to learn to express her milk after birth and then at least eight times every 24 hours. The volume of milk expressed should increase each day over the first two weeks as described in **Table 4.4** *(20).* Mothers of preterm infants who cannot suckle effectively should be supported to begin expressing their milk immediately after birth or immediately after it is ascertained that the baby cannot breastfeed. Mothers who cannot remain in the facility with their babies will need to bring it to their babies as soon as they can. Mothers who cannot bring their milk to their babies should be encouraged and supported to maintain this expressing schedule in order to establish their milk supply in preparation for breastfeeding.

Table 4.4 Average volume of breast milk needed per day

Time since birth	Volume (mL) each pumping (both breasts)	Volume per day[a] (mL)
Day 1–2	Drops to 20mL	Drops to 120mL
Day 3	25 to 45mL	160 to 360mL
Day 4–5	50 to 60mL	400 to 600mL
Day 6–9	75 to 90mL	600 to 720mL
Day 10 and beyond (to maintain supply)	90mL or more	720mL

[a] Volumes expected with pumping combined with hand expression after pumping at least 8 times in a 24-hour day.

Source: World Health Organization, United Nations Children's Fund. Protecting, promoting and supporting breastfeeding: the baby-friendly hospital initiative for small, sick and preterm newborns. Geneva: World Health Organization; 2020.

4.5.2 Donor human milk

When MOM is not available, or not available in sufficient quantity, parents should be offered DHM *(21)*. DHM provided through donor human milk banks is usually pasteurized to inactivate potentially pathogenic organisms and this alters the nutritional and functional properties of the milk. Infants fed infant formula may gain weight more quickly than infants fed DHM but feeding infant formula increases the risk of necrotizing enterocolitis. DHM does not provide the same nutritional or immunological support as MOM. Making DHM available to low birth weight babies is no substitute for providing timely, effective breastfeeding support *(20,21)*.

4.5.3 Kangaroo mother care

KMC is recommended as routine care for all babies and is especially beneficial for small, sick and preterm infants. KMC delivers outcomes similar to those provided by an incubator. KMC should be initiated as soon as possible after birth in a health-care facility or at home and should be given for 8–24 hours per day (as many hours as possible). KMC involves maintaining prolonged skin-to-skin contact between mother and baby.

Mothers should be supported to hold their babies, day and night, in an upright position between their breasts. The baby should wear only a diaper or nappy and there should be no clothing between the mother and the baby. A carefully fitted baby sling may be used to facilitate KMC. A binder may help to keep the infant in skin-to-skin contact with the mother's or caregiver's chest. The infant should also have a warm hat, socks and a diaper/nappy. The mother or caregiver should wear whatever is comfortable, provided the clothes accommodate the baby. Other arrangements can also make the baby and mother more comfortable, e.g. reclining beds and chairs. The baby should have as much skin-to-skin contact with the mother as possible, to help both breastfeeding and bonding. The baby is supported in this position by the mother's clothes, or by cloths tied around her chest. The baby's lower limbs are flexed and abducted to avoid compression on the abdomen. The baby's head is left free to breathe, and the face can be seen *(21, 40).*

Fig. 4.6 Baby in kangaroo mother care position

Kangaroo mother care helps the infant to initiate breastfeeding earlier and gain weight faster. Close contact with the mother means that the baby is kept very near to her breasts, where they can easily smell and lick milk that has been expressed onto her nipple. KMC should be provided for all babies, regardless of whether they are able to coordinate sucking and swallowing. Other methods of feeding can be used until the baby is able to breastfeed.

KMC has also been shown to keep the baby warm, to stabilize their breathing and heart rate, and to reduce the risk of infection. Most routine care can be carried out while the baby remains in skin-to-skin contact, either with the mother or another family member.

In a large, multi-site, multi-country study conducted by the WHO Immediate KMC Study Group in low-resource hospitals, continuous KMC initiated immediately after birth in infants with a birth weight between 1.0 and 1.799 kg resulted in a significantly lower risk of neonatal death than the currently recommended initiation of KMC after stabilization *(41)*. KMC should be continued until the baby can breastfeed without the need for alternative methods of feeding and is able to maintain their temperature and breathing without difficulty.

4.5.4 Comprehensive follow up of low birth weight babies

Small, sick and/or preterm infants have an especially high need for follow-up care, particularly for feeding support. Breastfeeding should be established before discharge unless personnel with specialized expertise in supporting transition to exclusive breastfeeding can be provided in an outpatient or community setting that is accessible to the mother *(37)*.

Discharge planning should take place with the mother to identify how she can be supported at home and in the community. Neonatal wards must be aware of, and refer mothers to, a variety of resources that exist in their communities. Early and frequent follow-up care, whether in the health facility or in the community by a community health worker and a breastfeeding counsellor, is especially important for small, sick and/or preterm infants. The most vulnerable period for breastfeeding progression and maintenance appears to be the first month after discharge. Mothers should be given the name and contact details of local breastfeeding support groups *(37)*.

Discharge strategies

- Post-discharge feeding plan developed with family, lactation consultants, and healthcare provider, including a feeding log that documents the number of feeds, urinary output, and stooling pattern.
- Follow-up visits with a healthcare provider within 24–48 hours of discharge and then weekly until the infant is exclusively breastfeeding.
- All mothers should be referred to breastfeeding counsellors in their communities. Breastfeeding counselling should be provided as often as it is needed and on at least six occasions.
- Regular monitoring should continue until the baby weighs more than 2.5 kg. Further follow-up can then continue monthly as for a term baby.
- Guidelines for the safe preparation of infant formula can be shared with caregivers to help reduce risks for infants who are fed with powdered formula *(38)*.

4.6 Supplementary feeding

If a baby is too ill or too small to feed from the breast immediately after birth, the mother should be encouraged and supported to express breast milk for her baby. Frequent expressing or breastfeeding is required to establish copious milk secretion as soon as possible. Hand expression is more effective than a breast pump before the onset of copious milk secretion known as lactogenesis II (see **Section 2**). Physiologic mechanisms for breast milk production can be affected if early stimuli and emptying of the breast is not initiated very soon after birth. Expressed breast-milk can be frozen for use when the baby is able to take oral feeds and colostrum can be used for oral care.

Supplementation with artificial formula at any time in the first six months of life increases the risk of infection, especially if given before the baby's microbiome is established. Supplementation deprives infants of the immunological protections in breast milk and alters their intestinal microflora, rendering infants more susceptible to infection while simultaneously increasing their exposure to pathogenic organisms. Even one dose of artificial formula disrupts the baby's microbiome. Supplementary feeds – defined as any solid or liquid foods other than breast milk given to a newborn before age six months of age – should not be given unless medically indicated *(33,42)*.

Routine supplementation is rarely necessary in the first few days and weeks of life. However, some infants will require supplementation. The Academy of Breastfeeding Medicine has developed a clinical protocol for supplementation *(43)*. Careful skilled assessment of a full breastfeeding session and of an infant's 24-hour feeding patterns should guide the need for any intervention.

Infants should be assessed for signs of inadequate milk intake and supplemented with the mother's own breastmilk or donor human milk when indicated. Infant formula should only be used when supplementation is indicated and human milk is not available. Lack of resources and lack of knowledge and skills among healthcare professionals are not clinical indications for supplementation.

Mothers who have made an informed decision not to breastfeed or not to breastfeed exclusively should be taught to mitigate the risks of infant formula feeding, including how to prepare, feed, and store infant formula properly *(38)*.

4.6.1 Cup feeding

Cup feeding is the safest feeding method for an infant who cannot feed from the breast or for whom a supplement is medically indicated and it does not interfere with the baby learning to suckle at the breast *(44)*. Cup feeding requires the active participation of the baby to lap the milk from the cup. When feeding a baby from a cup, ensure that the baby is alert and ready to feed. Their head must be supported in an upright position and the cup places so that it is contact with the baby's lower lip and both corners of their mouth (see **Fig. 4.7**). The cup should be tilted so that the milk reaches the rim but is not poured into the baby's mouth. The baby should be allowed to lap the milk from the cup with their tongue. **Box 4.6** describes how to cup-feed a baby. A video that demonstrates cup feeding is available from Global Health Media *(45)*.

Cup feeding is the safest feeding method for an infant who cannot feed from the breast.

Fig. 4.7 Cup-feeding

Box 4.6 Cup-feeding procedure

How to cup-feed a baby

1. Wash your hands.
2. Wrap the baby in a cloth to hold their hands by their side, and to support their back.
3. Hold the baby sitting upright or semi-upright on your lap.
4. Put a cloth in front to protect the baby's clothes from spilled milk.
5. Place the estimated amount of milk for one feed into the cup.
6. Hold the cup to the baby's lips.
   - Tip the cup so that the milk just reaches the baby's lips.
   - Rest the cup lightly on the baby's lower lip so that the edges of the cup touch the outer part of the baby's upper lip.
7. The baby starts to take the milk from the cup.
   - A low-birth-weight baby starts to take the milk into their mouth with their tongue.
   - A full-term or older baby may sip or suck the milk from the cup.
8. Do not pour milk into the baby's mouth. Just hold the cup to their lips and let them take it themself (sipping or lapping).
9. Expect some milk to spill from the sides of the baby's mouth.
10. When the baby has had enough, they will close their mouth and will not take any more. If the baby has not taken the whole amount, offer slightly smaller feeds more frequently
11. Measure the baby's intake over 24 hours rather than at a single feed.

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SECTION 5

Supporting infant and young child feeding in the community

Key messages

- Breastfeeding counselling improves infant and young child feeding practices and should be provided to all pregnant women and mothers.
- Breastfeeding counselling should be provided by appropriately trained and competent personnel.
- Routine growth monitoring helps identify children at risk of poor growth and development.

Section 5 outline

5.1 Supporting infant and young child feeding

Women who receive breastfeeding support breastfeed exclusively for longer and are less likely to stop breastfeeding prematurely than women who do not. Breastfeeding support can be provided by health professionals or trained lay or peer breastfeeding counsellors. WHO's recommendations for breastfeeding counselling are listed in **Box 5.1** *(1)*. While face to face counselling is most effective, counselling can be provided by telephone, digital technologies or a combination of these *(2)*.

Box 5.1 WHO Guideline: Counselling of women to improve breastfeeding practices

Recommendations

- Breastfeeding counselling should be provided to all pregnant women and mothers with young children *(recommended, moderate-quality evidence).*
- Breastfeeding counselling should be provided in both the antenatal period and postnatally, and up to 24 months or longer *(recommended, moderate-quality evidence).*
- Breastfeeding counselling should be provided at least six times, and additionally as needed *(recommended, low-quality evidence).*
- Breastfeeding counselling should be provided through face-to-face counselling *(recommended, low-quality evidence)*. Breastfeeding counselling may, in addition, be provided though telephone or other remote modes of counselling *(context-specific recommendation, moderate-quality evidence).*
- Breastfeeding counselling should be provided as a continuum of care, by appropriately trained health-care professionals and community-based lay and peer breastfeeding counsellors *(recommended, moderate-quality evidence).*
- Breastfeeding counselling should anticipate and address important challenges and contexts for breastfeeding, in addition to establishing skills, competencies and confidence among mothers *(context-specific recommendation, low-quality evidence).*

Common challenges and contexts include returning to work or school; the specific needs of mothers who are obese, adolescent girls, primiparous (first-time mothers) or carrying multiple pregnancies (when the mother is pregnant with two or more babies); mothers with mental health difficulties; mothers of infants with special needs, e.g. low birth weight or disability; mothers who delivers by caesarean section; breastfeeding in public spaces; and breastfeeding in humanitarian emergencies.

Source: WHO guideline: counselling of women to improve breastfeeding practices. Geneva: World Health Organization; 2018.

Every woman should be provided with at least six scheduled breastfeeding counselling contacts. These should include one contact during towards the end of the third trimester of pregnancy. In addition to scheduled contacts, breastfeeding counselling should be provided as needed. Consider key tasks for scheduled counselling contacts (see **Table 5.1**) and identify opportunities for combining them with routine pre- and post-natal care, immunization and well-care *(3)*.

Women who receive breastfeeding support breastfeed exclusively for longer.

Table 5.1 Key tasks for scheduled breastfeeding counselling contacts

Contact one: Before birth *(antenatal period)*

- Ask about plans for feeding, previous breastfeeding experience and concerns
- Give information about the benefits of breastfeeding and risks of artificial feeding
- Give anticipatory guidance for birth (vaginal and caesarean), such as uninterrupted skin-to-skin contact, self-attachment to the breast and the importance of colostrum
- Explain patterns of infant feeding (cluster feeding)
- Demonstrate positioning and attachment and encourage practice with a doll
- Demonstrate hand expression of breastmilk in the case of breastfeeding difficulties
- Reinforce the importance of exclusive breastfeeding
- Engage other health professionals as appropriate

Contact two: During and immediately after birth *(perinatal period up to the first two to three days after birth)*

- Explain importance of avoiding separation of mother and infant, regardless of mode of childbirth
- Explain importance of uninterrupted skin-to-skin contact for one hour immediately after birth
- Observe infant self-attach to the breast, assisting where needed
- Help to achieve good positioning and attachment and show alternative positions
- Explain infant feeding cues and behaviours, onset of lactation, responsive breastfeeding and signs of adequate transfer of milk
- Reinforce the importance of exclusive breastfeeding
- Counsel mothers on the use and risks of feeding bottles, teats and pacifiers
- Counsel on the importance of breastfeeding as part of continuous kangaroo care for low birthweight, small-for-age or preterm infants

Contact three: At one to two weeks after birth *(within the neonatal period)*

- Observe a breastfeed and ask how breastfeeding is going
- Reassure on what is going well, talk through any difficulties and offer help
- Explain good positioning and attachment and alternative positions
- Help with position and attachment as needed
- Reinforce the importance of exclusive breastfeeding
- Teach the skill of expressing milk, as needed
- Counsel on maternal or infant medical indications for supplementation following assessment and management, as needed

Contact four: In the first three to four months *(early infancy)*

- Reassure the mother on what is going well and ask if there are any difficulties
- Reinforce the importance of exclusive breastfeeding until 6 months of age. Discuss timely, adequate, safe and appropriate complementary feeding, and responsive complementary feeding, as anticipatory guidance
- Discuss any concerns about milk supply. Counsel on maternal or infant medical indications for supplementation following assessment and management, as needed
- Provide information and explain infant feeding behaviour

Contact five: At 6 months of age *(at the start of complementary feeding)*

- Ask how the mother is getting on, reassure her and help with any difficulties
- Discuss any challenges with feeding and growth spurts
- Discuss the benefits of continued breastfeeding up to 2 years of age
- Discuss timely, adequate, safe and appropriate complementary feeding, and emphasize responsive complementary feeding

Contact six: After 6 months of age *(late infancy and early childhood)*

- Discuss the benefits of continued breastfeeding up to 2 years of age and beyond
- Discuss the importance of timely, adequate, safe and appropriate complementary feeding, and emphasize responsive complementary feeding

Additional contacts as necessary

- Provide counselling when concerns or challenges arise, including feeding during illness
- Counsel when opportunities for breastfeeding counselling occur (e.g., postnatal check-ups, immunization contacts)
- Anticipate challenges for returning to work and practice breastmilk expression, as needed

Source: United Nations Children's Fund, World Health Organization. Implementation guidance on counselling women to improve breastfeeding practices. New York: United Nations Children's Fund; 2021:31–2.

Families and friends are often a mother's main source of information and advice about feeding her children. When counselling mothers about infant and young child feeding, health workers should include other caregivers and family members as appropriate.

5.2 Infant and young child feeding counselling

All health workers who care for women and children during the postnatal period and beyond have a key role to play in establishing and sustaining breastfeeding and appropriate complementary feeding. However, many health workers are not equipped to fulfil this role effectively because they have not been trained to do so.

Counselling differs in important ways from medical consultation. It involves engaging in a dialogue on breastfeeding with a woman and other family members as required. Effective breastfeeding counselling aims to identify feeding difficulties, provide practical support and help mothers manage and overcome difficulties.

The Assess, Analyse, Act (or Triple A) approach offers a three-step framework for structuring breastfeeding counselling *(3)*. The three steps are:

Assess – Listen to a mother's breastfeeding concerns and respond appropriately. Assess both the breastfeeding and the family's situation. It involves asking questions about the age of the baby, feeding history and observing a breastfeed. It also includes listening to the mother's concerns, observing the general condition of the mother and child, and eliciting the mother's breastfeeding plans.

Analyse – Identify feeding problems. Prioritize the most important issues, if there is more than one issue. Decide if there is a problem that needs referral.

Act – This may include providing reassurance, offering relevant information in small amounts, explaining the cause of breastfeeding difficulties and discussing how they can be addressed. The goal of this step is to guide the mother to develop a workable plan for continuing breastfeeding and/or addressing the presenting issue. This step may include offering anticipatory guidance or referring the mother and/or infant to a specialist for further assessment and/or treatment.

The specific actions that breastfeeding counsellors can take at each step are described in **Table 5.2** *(3)*.

Table 5.2 How to use a three-step counselling process

1. Assess the situation

- Greet the woman, mother and family in a friendly way.
- Use good communication skills through encouraging the mother to talk about her situation; listening and learning her history; and discussing present difficulties and potential future challenges.
- Observe the mother feeding and interacting with her infant(s).
- Look at the mother and the infant's general condition and growth chart, if the baby has been or can be weighed.
- Document maternal and infant history, any feeding difficulties, and any relevant observations.

2. Analyse the situation

- Decide if the care needed is within scope of practice.
- Determine if the mother needs a referral to another provider.
- Decide what preparation a pregnant woman may need to feed her baby.
- Analyse whether an infant is feeding appropriately and suckling effectively.
- Decide what help a woman may need for present feeding difficulties.
- Decide if anticipatory guidance is needed.

3. Act

- Use counselling skills to build confidence.
- Praise what a mother and her infant are doing well.
- Communicate relevant information, using simple language.
- Give practical help as needed, including with positioning and attachment at the breast.
- Teach relevant skills, such as expressing milk and cup-feeding.
- Suggest what a woman might do differently.
- Give anticipatory guidance as needed.
- Make a plan with the mother on steps to take moving forward.
- Document the care plan and make a plan for follow-up care as necessary.
- Determine a date and time for the next counselling contact, as necessary.
- Refer to breastfeeding counsellors with advanced competencies, as needed.
- Refer to breastfeeding counsellors in the community for follow-up, as needed.

Source: United Nations Children's Fund, World Health Organization. Implementation guidance on counselling women to improve breastfeeding practices. New York: United Nations Children's Fund; 2021:31–2.

Counselling is also different from educating, advising or sharing messages. A counsellor listens to understand a client's feelings, values, and goals in order to help them decide for themselves what is best for them. Rather than giving instructions, a counsellor offers suggestions or options and then supports their clients to create an action plan that aligns with their goals, values and circumstances. Counselling includes listening with respect, exploring and assessing infant feeding situations, and guiding mothers to prevent or resolve infant feeding issues in ways that are achievable and sustainable for them and their families. Counselling supports women in their own decision-making, provides anticipatory guidance (the right information, in the right dose, at the right time), and helps them to manage challenges as they arise. Effective counselling is both person-centred and quality focused.

5.3 Communication and support skills

Health care providers expected to provide breastfeeding counselling should be given opportunities to develop counselling skills. **Table 5.3** lists key skills required to provide breastfeeding counselling *(3)*, some of which may differ from the skills needed to perform other tasks required of health care workers. These may include:

Using helpful non-verbal communication. Non-verbal communication means how you communicate other than by speaking. Helpful non-verbal communication shows that the health worker respects the mother and is interested in her. It includes keeping your head about level with the mother's, and not towering over her; making eye contact, nodding and smiling; making sure that there are no barriers, such as a table or conspicuous papers, between you and the mother; making sure that you do not seem to be in a hurry; touching her or the baby in a culturally appropriate way.

Asking open questions. Open questions elicit a great deal of information and reduce the number of questions that need to be asked. They encourage a mother to talk expansively about her situation because they cannot be answered definitively or with 'yes' or 'no'. Open questions often ask the mother to describe something, such as "What happens when …?" or "What have you tried?", "What is happening?", or "How do you decide?". These questions allow the mother to share the information she thinks is important. In contrast, closed questions require short, definitive responses and often invite a "yes" or "no" answer. They may start with "Do you?", "Are you", "Is he?", "Has she?", "How many?", "When does?". Closed questions are useful for clarification and for eliciting specific information that is not provided in response to open questions. Closed questions should be used infrequently.

Using responses and gestures that show interest. Such responses include "Oh dear", "Really?", "Go on" or "Eeeeh". Gestures such as nodding and smiling are also responses that show interest. Showing interest encourages a mother to say more.

Reflecting back what the mother says. Reflecting is a very helpful way to show that you are listening and to encourage a mother to say more. It is best to reflect back using slightly different words from the mother, not to repeat exactly what she has said. You may only need to use one or two of the important words that she used to show that you have heard her.

Empathising. Showing that you understand how she feels lets the woman know that you understand her feelings from her point of view, using phrases such as "you are worried", "you were very upset" or "that is hard for you". You can also empathize with good feelings, for example, "you must feel pleased".

Avoiding words that sound judging. These are words such as "right", "wrong", "good", "well", "badly", "properly", "enough". For example, the care provider should not say "Are you feeding your baby properly?

Do you have enough milk?" This can make a mother feel doubtful, and that she may be doing something wrong. It is better to ask "How are you feeding your baby? How about your breast milk?" Sometimes asking "why" may sound judging, for example "Why did you give a bottle last night?" It is better to ask, "What made you give a bottle?"

Accepting what a mother thinks and feels. Accepting means not disagreeing with a mother or caregiver, but at the same time not agreeing with an incorrect idea. Disagreeing with someone can make her feel criticised, and reduce her confidence and willingness to communicate with you. Accepting involves responding in a neutral way. Later, you can give the correct information.

Recognizing and praising what a mother and baby are doing right. Health workers are trained to look for problems and may only see what is wrong and then try to correct it. Recognizing and praising a mother's good practices helps to reinforce them and build her confidence. You can also praise what a baby does, such as growing and developing well.

Giving practical help. Helping a mother or caregiver in other ways than talking, often quite simply, such as giving her a drink of water, making her comfortable in bed or helping her to wash are examples of practical help. When a mother has had a great deal of advice or has been struggling with her baby, this kind of practical help may be the best way to show that you understand, and she may be more receptive to new information and suggestions. Helping with her breastfeeding technique is also practical help, but of a different kind as it involves giving her information too. She may not be ready for that at first.

Giving a little relevant information. After you have listened to a mother or caregiver, think about her situation and decide what information is most relevant and useful at the time. You should avoid telling her too much, because she may become confused and forget what is most important. Sometimes the most useful information is a clear explanation of what she has noticed, for example the baby's behaviour, or changes in her breasts; or what to expect, for example how breast milk "comes in", or when and why the infant needs foods in addition to breast milk. Helping her to understand the process is better than immediately telling her what to do.

Using simple language. It is important to give information in a way that is easy for a person to understand, using simple, everyday words.

Making suggestions, not commands. If you tell a mother what to do, she may not be able to do it, but it can be difficult for her to disagree with you. She may just say "yes" and not come back. Giving a suggestion makes room for her to discuss whether or not she can follow it. You can make other suggestions, encourage her to think of more practical alternatives and help her to decide what to do. This is particularly important in the case of infant and young child feeding, when there often are different options.

Table 5.3 Key breastfeeding counselling skills

Basic breastfeeding counselling services
Listen and learn by: • using helpful non-verbal communication • asking open questions • using responses and gestures that show interest • reflecting back what the mother says • empathizing and communicating that you understand how the other person feels • avoiding words that sound judgemental. **Build confidence and give support by:** • accepting what a mother thinks and feels • recognizing and praising what a mother and child are doing right • giving practical help • giving relevant information • using simple language • making one or two suggestions, not commands. **Assess and document:** • feeding histories of healthy and sick infants or young children • the mother's history, present difficulties and possible future challenges • breastfeeding of healthy and sick infants and young children • the mother feeding and interacting with her baby • the infant's general condition, and measure and assess growth • the general condition of the mother, and check and palpate breasts as needed. **Assist mothers to:** • initiate breastfeeding within an hour of vaginal or caesarean birth • position themselves and their infant for breastfeeding • attach their infant to their breast • understand and practice responsive (unrestricted) breastfeeding • hand-express breastmilk • cup- or spoon-feed infants • manage flat or inverted nipples • manage sore or cracked nipples • manage engorgement and recognize signs of mastitis • recognize the signs of adequate milk transfer • manage perceived insufficient milk supply • manage low or excess milk supply • soothe infants who cry frequently or have sleeping difficulties. **Prepare:** • pregnant women for breastfeeding • mothers to breastfeed exclusively for six months • mothers who are returning to work or periods away from their infant to maintain breastfeeding • mothers with information on timely, adequate, appropriate and safe complementary feeding

- mothers who wish to wean their infants or young children from breastfeeding with information on how to do so.

Refer:

- mothers and infants to breastfeeding counsellors with advanced competencies or health care professionals with specialized skills, as needed.

Advanced breastfeeding counselling services

Assist mothers to:

- manage mastitis
- manage candida
- manage tongue tie (ankyloglossia)
- breastfeed a low birthweight, preterm or sick infant
- successfully breastfeed despite deeply inverted or very large nipples
- breastfeed infants through difficult personal circumstances or postpartum depression
- breastfeed infants who are refusing to breastfeed (nursing strike)
- induce or re-establish lactation (re-lactation).

Prepare:

- pregnant women at high-risk for low milk supply, including overweight or obese women
- pregnant women who are HIV-positive with information on how to safely breastfeed
- high-risk pregnant women to breastfeed a small, sick or preterm infant.

Refer:

- mothers and infants to counsellors with basic breastfeeding competencies for follow-up or to health care professionals with specialized skills, as needed.

Source: United Nations Children's Fund, World Health Organization. Implementation guidance on counselling women to improve breastfeeding practices. New York: United Nations Children's Fund; 2021:31–2.

5.4 Preparing families for breastfeeding

Breastfeeding should be discussed with pregnant women and their families during routine antenatal care counselling. **Box 5.2** contains a protocol for antenatal guidance on breastfeeding *(3)*. Initiating these conversations at the first or second antenatal visit ensures there is time to identify potential challenges and discuss effective management strategies. This is particularly important in settings where women have few antenatal visits and/or initiate visits late in their pregnancy. Women who give birth prematurely may not have adequate opportunities to discuss breastfeeding if these conversations are delayed until very late in pregnancy.

Box 5.2 Antenatal breastfeeding counselling topics

Anenatal breastfeeding counselling should include discussion of:

- immediate and sustained skin-to-skin contact after birth;
- early initiation of breastfeeding;
- rooming-in;
- exclusive breastfeeding;
- risks of giving formula or other breast-milk substitutes;
- continued breastfeeding after 6 months when other foods are given;
- positioning and attachment; and
- recognising feeding cues.

Source: Infant and young child feeding counselling: an integrated course: trainer's guide. 2nd ed. Geneva: World Health Organization; 2020.

5.5 Assessing an infant or child's growth

Assessing a child's growth provides important information on the adequacy of the child's nutritional status and health. Health care professionals who provide routine care and monitoring for mothers, infants and children should be trained to take anthropometric measures, plot them against the Standards, and respond appropriately *(4,5)*.

There are several measures that can be used to assess growth, including weight for age (WFA), weight for height (WFH), and height for age (HFA). Ideally, sequential measures of should be used to monitor infants' growth trajectories against WHO Growth Standards *(4,6,7)*.

There are multiple possible causes of slow growth, including infection (acute or chronic) as well as inadequate nutritional intake. When a child is ill their weight gain may temporarily slow or stop. These children usually experience a short period of rapid weight gain (catch-up growth) that restores their growth trajectories *(4).*

If sequential measures are not available and the child's age is known, a single measure of weight for age may be used to assess whether a child is underweight or severely underweight. Weight for age cannot be used where a child's age is not known and it does not discriminate between acute malnutrition and chronic low energy and nutrient intake. Any child with weight for age Z<-2 is at risk of poor growth and development and should be referred for comprehensive assessment and treatment if required *(6)*.

Mid upper arm circumference (MUAC) should be measured in all children who have a very low weight-for-age. Infants from six weeks to six months with MUAC <110 and children between six and 59 months with MUAC below 115 mm should be referred for full nutritional assessment and appropriate treatment *(6)*. MUAC can also be used for rapid screening of all children in a community for severe malnutrition *(6,8)*.

5.6 Taking a feeding history

During any contact with a mother and child, it is important to ask how feeding is progressing. Simple open questions can generate a great deal of *information (9).*

5.6.1 Taking a feeding history for infants less than months of age

When a child is not growing well or the mother has a feeding difficulty, it is useful to conduct a detailed feeding history *(9).* The Feeding history job aid for infants 0–6 months in **Box 5.3** summarizes key topics to cover in a counselling session with a mother with an infant less than 6 months of age. The form is not a questionnaire, and it may not be necessary to ask the mother about every topic listed. Rather, it should be used as a memory aid or prompt to ensure that the mother is asked about everything that could be relevant to the child's situation. Concentrate on those issues that are relevant for the child's age and situation. Ask the mother about the child, how they are fed, about herself, the family, and their social situation using the listening and learning skills described in **Section 5.3**, above *(3, 9).*

Box 5.3 Feeding history for infants less than 6 months

Key topics to explore, depending on the infant's age and situation, include:

- Age of child
- Particular concerns about feeding of child

Feeding
- Milk (breast milk, formula, cow milk, other)
- Frequency of milk feeds
- Length of breastfeeds/quantity of other milks
- Night feeds
- Other foods in addition to milk (when started, what, frequency)
- Other fluids in addition to milk (when started, what, frequency)
- Use of bottles and how cleaned
- Feeding difficulties (breastfeeding/other feeding)

Health
- Growth chart (birth weight, weight now, length)
- Urine frequency per day (6 times or more, if less than 6 months)
- Stools (frequency, consistency)
- Illnesses

Pregnancy, birth, early feeds (where applicable)
- Antenatal care
- Feeding discussed at antenatal care
- Delivery experience
- Rooming-in
- Pre-lacteal feeds
- Postnatal help with feeding

Mother's condition and family planning
- Age
- Health (including nutrition and medications)
- Breast health
- Family planning

Previous infant feeding experience
- Number of previous babies
- How many breastfed and for how long
- If breastfed – exclusive or mixed-fed
- Other feeding experiences

Family and social situation
- Work situation
- Economic situation
- Family's attitude to infant feeding practices

Source: Infant and young child feeding counselling: an integrated course: trainer's guide. 2nd ed. Geneva: World Health Organization; 2020.

5.6.2 Taking a feeding history for infants or children 6–23 months of age

A feeding history taken for an infant or young child 6–23 months of age includes the topics listed in **Box 5.4** below *(9)*. Again, this is not a questionnaire, but a reminder about the important things to learn from the mother when assessing feeding to help the mother develop a plan for improving the child's nutrition.

Box 5.4 Feeding history for children 6-23 months

- Age of child
- Particular concerns about feeding of child
- Growth chart (birth weight, weight now, length)

Mother's Health

- Age
- Health – including nutrition and medications
- Mental health
- Family planning
- Work situation
- Economic situation
- Family's attitude to infant feeding practices

Feeding

- Breastfeeding?
  - How many times per day?
  - Overnight?
  - If using expressed breast milk, how is the milk stored and given?
- Other milk (including infant formula)?
  - How much?
  - How often?
- Bottle fed?
  - Cleaning and hygiene
  - Feeding difficulties (breastfeeding/other feeding)?
  - At what age was complementary feeding started?
- How many meals and snacks each day?
- How much food at each meal?
- What is the consistency of the main meals?
- Do meals include: animal-source foods, dairy products, dark green vegetables or red or orange fruits or vegetables, pulses (beans, lentils, peas, nuts), oil?
- Who helps the child to eat?
- What bowl does the child get food from (his or her own bowl, or the family pot)?
- Is the child given any vitamin or mineral supplements?
- How does the child eat during sickness?

Source: Infant and young child feeding counselling: an integrated course: trainer's guide. 2nd ed. Geneva: World Health Organization; 2020.

5.7 Observing a breastfeed

Observation is an important tool for assessing breastfeeding. **Box 5.5** describes the elements that should be observed during breastfeeding *(10)*. Breastfeeding should be observed whenever an infant less than 2 months of age is presented for routine screening, assessment for, or treatment of illness. For infants from 2 months breastfeeding should be observed whenever a mother reports difficulty feeding or an infant is not growing as expected.

The Breastfeeding Observation Checklist (**Box 5.6**) may be used to support breastfeeding assessment *(11)*. If the baby has just breastfed or is fast asleep, wait until ready to breastfeed again. To initiate an observation, ask the mother to offer her baby a breastfeed as she usually does and observe a complete feed (to see how long the baby suckles for and whether they release the breast spontaneously). If the mother has obvious difficulties, it may be appropriate to interrupt the feed in order to help her to improve positioning and attachment while the baby is still hungry.

The signs of good positioning and attachment are described in **Section 2.3.3**.

Box 5.5 Elements to observe during breastfeeding*

- infant is able to latch and transfer milk;
- infant has rhythmic bursts of suckling with brief pauses;
- infant releases the breast at the end of feed in obvious satiety;
- infant shows similar behaviours if he takes the second breast;
- mother's hand supports the baby's neck and shoulders, without pushing the baby's head onto the breast;
- mother ensures the baby's postural stability;
- mother's breasts and nipples are comfortable and intact after the feed;
- mother reports no breast or nipple pain;
- any signs/symptoms that could require further assessment or treatment.

*Adapted from Competency verification toolkit: ensuring the competency of direct care providers to implement the Baby-Friendly Hospital Initiative (*10*)

Source: World Health Organization, United Nations Children's Fund. Baby-friendly hospital initiative training course for maternity staff: trainer's guide. Geneva: World Health Organization; 2020.

Box 5.6 Breastfeeding Observation Checklist

adapted from Baby-friendly hospital initiative training course for maternity staff *(11)*

Breastfeeding Observation Checklist

Mother's name________________________ **Date** ______________________________

Baby's name __________________________ **Baby's age** __________________________

Signs that breastfeeding is going well:	**Signs of possible difficulty:**
GENERAL	
Mother	**Mother**
☐ Mother looks healthy	☐ Mother looks ill or depressed
☐ Mother relaxed and comfortable	☐ Mother looks tense and uncomfortable
☐ Signs of bonding between mother and baby	☐ No mother-baby eye contact
Baby	**Baby**
☐ Baby looks healthy	☐ Baby looks sleepy, ill or listless
☐ Baby calm and relaxed	☐ Baby is restless or crying
☐ Baby reaches or roots for breast if hungry	☐ Baby does not reach or root
BREASTS	
	☐ Breasts look red, swollen or sore
☐ No pain or discomfort	☐ Breast or nipple painful
☐ Breast well supported with fingers away from nipple	☐ Breast held with fingers on areola
BABY'S POSITION	
☐ Baby's head and body are in line	☐ Baby's neck and head twisted to feed
☐ Baby is held close to mother's body	☐ Baby not held close
☐ Baby's whole body is supported	☐ Baby supported by head and neck only
☐ Baby approaches breast nose to nipple	☐ Baby approaches breast with lower lip/ chin to nipple
BABY'S ATTACHMENT	
☐ More areola seen above baby's top lip	☐ More areola seen below bottom lip
☐ Baby's mouth open wide	☐ Baby's mouth not open wide
☐ Lower lip turned outwards	☐ Lips pointing forward or turned in
☐ Baby's chin touches breast	☐ Baby's chin not touching breast
SUCKLING	
☐ Slow, deep sucks with pauses	☐ Rapid shallow sucks
☐ Cheeks round when sucking	☐ Cheeks pulled in when suckling
☐ Baby's hand relaxes and opens during feed	☐ Baby's hands remain closed or fisted
☐ Baby releases breast when finished	☐ Mother takes baby off the breast
☐ Mother notices signs of oxytocin reflex	☐ No signs of oxytocin reflex noticed

NOTES

Source: WHO guideline: counselling of women to improve breastfeeding practices. Geneva: World Health Organization; 2018.

5.8 Assessing infant and young child feeding

Assessment should include:

- identifying IMCI danger signs *(11,12)*
- assessing the general health of the mother and the infant
- taking a feeding history
- mother's experience with suggestions discussed at any previous contact
- how baby has been feeding, any changes or improvements
- signs of sufficient milk intake (see **Section 7.4**)
- observing a breastfeed
- measuring weight and height/length to assess growth *(7).*

The results of the clinical assessment are used to classify the infant and to decide on management. **Fig. 5.1** summarizes three categories of actions that may be required:

Pink – Refer urgently *(8,12,13)*

Purple – Help with difficulties and poor practices and refer if necessary *(9)*

Blue – Support for good feeding practices *(9).*

5.8.1 Pink – refer urgently

Refer the infant or young child to an inpatient facility if they have any of the IMCI danger signs *(11,12)* or are severely malnourished (see **Section 6.1**). It may be necessary to give one or more treatments in the clinic before the infant or child leaves for the hospital.

5.8.2 Purple – assist with breastfeeding difficulties

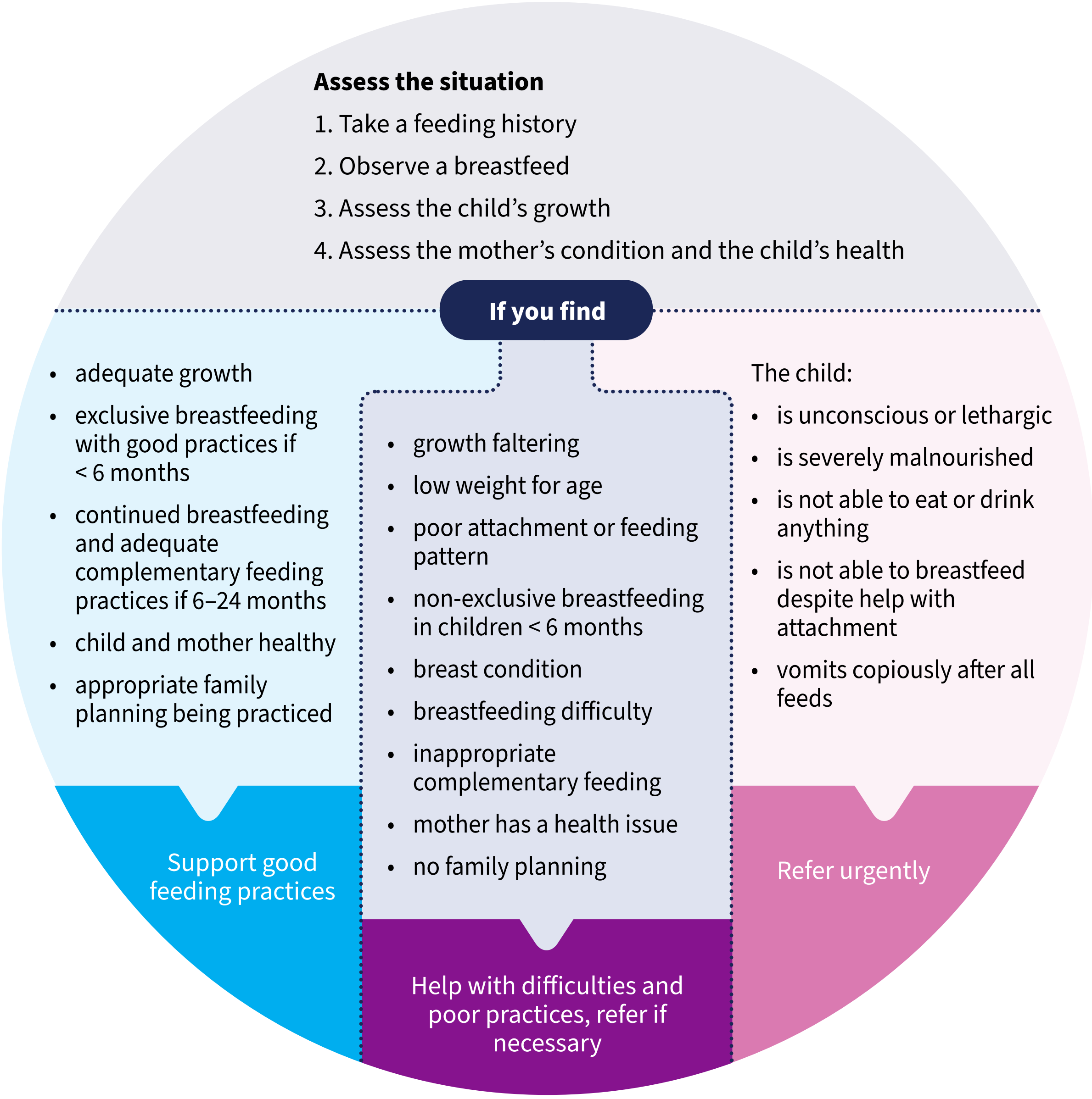
Infant and young child feeding counselling

IYCF counselling should be provided to mothers of infants and young children up to 24 months and beyond.

Infants who are not growing well, including infants less than six months of age at risk of poor growth and development *(6),* should be assessed to determine whether their breast milk intake is adequate, and supported to increase breast milk intake. Insufficient breast milk intake is discussed in **Section 7**. Infants less than six months who are not breastfed are particularly vulnerable and should receive special attention from the health and social welfare system *(14).*

Many social and environmental factors can affect feeding, care and resulting growth. Therefore, it may be necessary to address underlying causes of undernutrition in the home environment, such as the absence of a responsible adult to care for the child during the day, poor sanitation or contaminated water. It is not always possible to resolve these causes, but the health care provider can help the mother to understand them and think of positive actions to take.

Fig. 5.1 Assessing and classifying infant and young child feeding: a decision tree



During the counselling session it is important to help the mother or caregiver identify actions she can take to improve the child's growth that are feasible given her situation. Giving relevant information, offering suggestions and inviting the mother to make her own suggestions (rather than giving instructions), and supporting the mother to develop her own, achievable action plan may improve adherence. Giving too much information or too many suggestions can be overwhelming, which can cause decisional paralysis. Offer small pieces of information and make two or three suggestions. If there is more than one feeding issue to address, choose the most important or encourage the mother to choose the issue she would like to tackle first and recommend appropriate follow-up visits. Follow-up visits provide opportunities for continued monitoring and counselling, which should include reassurance and validation. Change takes time and the underlying causes of poor growth are unlikely to be resolved in a single counselling session.

Poor attachment or feeding pattern, including non-exclusive breastfeeding

If a baby is not transferring milk effectively from the breast or is not breastfed frequently enough, it may be necessary to help a mother to position and attach her baby at the breast to establish optimal and effective breastfeeding (see **Section 2.3** and **Section 7**) and discuss with her how to improve her breastfeeding pattern.

Breastfeeding difficulties

Most feeding difficulties can be managed in the community. **Section 7** describes the most common feeding difficulties and summarizes key steps in their management.

Inappropriate Complementary Feeding

Sometimes a child over 6 months of age may be malnourished or growing poorly or may not be eating well. Mothers and other caregivers may not complain of difficulties with complementary feeding, but their practices are not optimal. In either situation, you should recognise the need to counsel them about improving the way in which they feed the child. Complementary feeding is discussed in **Section 3**.

Mother's health

Mothers may need help to adopt better practices, or to overcome difficulties with their own health, nutrition, or family planning. These issues are discussed in **Section 7** and **Section 8**.

Non-urgent referral may be necessary if more specialized help is needed than is available at your level. Refer mothers who have:

- breastfeeding difficulties that do not respond to the usual management;
- babies with poor growth that persists despite health centre or community care;
- a health condition that cannot be managed at the health facility;
- special needs where infant feeding counselling for their situation cannot be provided at health facility; or
- any condition that creates disability (e.g. cleft lip and palate, tongue tie, Down syndrome, cerebral palsy).

Child's health

When assessing a child, use Integrated Management of Childhood Illness *(12,13)* to identify infants and young children who need urgent referral or treatment, vaccination or micronutrient supplementation.

5.8.3 Blue – support effective feeding practices

An important part of counselling for every mother and child is active support and reinforcement of good feeding practices. Mothers may not be aware how important and valuable it is to continue them. Support and reinforcement are equally important if all her practices are already going well, or if encouraging her to improve some which are not optimal.

5.9 Follow-up infants with feeding difficulties

Infants less than one month old with any feeding difficulties can deteriorate rapidly and should be monitored and reassessed daily until the difficulty is resolved and indicators of adequate milk intake are present (see **Section 7.4**).

Infants less than one month old with any feeding difficulties can deteriorate rapidly and should be monitored and reassessed daily until the difficulty is resolved

From one month, infants with feeding difficulties should be monitored and re-assessed at least weekly and, depending on the severity of the infant's condition, as often as every second day.

If the infant has persistent feeding difficulties or is not gaining weight, offer additional assistance, follow up again after a few days (the exact interval will depend on the age of the infant and their condition), or consider referral.

If feeding difficulties are resolved and the infant has gained weight, continue scheduled well-child follow up *(15)*.

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SECTION 6

Special situations

Key messages

Intensive breastfeeding support is critical for mothers with:

- infants at risk of poor growth and development
- infants and young children with severe malnutrition
- infants and young children living in emergency situations
- exposure to infectious disease
- a history of breast surgery or breast cancer
- infants and young children or other caregivers who need to relactate.

Section 6 outline

6.1 Malnutrition

6.1.1 Infants at risk of poor growth and development

Infants under 6 months old are at risk of poor growth and development if **one or more** of the following applies *(1)*.

- They lose or fail to gain weight between sequential measures**.**
- They cross weight-for-age centile lines[1] in a downward direction**.**
- They do not gain around 500g over a month.
- They are 0–3 months of age and do not gain approximately 150–200g over a week.
- They are 3–6 months of age and do not gain approximately 100–150g over a week.
- They do not maintain expected growth velocity.[2]
- They have a weight-for-age z-score < -2 SD.
- They have a weight-for-length z score < -2 SD.
- They have nutritional oedema.
- They are between 6 weeks and 6 months of age with mid upper arm circumference (MUAC) < 110 mm.

Infants of any age are known to be at risk of poor growth and development if **one or more** of the following applies.

- They have neurodevelopmental concerns.
- They are not feeding well.
- Their mothers have physical or mental health problem(s) that affect their capacity.
- They have been hospitalized for any reason.
- They were born before 37 completed weeks gestation.
- Their birth weight was <2500g.
- Their birth weight was lower than the 10th centile of a birthweight-for-gestational-age based on a gender-specific reference population.

Sequential anthropological measures (WFL/H, WFA) are more accurate indicators of nutritional status than single measures but an infant who has any one or more of these is at risk of poor growth and development.

Treatment recommendations

Whenever an infant less than six months of age at risk of poor growth and development is identified health care providers should conduct comprehensive assessments of the mother/caregiver-infant pair and follow best practices for the management of breastfeeding/lactation challenges and underlying factors contributing to these challenges.

1 ≥ 1 growth centile space if birth weight < 9th centile; ≥ 2 centile spaces if birth weight 9th–91st centile; ≥ 3 centile spaces if birth weight > 91st centile.

2 Less than 2 standard deviations (SD) below the median on the WHO growth velocity standards from one measurement to the next.

Decisions about whether an infant less than 6 months of age at risk of poor growth and development needs a supplementary milk in addition to breastfeeding must be based on a comprehensive assessment of the medical and nutritional/feeding needs of the infant, as well as the physical and mental health of the mother/caregiver.

Infants less than 6 months of age at risk of poor growth and development should be referred and admitted for immediate inpatient care if they have:

– one or more danger signs listed in the Integrated Management of Childhood Illness;
– acute medical problems or conditions under severe classification as per Integrated Management of Childhood Illness;
– oedema (nutritional); or
– recent weight loss.

Infants who are less than 6 months of age with severe wasting and/or nutritional oedema who are admitted for inpatient care should be breastfed where possible and their mothers supported to breastfeed and express extra breastmilk (see **Section 8.5.1**) for supplementary feeding (see **Section 6.3.3**). If an infant is not breastfed, the mother should be supported to relactate and if this is not possible, wet nursing should be encouraged. These infants should also be provided with a supplementary feed using a supplementary feeding system (see **Section 6.3.4**) or cup feeding (see **Section 4.6)**.

Infants with severe wasting but no oedema should be supplemented with expressed breast milk, either (see **Section 6.3**) by the mother or a wet-nurse. Where expressed breast milk is not available or not available in sufficient quantities, commercial (generic) infant formula, F-75, or diluted F-100 may be given in addition to expressed breast milk or as the sole supplement if no breast milk is available. Infants less than six months of age with nutritional oedema should be given commercial (generic) infant formula or F-75 in addition to breast milk if necessary.

Full-strength F-100 should not be given to infants if they are clinically unstable and/or have diarrhoea or dehydration and/or nutritional oedema. Undiluted F-100 can cause hyponatraemic dehydration.

Infants who are not exclusively breastfed should be given appropriate and adequate replacement feeds such as commercial (generic) infant formula and supported to ensure they are able to maintain safe preparation and use at home when transferred from inpatient care.

Infants less than 6 months of age at risk of poor growth and development should be assessed (including assessment of their anthropometry) once they reach 6 months of age to determine if they need ongoing follow-up and referral to services for infants 6 months of age and older (including for nutritional treatment/supplementation) as appropriate to their clinical and nutritional status *(1)*.

6.1.2 Moderate wasting (moderate acute malnutrition)

Criteria

Infants and children 6–59 months whose weight-for-height or weight-for-length z-score is ≥ -3 and < -2 SD) or whose MUAC ≥ 115 mm to < 125 mm and who do not have nutritional oedema are moderately wasted *(1)*. Moderate wasting can escalate to severe wasting if it is not addressed with appropriate nutrition counselling and, if indicated, specially formulated foods.

Treatment recommendations

Infants and children aged 6–59 months of age with moderate wasting should have access to a nutrient-dense diet to fully meet their extra needs for recovery of weight and height and for improved survival, health and development.

Breast milk is a nutrient dense food and mothers or caregivers of moderately malnourished children should be encouraged and supported to maintain breastfeeding *(2,3)*.

Specially formulated foods should be considered in addition to nutrition counselling for infants and young children 6–59m who are moderately wasted when they:

Moderate wasting can escalate to severe wasting if it is not addressed with appropriate nutrition counselling and, if indicated, specially formulated foods.

- have MUAC 115–119mm
- have weight-for-age z-score < -3 SD
- are < 24m
- are not improving with counselling alone
- have relapsed after recovering from moderate wasting
- have a history of severe wasting
- have a medical condition that requires ongoing care and with significant association with nutritional status such as HIV and tuberculosis or a physical or mental disability
- have severe personal circumstances (such as poor maternal physical or mental health or mortality), or
- are affected by a recent or ongoing humanitarian crisis.

Lipid-based nutrient supplements (LNS) are the preferred choice of specially formulated foods for infants and children with moderate wasting. When these are not available, Fortified Blended Foods with added sugar, oil, and/or milk are preferred over Fortified Blended Foods with no added sugar, oil, and/or milk.

Infants and children 6–59 months of age with moderate wasting who require specially formulated foods should be given enough to provide 40–60% of the total daily energy requirements needed to achieve anthropometric recovery. Total daily energy requirements needed to achieve anthropometric recovery are estimated to be around 100–130 kcal/kg/day.

6.1.3 Severe wasting and nutritional oedema (severe acute malnutrition)

Children 6–59 months of age are considered to have severe acute malnutrition (or SAM) if they have *(1)*:

- WFHZ <-3,
- weight-for-age z-score (WFAZ) <-3,

– MUAC <115mm, OR
– nutritional oedema.

Infants 6–59 months may be given specially formulated foods in addition to breastfeeding and complementary foods. The preferred choice of specially formulated foods is lipid-based nutrient supplements. When these are not available, fortified blended foods with added sugar, oil, and/or milk are preferred over fortified blended foods with no added sugar, oil, and/or milk.

Infants 6–59 months who are managed as outpatients should be given specially formulated foods in addition to nutrition counselling if they:

– have MUAC 115–119 mm, or
– have a WFAZ < -3, or
– have a WFHZ < -3, or
– are aged < 24 months, or
– are not recovering from moderate wasting after receiving other interventions (for example, counselling alone), or
– have been at risk of poor growth and development before reaching 6 months of age, or
– have ever been severely wasted, or
– have severe personal circumstances, such as mother died or poor maternal health and well-being.

In infants and children 6–59 months of age with severe wasting and/or nutritional oedema who are enrolled in outpatient care, ready-to-use therapeutic food should be given in a quantity that will provide:

- 150–185 kcal/kg/day until anthropometric recovery and resolution of nutritional oedema, or
- 150–185 kcal/kg/day until the child is no longer severely wasted and does not have nutritional oedema, and then 100–130 kcal/kg/day, until anthropometric recovery.

Any breastfed child admitted for inpatient care should continue breastfeeding. Mothers of breastfed infants and children admitted for inpatient care should be supported to continue breastfeeding, especially if their infants are <6m. Breastfeeding should be protected when infants and young children require supplementary feeding. Mothers and caregivers who have stopped breastfeeding should be encouraged and supported to relactate (see **Section 6.3**).

6.2 Infants and young children living in emergency situations

An emergency is an event or series of events that represents a critical threat to the health, safety, security or wellbeing of a community or other large groups of people, usually over a wide area.

Emergencies can be natural disasters (such as earthquake, drought, epidemic/pandemic) or human-made (such as civil war, explosion, nuclear spill), and may be sudden or slow onset,

Babies born during and shortly after acute emergencies depend on breastfeeding to prevent infectious diseases that threaten survival.

short-term or protracted. In natural disasters and other humanitarian crises, breastfeeding can be lifesaving.

In emergency situations, it is essential to protect, promote and support breastfeeding, and ensure timely, safe and appropriate complementary feeding in accordance with international guidance *(4).*

In emergencies, infants and young children are more likely than older children or adults to become ill and die from malnutrition and disease *(4)*. Optimal feeding is often disrupted due to a lack of basic resources such as shelter and water, and the physical and mental stress on families. Breastfeeding may stop because mothers are ill, traumatized, or separated from their babies – and yet in emergency situations breastfeeding is especially valuable. Artificial feeding becomes more dangerous because of poor hygiene, lack of clean water and fuel, and unreliability of supplies. There may be no food suitable for complementary feeding, or facilities for preparing feeds and storing food safely.

Discharging babies from maternity services before breastfeeding is established is especially dangerous during emergencies and should be avoided.

Babies born during and shortly after acute emergencies depend on breastfeeding to prevent infectious diseases that threaten survival.

During emergencies, breast-milk substitutes cannot usually be used safely outside clinical settings and so should be avoided *(4,5)*.

Donors may send breast-milk substitutes and feeding bottles to emergency situations in inappropriate amounts, believing that they are urgently required. Without proper controls, these supplies are often given freely to families who do not need them, and stocks run out before more arrive for those who might have a genuine need. The result is inappropriate and unsafe use of breast-milk substitutes, and a dangerous and unnecessary increase in early cessation of breastfeeding. Babies may be given unsuitable foods, such as dried skimmed milk, because nothing else is available.

Discharging babies from maternity services before breastfeeding is established is especially dangerous during emergencies and should be avoided.

During emergencies, breast-milk substitutes cannot usually be used safely outside clinical settings and so should be avoided .

6.2.1 Counselling in emergencies

The principles and recommendations for feeding infants and young children in emergency situations are exactly the same as for infants in ordinary circumstances (see **Table 6.1**). Therefore, breastfeeding counselling should be an integral part of emergency preparedness plans for infant and young child feeding, and both initial and sustained responses *(5,6)*. Most malnourished mothers can continue to breastfeed while they are being fed and treated themselves.

Table 6.1 Key recommendations for breastfeeding counselling in emergencies

Recipients	Timing	Frequency	Mode	Provider	Qualities
Key recommendations for counselling women to improve breastfeeding practices					
ALL pregnant women and mothers with infants and young children	Pregnancy Directly after birth and up to 2-3 days after birth Up to 28 days after birth (neonatal period) First 3-4 months of age At 6 months of age (start of complementary feeding) After 6 months (late infancy and early childhood) Any age, as needed	At least six contacts and additional contacts as needed	In-person counselling Individual counselling is preferred Group counselling is useful as a complement to individual counselling Remote counselling may complement but not replace individual face-to-face counselling	Healthcare professionals • Physicians • Nurses • Midwives • Lactation consultants Para-professionals • Peer counsellors • Community-based health workers	Person-centred • Sensitive • Culturally appropriate • Trauma-informed • Participatory • Anticipatory Quality-focused • Timely • Efficient • Equitable

Source: Operational Guidance on Breastfeeding Counselling in Emergencies. London: IFE Core Group; 2021.

6.2.2 Specific help with feeding in emergencies

An emergency response should aim to include:

Baby-friendly maternity care: The Ten Steps for Successful Breastfeeding (see **Section 4.1**) should be implemented both at health facilities and for home deliveries *(7,8)*. All mothers should be provided with breastfeeding counselling by trained breastfeeding counsellors *(6,9)*.

Availability of suitable complementary foods: In addition to breast milk, infants and young children from six months onwards need complementary foods that are hygienically prepared and easy to eat and digest *(10).* Blended foods, especially if they are fortified with essential nutrients, can be useful for feeding older infants and young children. The use of feeding bottles should continue to be discouraged.

Adequate health services to:

- support breastfeeding and complementary feeding
- help mothers to express their milk (see **Section 8.5.1**) and cup feed (see **Section 4.6**) any infant who is too small or sick to breastfeed
- search actively for malnourished infants and young children so that their condition can be assessed and treated

- admit mothers of sick or malnourished infants to the health or nutrition rehabilitation clinic with their children
- help mothers of malnourished infants to relactate and achieve adequate breastfeeding before discharge from care, in addition to necessary therapeutic feeding.

Skilled help in the community to:

- teach mothers how to breastfeed and continue to support them until their infant reaches 24 months
- teach mothers about adequate complementary feeding from six months of age using available ingredients
- support mothers to practice responsive feeding
- identify and help mothers with difficulties, and follow them up at home if possible
- monitor the growth of infants and young children, and counsel the mother accordingly.

6.2.3 Controlled use of breast-milk substitutes

In emergencies, the use of breast-milk substitutes requires a context-specific, coordinated package of care and skilled support in order to ensure that the nutritional needs of non-breastfed children are met, but the risk posed to all children by inappropriate use of breast-milk substitutes is minimized.

Donations of breast-milk substitutes, complementary foods and feeding equipment should therefore not be sought nor accepted in emergencies. Instead, supplies should be purchased based on assessed need. Breast-milk substitutes can be distributed as part of the regular inventory of foods and medicines, in quantities only as needed *(4)*. Do not send supplies of donor human milk to an emergency that is not based on identified need and part of a coordinated, managed intervention. Breast-milk substitutes, other milk products, bottles and teats should not be included in a general or blanket distribution *(4)*.

A minority of infants less than 6 months of age will need to be fed using breast-milk substitutes, either short-term or long-term *(9)*. Replacement feeding with commercial infant formula should be supported until the infant is 6 months of age or until relactation (see **Section 6.3**) is established *(4)* if:

- a mother has died or is unavoidably absent;
- a mother is very ill (temporary use may be all that is necessary);
- a mother is relactating (temporary use);
- a mother tests HIV-positive and chooses to use a breast-milk substitute;
- a mother rejects the infant, for example after rape (temporary use may be all that is necessary); or
- an infant (born before the emergency) is already dependent on breast milk substitutes.

Caregivers will need guidance about hygienic and appropriate feeding with breast-milk substitutes, but every effort should be made to prevent "spill over" of artificial feeding to mothers and babies who do not need it, by teaching the caregiver privately to prepare feeds, and by taking care not to display containers of breast-milk substitutes publicly.

From 6 months, infant formula may be replaced with whole milk that has been pasteurised, boiled or reconstituted using safe drinking water *(10).*

6.3 Relactation

Relactation means re-establishing lactation sometime after it was previously ceased. A woman can relactate for a child she gave birth to or for another child. Relactation is an important management option in emergency situations, and for infants who are malnourished or ill *(9, 11, 12, 14)*.

Most women can relactate any number of years after their last child. Relactation is easier for women who stopped breastfeeding recently and for those who have an infant who still suckles sometimes. To relactate successfully, a woman needs to be well supported by others, including healthcare professionals, community health workers, mother support groups, women friends, older women and traditional birth attendants.

To relactate successfully, a woman needs to be well supported by others, including healthcare professionals

Physiologically, re-establishing a milk supply requires *(11,12,14):*

i) frequent breastfeeding and/or breast massage with nipple stimulation

ii) removing breast milk from the breasts frequently.

6.3.1 Frequent suckling

The mother and infant must stay together as much as possible. Skin-to-skin contact and KMC are helpful *(11,12,14)*.

To increase milk supply, the mother offers a breast frequently, at least eight to 12 times every 24 hours. The baby attaches properly and suckles well from the breast often and for as long as possible *(11,12,14)*.

Suckling causes the release of prolactin, which stimulates growth of alveoli in the breast and the production of breast milk. Because prolactin levels are higher at night, breastfeeding at night helps to build a milk supply *(11,12,14)*.

If the infant is not willing to suckle, a woman can start the relactation process with gentle breast massage eight to 12 times a day followed by 20 to 30 minutes of hand expressing (see **Section 8.5.1**) *(11,12,14)*.

Anything that makes a baby suckle less will make it harder for a mother to increase her milk supply. Therefore, minimize separation of the mother and child *(11,12,14)*.

6.3.2 Removing breast milk

If a child is not yet willing to breastfeed, or a mother is preparing to have a child with her, she can use a breast pump and/or hand express (see **Section 8.5.1**) to remove milk from her breasts and build up a milk supply *(12,14).* Frequent expressing for short periods is more effective than expressing less often for longer sessions. Hand expressing after expressing with a pump can yield more milk and therefore boost milk supply.

6.3.3 Supplementary feeds

While a milk supply is being re-established, the infant needs a temporary supplement, which can be expressed breast milk, donor milk, or a commercial milk formula.

The full amount of milk normally required by a healthy term baby is estimated to be 130 ml/kg body weight per day. At the start of relactation, the infant may need to receive the full amount as a supplement each day. Divide this into six to 12 feeds depending on the infant's age and condition. Young, weak or sick infants will need more frequent, smaller feeds.

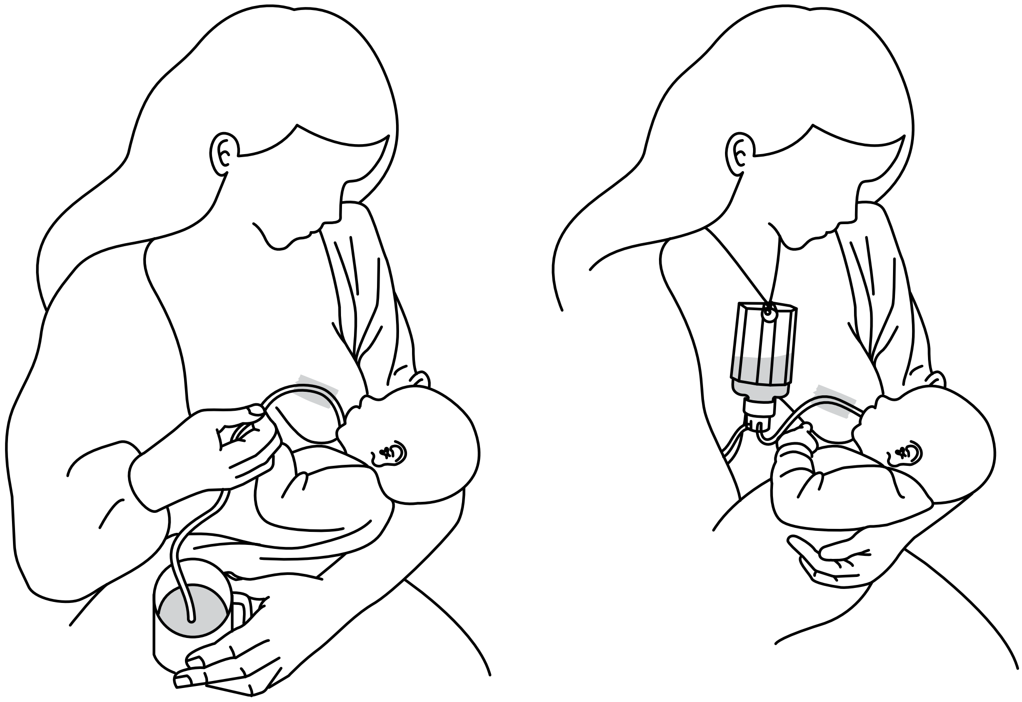
Whenever the baby wants to suckle, he or she should do so from the breast. The supplementary suckling technique described below may encourage an infant to suckle or to suckle for longer. If the supplementary suckling technique is not available, milk may be dripped from a spoon or paladai over the breast so that it runs into the corner of the baby's mouth. Dripping a few drops of milk onto the nipple before offering the breast may entice the baby to latch. This is known as the Drip and Drop Method *(13)*. Alternatively, the supplement can be given from a cup.

Some babies will latch and suckle more enthusiastically when they are most hungry. Giving these babies supplementary feed after breastfeeding ensures that they receive as much breast milk as possible. Some babies become too frantic to latch and suckle when they are most hungry. These babies may be more willing to latch and suckle after they are given a small amount of supplementary feed. Give the majority of the supplementary feed after a breastfeed has been attempted.

6.3.4 Feeding supplements at the breast

A supplemental nursing system, also known as a breastfeeding supplementer or supply line, is a device that makes it possible for a child to be fed a supplemental feed at the breast. It consists of a tube which leads from a cup of supplement to the breast, and which goes along the nipple and into the infant's mouth. The infant suckles and stimulates the breast at the same time as drawing the supplement through the tube. This technique usually needs to be practised under the supervision of a healthcare professional in a facility. *(12,14)*

Fig. 6.1 Using the supplementary suckling technique to assist relactation



Key practices:

- Use a fine nasogastric tube (gauge 8) or other fine plastic tubing.
- The mother can control the flow by raising or lowering the cup so that the infant suckles for about 30 minutes at each feed.
- If the tube is wide, a knot can be tied in it, or it can be pinched.
- Clean and sterilize the cup and tube after every feed.
- Encourage the mother to let the infant suckle on the breast as frequently as possible, not just when she is giving the supplement.

Once the infant is able to suckle at the breast without the supplemental nursing system, the mother can start giving supplementary feeds by cup instead, as cup feeding is more feasible in home conditions.

6.3.5 Galactagogues

If the above measures are not effective by themselves, medications called galactagogues can be prescribed to stimulate increased lactation. Commonly used galactagogues include domperidone, fenugreek, metoclopramide, and silymarin *(15)*.

Medications are only effective if the woman receives adequate help and her breasts are stimulated by a suckling baby or with a breast pump at least eight times a day.

6.3.6 Signs that breast milk is being produced

Breast-milk production may start in a few days or a few weeks. Signs of milk production include:

- breast changes – the breasts feel fuller or firmer, or milk leaks or can be expressed;
- less supplement consumed – the infant takes less supplement while continuing to gain weight;
- stool changes – the infant's stools become softer, more like those of a breastfed infant; and
- swallowing – the infant swallows while breastfeeding when the supplemental nursing system is not in use.

It is also important to know the signs of too little milk intake. These are described in **Section 7.4**.

6.3.7 Follow-up care

During the relactation process, closely monitor the infant's weight as well as output of urine and stool. As the infant gains weight and there are signs of breast-milk production, the volume of supplement can be gradually reduced every few days.

Once relactation has been established, the mother and baby can be discharged for community-level follow-up, with checks as often as possible from community health and nutrition workers.

6.4 Infectious disease

Breastfeeding protects infants from the vast majority of infectious diseases and can help speed children's recovery from illness by providing immunological support in addition to necessary fluids and nutrients. Very few infectious diseases are transmissible via breastfeeding and fewer still present health risks to breastfed infants that outweigh the risks of morbidity and mortality associated with withholding breastfeeding and using breast-milk substitutes.

When deciding whether or not to recommend cessation (temporary or permanent) of breastfeeding, health professionals must weigh up the risk of transmission, the means of transmission (e.g. milk, blood, skin, breathing), severity of illness, and risk of transmission through other exposures on the one hand, against, on the other, the availability of alternative milks, their safety, and the risks of morbidity and mortality associated with not breastfeeding.

WHO has not developed specific guidance on all infectious diseases but does have guidance on feeding in the context of HIV (see 6.4.1 HIV), Ebola (see 6.4.2 Ebola virus), Zika (see 6.4.3 ZIKA), human T-lymphotropic virus type 1 (HTLV-1) (see 6.4.4 HTLV-1), SARS-COV2 (COVID19) (see 6.4.5 SARS-COV2 (COVID19)), Hepatitis (see 6.4.6 Hepatitis A, B or C), and Tuberculosis (see 6.4.7 Tuberculosis).

6.4.1 HIV

Receiving antiretroviral therapy (ART) during pregnancy and while breastfeeding reduces the likelihood of mother-to-child transmission from up to 45% to less than 5% *(16)*. All infants born to mothers living with HIV should receive prophylactic antiretroviral therapy (see the WHO *Consolidated guidelines on the use of antiretroviral drugs for treating and preventing HIV infection) (17)*.

Mothers living with HIV and receiving ART should breastfeed exclusively for the first six months of life, introduce complementary foods from six months and continue breastfeeding alongside complementary feeding for at least 12 months and up to 24 months or beyond. Infants and young children known to be living with HIV should be exclusively breastfed for the first six months of life and continue breastfeeding alongside complementary feeding, in accordance with the recommendations for the general population, for up to two years or beyond *(16)*.

ART also reduces the risk of postnatal mother-to-child transmission in the context of mixed feeding. Although exclusive breastfeeding is recommended, practicing mixed feeding is not a reason to stop breastfeeding in the presence of ART. Mothers living with HIV who are receiving ART and do not intend to breastfeed for at least 12 months can also be encouraged and supported to breastfeed. Breastfeeding should only stop once a nutritionally adequate and safe diet can be provided in the absence of breast milk.

When antiretroviral drugs are not available, or not immediately available, breastfeeding may still provide infants born to mothers living with HIV a greater chance of HIV-free survival. In circumstances in which ART is unlikely to be available, such as acute emergencies, breastfeeding is still recommended *(18).*

Mastitis, abscess, and nipple fissure in a mother living with HIV

Conditions of the breast or of the child's mouth are factors that can affect mother-to-child transmission. Nipple fissure (particularly with bleeding), mastitis or breast abscess may increase the risk of HIV transmission. Therefore, prioritizing access to skilled breastfeeding support is particularly important for this group.

Mastitis, breast abscess and nipple fissure (especially if the nipple is bleeding or oozing pus) may increase the risk of mother-to-child transmission. Mothers living with HIV who develop mastitis should:

- avoid breastfeeding on the affected side while the condition persists
- express milk by hand (see **Section 8.5.1**) from the affected breast by for comfort
- continue to breastfeed on the unaffected breast, and breastfeed more frequently from that side to increase production and ensure an adequate intake
- initiate antimicrobial treatment and continue for 10–14 days
- rest and use analgesics as required
- incise and drain an abscess as required
- resume breastfeeding from the affected breast or breasts when the condition subsides.

Replacement feeding

Replacement feeding is NOT recommended even where ART is not available unless all the following conditions are met:

1. safe water and sanitation are assured at the household level and in the community
   and
2. the mother or other caregiver can reliably provide sufficient infant formula milk to support the normal growth and development of the infant
   and
3. the mother or caregiver can prepare it cleanly and frequently enough so that it is safe and carries a low risk of diarrhoea and malnutrition
   and
4. the mother or caregiver can exclusively give infant formula milk in the first six months
   and
5. the family is supportive of this practice
   and
6. the mother or caregiver can access health care that offers comprehensive child health services.

Mothers known to be living with HIV who decide to stop breastfeeding and transition to replacement feeds at any time should reduce breastfeeding frequency gradually and stop completely within one month.

Infants who have been receiving ART prophylaxis should continue prophylaxis for one week after breastfeeding is stopped altogether. Stopping breastfeeding abruptly is not advisable.

Heat-treated breast milk

Mothers living with HIV may choose to express and heat treat their milk *(18, 19)* as an interim feeding strategy in special circumstances, such as:

- when the mother is unwell and temporarily unable to breastfeed or has a temporary breast health problem such as mastitis or
- if ART is temporarily not available or
- when the infant is low birth weight or is otherwise ill in the neonatal period and unable to breastfeed or
- to assist mothers in stopping breastfeeding.

These mothers will need guidance on expressing and storing breast milk (see **Section 8.5**), heat treating expressed breast milk, cup feeding and the volume of expressed breast milk their infants need each day.

6.4.2 Ebola virus

If the mother or the child are suspected or known to have Ebola virus disease, breastfeeding should be suspended *(20)*. The child should be separated from the breastfeeding mother and provided an appropriate breast-milk substitute as needed. Infants and children of mothers suspected or known to have Ebola virus disease should undergo close monitoring for signs and symptoms of Ebola virus disease for at least 21 days. Post-exposure prophylaxis for Ebola virus disease can be considered for children exposed to the breast milk of women with Ebola virus disease on a case-by-case basis.

A woman who has recovered from Ebola virus disease should wait to restart breastfeeding until she has had two consecutive negative reverse transcription polymerase chain reaction (RT-PCR) breast milk tests for Ebola virus separated by 24 hours. During this time, the child should be provided with an appropriate breast milk substitute as needed.

Only if there is no other option – an appropriate and safe breast milk substitute is not available, the child is less than six months of age, and there is no one other than the mother who can provide appropriate care – may breastfeeding continue.

6.4.3 ZIKA

Infants born to mothers with suspected, probable, or confirmed Zika virus infection, or who reside in or have travelled to areas of ongoing Zika virus transmission, should be fed according to normal infant feeding guidelines *(21)*. They should start breastfeeding within one hour of birth, be exclusively breastfed for six months and have timely introduction of adequate, safe, and properly fed complementary foods, while continuing breastfeeding up to two years of age or beyond.

Babies whose mothers were infected with Zika virus during pregnancy are at increased risk of being born with congenital anomalies, including microcephaly. These infants may have difficulty breastfeeding and are likely to require intensive, sustained support from multi-disciplinary teams that include breastfeeding counsellors.

6.4.4 HTLV-1

HTLV-1 may be transmitted through breastfeeding. The estimated mother-to-child transmission rate has ranged from 3.9% to 27%. Most people who have HTLV-1 do not know they are infected. Screening tests are available but it can take several months or even years for a person who has been exposed to HTLV-1 to return a positive test result *(22)*.

Based on observational studies of mother-to-child transmission, shortening the duration of breastfeeding or even eliminating it altogether could enable women with HTLV-1 to limit the extent of exposure to their infants *(22)*. A recent systematic review found that infants breastfed for 3 months or less are no more likely to be infected with HTLV-1 than infants who are not breastfeed at all *(23)*.

The freeze-thaw method effectively eliminates the cells in breast milk that are infected with HTLV-1 and hence the source of transmission *(22)*. A recent systematic review found that freezing expressed breast milk at -20 degrees Celsius for 12 hours before thawing eliminates HTLV-1 *(23)*.

6.4.5 SARS-COV2 (COVID19)

Mothers with suspected or confirmed COVID-19 should be encouraged to initiate or continue to breastfeed *(24)*. All mothers and infants, regardless of whether they or their infants have suspected or confirmed COVID-19, should be accommodated together (known as "rooming-in") throughout the day and night and be supported to provide immediate and sustained skin-to-skin contact, including KMC, after birth and until breastfeeding is established *(24)*. Mothers who receive SARS-COV2 vaccines should be encouraged and supported to continue breastfeeding to protect their infants *(25)*.

6.4.6 Hepatitis A, B or C

If a mother has hepatitis (A, B, or C) breastfeeding can continue normally as the risk of transmission by breastfeeding is very *low (26).*

6.4.7 Tuberculosis (TB)

The risk of TB transmission through breast milk is negligible and mothers with TB should be supported to breastfeed as recommended for all infants *(27).* Separation from the mother is not advised, especially in resource-limited settings, where establishing breastfeeding can be critical for the child's survival *(28).*

Although the drugs most commonly used to treat TB are excreted into breast milk in small amounts, there is no evidence that this harms the infant or induces drug resistance *(29–31).* If TB is presumed or confirmed in the mother of an acutely ill neonate, the mother and her baby should be removed from the neonatal unit as soon as possible to prevent transmission to other neonates. Mothers should be supported to initiate and maintain breastfeeding and encouraged to wear a surgical mask when caring for the baby *(27).* Information about second-line TB medication use in breastfeeding can be found in the medicine information in the *WHO consolidated guidelines on tuberculosis. Module 4: treatment – drug-resistant tuberculosis treatment, 2022 update (31).*

6.5 Breast Surgery

Breast surgery has the potential to disrupt breast anatomy and thus negatively impact lactation. However, having had breast surgery is not a contraindication to breastfeeding *(32)*.

Regardless of the original indication, to successfully breastfeed, people who have had breast surgery will need additional support from skilled healthcare professionals to increase and maintain their milk production. Their babies require close monitoring for adequate weight gain.

Breast surgeries that include peri-areolar incisions, such as some augmentation, lift, and reduction procedures, have the potential to disrupt the areolar-nipple complex, and thus negatively impact lactation.

Mothers with breast implants or a prosthesis can produce some milk but are less likely to establish breastfeeding, especially exclusive breastfeeding. Implants below the muscle usually affect milk production less than implants above the muscle. Depending on the exact procedure and with the appropriate support, some people can successfully breastfeed after breast surgery *(33).*

6.6 Breast cancer

Breast cancer, including its treatment, may affect lactation *(34)* and breastfeeding dyads require close monitoring to ensure adequate infant growth.

Although women who have had a total mastectomy cannot lactate from the affected breast, these mothers may choose to breastfeed from the unaffected breast. Women who have had a partial mastectomy may be able to produce some breast milk from the affected breast. These mothers can be encouraged to offer the affected breast first to ensure that the remaining mammary tissue is maximally stimulated. Diminished milk production from both breasts may occur after chemotherapy *(33).*

Ongoing research is examining the oncologic safety of interrupting adjuvant endocrine therapy for childbearing with and without breastfeeding *(34).*

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SECTION 7

Common breastfeeding challenges

Key messages

- Nipple pain is not normal and indicates poor attachment.
- Nipple shape and size do not usually affect ability to breastfeed but mothers with nipples that are very flat, inverted or very large or long may need extra support to achieve proper attachment.
- Crying is normal and tends to peak at around 6-8 weeks, but investigation is warranted when a parent reports a sudden increase in crying.
- It is very common for mothers and other caregivers to worry that their baby is not getting enough breast milk; teach families the signs of sufficient milk intake.
- The most common causes of insufficient breast milk intake are poor attachment and infrequent breastfeeding frequency.
- It is possible to minimise these challenges by:
  - Providing extra support to mothers affected by engorgement, oversupply, blocked ducts, mastitis, candidiasis (thrush) and postnatal mood disorders.
  - Referring mothers for ongoing breastfeeding counselling in the community to increase their confidence in breastfeeding and encourage appropriate help-seeking.

Section 7 outline

7.1 Nipple pain

Nipple pain during breastfeeding is not normal. If a mother is experiencing nipple pain, it is very likely that her baby is not attaching properly to the breast, even if there are no other signs of poor attachment. A baby who is not attaching properly at the breast may not be transferring milk effectively.

A baby who is not attaching properly to the breast may cause damage to the mother's nipples that ranges from a mild graze to deep fissures. Fissures may occur across the tip of the nipple or around the base. The nipple can look discoloured or squashed from side-to-side at the end of a feed or may have a white pressure line across the tip.

The mother may require assistance to improve her baby's position and attachment. Once the baby is well attached and the nipple is no longer being damaged, the pain will decrease, and the nipple will heal quickly.

Persistent nipple pain suggests anatomical or infectious concerns that require further investigation and treatment *(1,2)*.

7.2 Flat, inverted or large nipples

Nipples occur naturally in a wide variety of shapes that usually do not affect a mother's ability to breastfeed successfully. Nipples that are very flat, inverted or very large or long can make it difficult to achieve proper attachment, however.

Most nipples that look flat are protractile, which means that if a mother pulls them out with her fingers, they will stretch, in the same way that they stretch in a baby's mouth during breastfeeding. A baby should have no difficulty suckling from a nipple that looks flat if it is protractile.

A nipple that is inverted retracts in response to stimulation and may not protract. This anatomy makes it more difficult for the baby to attach. A mother may experience pain during the initial feeds. Protractility often improves during pregnancy and in the first few weeks after a baby is born. An inverted nipple bound by tight connective tissue may gradually stretch over time in response to repeated suckling or regular application of negative pressure using a modified syringe as illustrated in **Fig. 7.1** *(3,4)*.

A large or long nipple may make it difficult for a baby to take enough breast tissue into their mouth to reach the larger ducts and suckle effectively. Sometimes the base of the nipple is visible even though the baby has a wide-open mouth. As a baby grows, their mouth becomes larger and it may become easier to attach.

Flat, inverted, large or long nipples can all be managed by applying the following principles *(3–5)*.

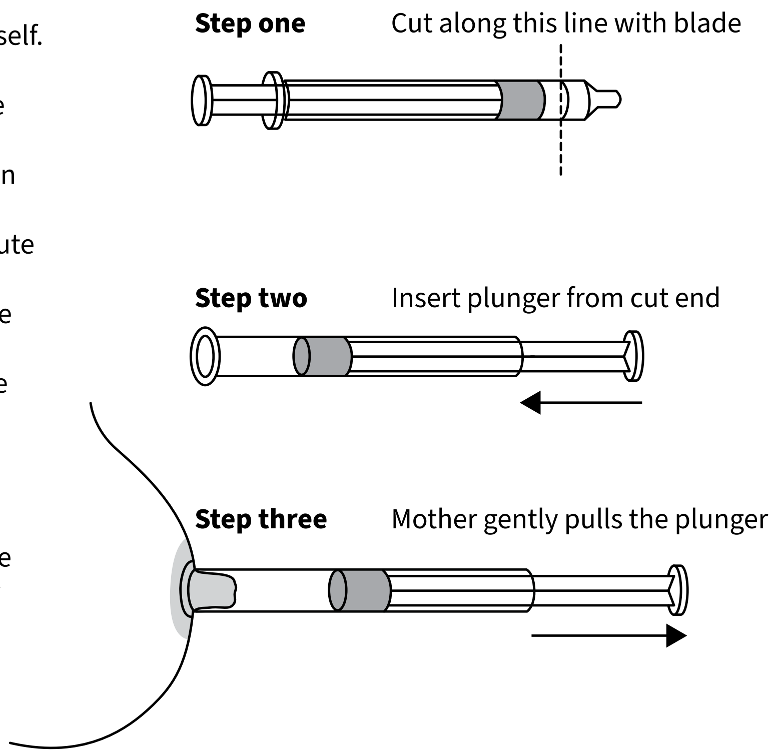
- Antenatal education is helpful. Reassure pregnant women who are worried about the shape of their nipples that most babies can suckle without difficulty from nipples of any shape.
- Preparing nipples during pregnancy is not necessary and can negatively affect the mother's feeling of self-efficacy.
- Early skilled breastfeeding support soon after birth is important.
- As soon as possible after birth, the mother should be assisted to position and attach her baby.

- The mother should give the baby plenty of skin-to-skin contact near the breast, and let the baby try to find their own way of taking the breast.
- The mother should keep putting her baby to the breast in different positions until the baby can attach.
- The mother can express milk into the baby's mouth and touch the baby's lips to stimulate the rooting reflex. These actions will encourage the baby to open their mouth wider.
- While her baby is still learning to suckle from the breast and cannot yet attach well, the mother is encouraged to express milk and feed it to her baby with a cup.
- Avoid using feeding bottles or dummies, which can alter the way a baby suckles at the breast.
- For flat or inverted nipples, a mother can use a 20 ml syringe with the adaptor end cut off and the plunger put in backwards (as shown in **Fig. 7.1**) to stretch out the nipple just before a feed *(3–5).*

Fig. 7.1 Syringe method for treatment of inverted nipples

The mother must use the syringe herself. Teach her to:

- put the smooth end of the syringe over her nipple;
- gently pull the plunger to maintain steady but gentle pressure;
- do this for 30 seconds to one minute several times a day;
- push the plunger back to decrease the suction if she feels pain; this prevents damaging the skin of the nipple and areola;
- push the plunger back to reduce suction when she removes the syringe from her breast;
- use the syringe to make her nipple stand out just before she puts her baby to the breast.



7.3 Crying baby

Crying is normal and tends to increase until it peaks at around 6–8 weeks, when many babies cry for 2–3 hours a day for no apparent reason. Most babies experience a period of crying in the late afternoon or evening when they may want to breastfeed very frequently and be difficult to console for some hours. However, when a parent reports a sudden increase in crying accompanied by unusual irritability or the baby's crying is causing acute distress to the parents, investigation is warranted. Acute onset crying can be an indication that the baby is in pain and should be thoroughly assessed *(6,7)*. In the absence of a clinical cause, parents need support and reassurance that crying behaviour is likely to improve or resolve over time, usually after the first three months of age.

7.4 Assessing breast milk intake

It is very common for mothers and other caregivers to worry that their baby is not getting enough breast milk. This concern often arises from misinterpretation of normal infant behaviour or unrealistic expectations of normal infant behaviour, such as crying, long or frequent breastfeeds, proximity seeking (unsettled unless held), difficulty settling the baby to sleep especially in the afternoon or evenings or waking to breastfeed during the night. As the baby grows, breastfeeds may become shorter and this can lead some mothers to worry that they are not making enough breast milk. Mothers who express their milk may notice that their milk is opalescent or translucent and worry that their milk is deficient in fat or other nutrients. None of these signs alone indicate that that the baby's milk intake or the mother's milk is inadequate *(8)*.

Mothers may worry that they are "losing their milk" when their breasts no longer feel firm or full. This usually occurs when the baby is around six weeks of age and is an indication that autocrine control of milk synthesis is established. Once autocrine control is established milk production is more closely matching the baby's intake than it is during early lactation when milk production is driven by the endocrine system. Some mothers worry that they are not producing sufficient breast milk if they cannot express any milk before or after a breastfeed. However, it is not unusual for a woman to be unable to express breast milk, despite breastfeeding successfully *(8)*.

To assess breastfeeding, healthcare workers should:

- identify the source of concern
- assess the general health of the mother and the child
- take a feeding history
- observe a full breastfeed, including attachment and suckling
- assess the infant/child's growth against WHO Growth Standards *(9)*.

Mothers of infants more than five days of age can be reassured that their baby is receiving sufficient breast milk if:

- they are breastfeeding between eight to twelve times in 24 hours;
- they can see and hear the baby swallowing during breastfeeding;
- the baby seems relaxed and satisfied (sometimes described as "milk drunk") after breastfeeding;
- the baby is passing pale urine at least six times per day;
- the baby is passing mustard-yellow (sometimes pale green, brown or orange), unformed stool at least the size of the palm of their hand at least three times per day until six weeks;
- the baby is passing copious, unformed stool every 1–10 days from six weeks; and
- the baby regains their birth weight by the end of the second week of life and their growth trajectory reflects the WHO Growth Standards *(9).*

Health care workers can work with mothers and other household members to help them understand that their baby's behaviours do not indicate that they are hungry and that crying, frequent breastfeeding, and proximity-seeking are normal, especially in the early weeks. Referring mothers for ongoing breastfeeding counselling in the community can increase their confidence in breastfeeding and encourage appropriate help-seeking.

There are two reliable signs that a baby is not getting enough milk:

i. Low output of urine and/or stools

There are two reliable signs that a baby is not getting enough milk – low output of urine and/or stools, and/or inadequate weight gain.

After the first week of life, a baby who is passing urine fewer than six times a day or has urine that is concentrated (dark yellow or red and strong smelling) is not getting enough breast milk. A baby who is more than four days old and continuing to pass meconium stool may not be receiving enough milk and needs additional support. Once breastfeeding is well established, there is a wide variety of normal stooling patterns amongst exclusively breastfed babies. After the first 20–30 days of life, an exclusively breastfed infant may stool several times a day or as infrequently as once every 1–10 days or even longer *(8)*. As long as the stool is unformed or liquid and copious when it is passed, there is no cause for concern.

ii. Inadequate weight gain

The WHO Growth Standards *(9)* provide a single international standard that represents the best description of physiological growth for all children from birth to five years of age. Infants whose weight gain does not conform to the weight-for-age standards need further assessment.

Poor weight gain is not a diagnosis but an indicator of some underlying problem. Babies can gain weight in variable patterns and each infant may follow their own pattern but a baby whose weight for age crosses two centile lines in a downward trajectory is not growing well. It can be difficult to determine from a single weight if a baby is growing satisfactorily.

By two weeks of age, babies should have recovered their birth weight and begun to follow a defined growth trajectory. From two weeks, babies who are exclusively breastfed should gain between 150 and 200 grams per week and at least 500 grams each month, depending on their birth weights.

7.5 Causes of insufficient breast milk intake

The most common causes of insufficient breast milk intake are poor attachment (which impedes milk transfer) and infrequent breastfeeding frequency. Poor attachment and infrequent breastfeeding can usually be corrected with skilled support and assistance. If milk intake does not improve in response to correcting attachment and or increasing breastfeeding frequency or duration, the mother and the infant should be assessed to identify underlying physical or psychological concerns that may be interfering with breastfeeding. Persistent difficulty with milk production is not common.

Uncommonly, a baby cannot suckle effectively due to illness or congenital abnormality. Rarely, physical or mental illness or physiological anomaly in the mother limits her capacity to produce sufficient breast milk *(8)*.

If not corrected, insufficient breast milk intake can cause a reduction in the volume of breast milk a woman produces. Milk production increases in response to efficient removal of milk from the breast. When a baby only partially removes milk from the breast, milk production decreases. When a baby removes more milk, milk production increases. As the mother's breastfeeding technique or baby's suckling pattern improves, the baby removes more milk, more milk is produced, and the baby's cumulative intake increases (see **Section 7.4**).

The most common causes of insufficient breast milk intake are poor attachment and infrequent feeding

Poor attachment (see **Section 2**) is likely to be the cause of inadequate milk intake if the mother experiences pain while breastfeeding or the baby:

- is generally unsettled
- wants to feed more than 12 times per day
- suckles for a long time at each feed after the first four weeks
- sleeps easily during the feed but then seems hungry at the end.

Health care providers can use the Breastfeeding Session Observation Job Aid to identify and address poor attachment.

Infrequent feeding is likely to be the cause of insufficient breast milk intake if the baby is:

- breastfed according to a schedule rather than in response to hunger cues
- breastfeeding fewer than 8–12 times in 24 hours
- breastfeeding for more than an hour at a time
- not breastfeeding at least once over night
- using a pacifier or dummy between breastfeeds.

Health care providers can show mothers how to recognise their baby's hunger cues and encourage them to breastfeed as often as their baby is hungry.

Most babies breastfeed between 8 and 12 times in 24 hours and often even more frequently than this. Babies usually breastfeed more frequently during the first four weeks of life and it may take them 30 minutes or more to finish a feed. Babies should be allowed to continue breastfeeding until they stop suckling and/or spontaneously release the breast. As infants grow, they become more efficient at the breast and may be able to complete some feeds in 5–10 minutes. Other feeds may be longer, especially if the baby is tired or breastfeeding to sleep *(8)*.

Restricting the duration of breastfeeds can also cause insufficient breast milk intake. During the first days and weeks of life, both breasts should be offered at every breastfeed to maximise nipple stimulation. If the baby falls asleep after feeding from the first breast, the mother can offer the other breast when the baby next displays hunger cues.

From one month of age, around 30% of infants always feed from one breast only at each breastfeed, around 13% always feed from both breasts at each breastfeed and most infants sometimes feed from one breast and at other times feed from both *(10,11)*. If a baby rejects one breast or a mother is only able to feed from one breast, her baby may breastfeed more frequently until that breast adjusts. Each breast functions independently and is usually capable of producing sufficient breast milk to exclusively breastfeed a baby. This can be reassuring for mothers of twins.

Babies who were born small (low birth weight), preterm or sick may not be strong enough to display clear hunger cues. These babies may need to be woken to breastfeed at least every three hours until they are strong enough to communicate their needs. More information about feeding these babies can be found in **Section 5** and **6**.

7.6 Increasing breast milk intake

Healthcare workers can support mothers to increase their baby's breast milk *(8,11)* intake by:

- helping the mother improve the baby's positioning and attachment;
- teaching the mother to recognise her baby's hunger cues;
- encouraging the mother to reduce or eliminate the use of pacifiers or dummies and use breastfeeding to soothe their babies;
- encouraging mothers to offer more frequent breastfeeds (at least 8–12 times per day) including at least one breastfeed overnight;
- identifying any signs of illness or abnormality in the baby or the mother and treating or referring them for treatment appropriately; and
- expressing milk from her breasts (see **Section 8.5.1**) the baby is unable or unwilling to remove milk effectively while breastfeeding (this can be after breastfeeding, between breastfeeds or instead of breastfeeds if the baby.

Health care workers should evaluate whether supplemental feeding is necessary (see **Section 5**). If it is, the healthcare worker can minimize the duration and impact of supplementation on breastfeeding by *(8,11):*

- teaching the mother to express breast milk by hand or with a breast pump (see **Section 8.5.1**);
- teaching supplemental feeding techniques to minimize impact on breastfeeding (for example, offering both breasts twice before supplementing; using expressed breast milk before offering supplemental milk; recognising and responding to satiety cues instead of encouraging the baby to finish a cup feed); and
- supporting the mother to protect and increase her breast milk production if supplementary feeding is necessary.

7.7 Feed Refusal

Babies may refuse to breastfeed by crying, arching their backs, and turning away when put to the breast. This behaviour can be observed during the first days of life or it can happen suddenly at any time in the first several months. Mothers may feel rejected and be distressed. The cause of refusal to feed must be identified and addressed as early as possible.

Babies may refuse to breastfeed for a variety of reasons *(8,11).* In the postnatal period feed refusal can be the result of:

- sedation caused by analgesics given to the mother during labour
- pain due to bruising or other trauma sustained during birth
- stiffness in the neck or body as a result of positioning in the uterus
- difficulty attaching to the breast or frustration due to poor milk transfer
- or becoming accustomed to the different sucking action required when bottle feeding.

In older infants, feed refusal can be caused by pain associated with illness such as oral candidiasis, otitis media, gastro-enteritis, gastro-oesophageal reflux disease. It can also be an indication of infective mastitis, which can change the flavour of the milk *(11).*

Difficulty attaching to the breast may be the result of the mother or someone helping with positioning or attachment applying pressure to the back of the baby's head or using force to encourage the baby to take the breast. Babies resist this pressure by pulling away from the breast, impeding attachment *(8).*

7.7.1 Management of feed refusal

Taking a thorough history and performing a physical examination can identify causes of breast refusal. Once identified, the cause of breast refusal should be treated or eliminated *(8,11).*

Aim for breastfeeding to be enjoyable for both the mother and the baby. Reduce separation of mother and baby and avoid other changes that may be upsetting. Breastfeeding technique can be improved by:

- limiting the number of people involved with caring for the baby for a time;
- keep the baby close and practicing frequent skin-to-skin contact;
- offering a breast whenever the baby shows signs of interest in suckling (responsive feeding);
- avoiding taking the breast towards the baby and pressing the baby's head to force the baby to the breast;
- breastfeeding in a comfortable position and setting;
- expressing breast milk directly into the baby's mouth to encourage suckling;
- expressing breast milk and feeding it to the baby until the baby resumes breastfeeding;
- feeding the baby with a cup and not a bottle; and
- using at-breast supplementation (see **Section 6.3.4**) to encourage a baby to suckle and stimulate relactation.

7.8 Engorgement

Engorgement commonly occurs when milk production increases quickly during the first few days after birth and is exacerbated when breast milk is not removed effectively. Blood flow to the breasts also increases in the days after birth. This can cause swelling (oedema), which can inhibit milk flow and make it more difficult to remove milk from the breast. Engorgement is usually temporary. Delayed initiation of breastfeeding, infrequent or scheduled feeds, poor attachment, newborn separation, and ineffective suckling increase the likelihood and severity of engorgement.

The mainstay of treatment for engorgement is removal of milk from the breasts.

Typically, engorgement affects both breasts. The most important symptom is pain. A mother may have a fever during the first 24 hours, which usually subsides. The breasts feel swollen and oedematous, and the skin looks shiny and diffusely red. The breasts become hard and tight. The nipples may not protract, which makes it difficult for a baby to attach and remove the milk.

How to alleviate engorgement

- The mainstay of treatment for engorgement is removal of milk from the breasts.
- Before feeding or expressing, apply warm compresses to the breast or take a warm shower, which helps the milk to flow.
- Assist the mother to remove the breast milk by breastfeeding her baby and/or expressing milk from her breasts.
- If the baby can be assisted to attach well and suckle, then the mother should breastfeed as frequently as the baby needs.
- If the baby is not able to attach and suckle effectively, encourage the mother to express her milk, by hand or with a pump, a few times. Then when her breasts are softer and the baby can attach well increase breastfeeding frequency.
- Reverse pressure softening may be used to reduce oedema and evert the nipple *(14)*.
- Engorgement can be alleviated with therapeutic breast massage. Lymphatic drainage massage, from the nipple to the armpit, can relieve oedema *(14,15)*.
- After feeding or expressing, use cool compresses to reduce the oedema *(14,15)*.

7.9 Oversupply

Oversupply, also known as hyperlactation or hypergalactia, is when a mother produces much more milk than is needed to feed her infant. Hyperlactation is often accompanied by over-active milk ejection reflex or a fast flow of milk, and so can lead to discomfort for the mother and the baby in the short term. If not addressed, hyperlactation with over-active milk ejection reflex can prevent the baby from learning to actively remove milk from the breast (because the milk flows independently) and interfere with the establishment of an equilibrium between supply and demand. Paradoxically, this can lead to late-onset of insufficient milk intake *(14)*. Signs and symptoms associated with hyperlactation are described in **Table 7.1**.

Table 7.1 Signs and symptoms that may be associated with hyperlactation in the breastfeeding dyad

Signs and symptoms that may be associated with hyperlactation in the breastfeeding dyad	
Maternal signs/symptoms	**Infant signs/symptoms**
• Excessive breast growth during pregnancy >2 cup sizes • Persistent or frequent breast fullness • Breast and/or nipple pain • Copious milk leakage • Recurrent plugged ducts • Recurrent mastitis • Nipple blebs • Vasospasm	• Excessive weight gain • Difficulty achieving a sustained, deep latch • Fussiness at the breast • Choking, coughing or unlatching during feeds • Breast refusal • Clamping down on the nipple/areola • Short feedings • Gastrointestinal symptoms (e.g. spitting up, gas, reflux or explosive green stools)

Source: Johnson H, Eglash A, Mitchell KB, Leeper K, Smillie CM, Moore-Ostby L, et al. ABM Clinical Protocol #32: Management of hyperlactation. Breastfeed Med. 2020;15(3):128–132.

When managing oversupply, the aim is to gradually reduce the volume of milk taken from the breasts so that autocrine control inhibits the rate of milk synthesis. A number of strategies can be used alone or in concert to manage oversupply *(8, 16)*.

- Avoiding expressing breast milk between breastfeeds.
- Limiting nipple stimulation apart from during breastfeeding (e.g. reduce sexual stimulation, wear well fitted tops that do not allow the breasts to brush against clothing, sleep on side or back);
- Feeding the baby from only one breast at each feed, alternating breasts at each feed;
- Feeding the baby from the same breast at every feed (block feeding) for 3–4 hours or more before feeding from the other breast for a few days. The duration of each block feeding will depend on the severity of the oversupply (only express a small amount of milk from the other breast if necessary to relieve initial discomfort);
- Using a reclining position to manage forceful milk ejection reflex.
- Using cool compresses for comfort and analgesia to relieve pain.

If these strategies are ineffective, consider using a breast pump to express the milk from both breasts as completely as possible and then block feeding for a few days.

7.10 Blocked ducts

Blocked ducts cause localized inflammation accompanied by tenderness and a small lump in one breast with or without redness in the skin. Blocked ducts occur when milk is not removed effectively from part of the breast and may be caused by Infrequent breastfeeds, poor attachment, tight clothing or trauma to the breast. Sometimes, a duct is blocked by thickened milk. To address blocked ducts *(12)*:

- improve the removal of milk and correct the underlying cause;
- use a different position to breastfeed the baby, with baby's chin toward the blocked duct;

- feed the baby from the affected breast frequently with very gentle massage over the lump, only while the baby is suckling;
- very gently massage the lump towards the nipple during feeds and discouraged from massaging the lump when breast milk is not flowing; and
- apply cold compresses after feeding or expressing to reduce aoedema.

7.11 Mastitis

A mother with mastitis has a fever and feels ill. The breast is swollen and red. The mother usually experiences pain that is often severe. Mastitis may be limited to part of the breast or affect the whole breast. Mastitis is most common in the first two to three weeks after birth but can occur at any time.

Breastfeeding should not be withheld during mastitis treatment

The most common cause of mastitis is milk stasis, or lack of milk flow *(17).* Milk stasis can be caused by poor attachment where this restricts milk removal; unrelieved engorgement; an unresolved blocked duct; frequent pressure on one part of the breast from fingers or tight clothing; and trauma to the breast or nipple, especially if there is a wound that becomes infected. Long gaps between feeds, for example when breastfeeding is delayed because the mother is busy or resumes employment outside the home, or when the baby starts sleeping through the night.

Mastitis usually begins as non-infectious inflammation. If the cause is addressed within 24 hours, the mother feels relief and no other treatment is necessary.

If stasis persists, infectious mastitis may result. Infectious mastitis is commonly caused by penicillin-resistant *Staphylococcus aureus*. If symptoms persist for more than 24 hours with adequate drainage and anti-inflammatory measures, antibiotics treatment is indicated.

Breastfeeding should not be withheld during mastitis treatment. Removing milk from the breast is critical to the management of mastitis. If breastfeeding is not possible, milk should be expressed regularly to avoid milk stasis *(17).* In addition, mothers with mastitis should be advised to:

- Rest. Family members should be encouraged to take care of the mother's household and caring responsibilities and mothers who are employed should be encouraged take leave so they can rest in bed and feed the baby frequently.
- Continue to breastfeed the baby frequently. Start breastfeeding the baby with the unaffected breast, which will stimulate the oxytocin reflex and help milk to flow.
- Vary the position of the baby.
- Apply warm compresses to the breast before a breastfeed to stimulate the oxytocin reflex and milk flow.
- Apply cold compresses to the affected breast(s) to reduce pain and swelling after breastfeeding.
- Avoid long gaps between feeds.
- Express milk by hand (see **Section 8.5.1**) or with a breast pump if feeding from the affected breast is too painful.
- Use analgesics such as ibuprofen, which also reduces inflammation of the breast, or paracetamol.

If symptoms are severe, if there is an infected nipple fissure, or if no improvement is seen after 24 hours of improved milk removal, the treatment should then include penicillinase-resistant antibiotics. However, antibiotics must be accompanied by effective milk removal *(17)*.

7.12 Breast abscess

Untreated mastitis can cause a breast abscess. A breast abscess is a localized accumulation of pus in the breast caused by infection. It is usually very painful and mothers with breast abscesses feel unwell and febrile. There may be discoloration of the skin at the point over the swelling.

A breast abscess is an emergency. Untreated, it can lead to sepsis.

A breast abscess is an emergency. Untreated, it can lead to sepsis. Breast abscesses must be drained by a skilled healthcare professional in a clinical setting and treated with penicillinase-resistant antibiotics. When possible, drainage should be either by catheter through a small incision or by needle aspiration, guided by ultrasound if available. Drainage may need to be repeated. Large surgical incisions damage the areola and milk ducts and interfere with continued breastfeeding and are best avoided.

As with mastitis, milk removal is a critical feature of treatment for breast abscess. Breastfeeding should be continued from the affected breast during treatment unless the mother is living with HIV (see **Section 6.4.1**). Feeding a baby from an infected breast does not affect the infant. Mothers can continue to feed from the other breast. If breastfeeding is not possible, breast milk must be expressed to prevent or relieve milk stasis. Hand expressing (see **Section 8.5.1**) is usually more comfortable than using a breast pump. Sometimes breast milk drains from the incision during breastfeeding or expressing. This is temporary and it is not a reason to stop removing milk from the breast. Encourage her to take adequate analgesia *(18)*.

7.13 Candidiasis (thrush)

Candidiasis, or thrush, causes nipple and breast pain that persists between feeds. Candidiasis causes pain that is often described as feeling like sharp needles penetrating deep into the breast or razor blades being drawn across the nipples that is not relieved by improved attachment. There may be a red or flaky rash on the areola, itching and depigmentation.

The baby might have white spots inside the cheeks or over the tongue that look like milk curds but cannot be removed easily. Some babies breastfeed normally. Some feed for a short time and then pull away. Some refuse to feed altogether and some are distressed when they try to attach and feed, suggesting that their mouth is sore. There may be a red rash over the diaper/nappy area ("diaper dermatitis"). However, babies can also be asymptomatic.

Thrush is caused by a fungal (or yeast) infection, usually *Candida albicans*. Thrush may follow the use of antibiotics in the baby or in the mother. There is emerging evidence that maternal symptoms are caused by infections other than *Candida*, such as coagulase-negative *staphylococci* and *streptococci* (mainly from the mitis and salivarius groups) *(19)*.

If the mother has symptoms, both mother and baby should be treated. If only the baby has symptoms, it is not necessary to treat the mother.

First-line treatment is topical nystatin. Mothers may be treated with nystatin cream 100 000 IU/ml applied to the nipples four times daily after breastfeeds and continued for seven days after lesions have healed. Babies may be treated with nystatin suspension 100 000 IU/ml, 1 ml applied by dropper to child's mouth four times daily after breastfeeds for at least seven days or as long as the mother is being treated.

Some strains of thrush are becoming resistant to nystatin. When there is a definite diagnosis of thrush and topical treatments have been ineffective, oral antifungal medicine be used. Topical gentian violet is no longer recommended *(20)*.

7.14 Postnatal mood disorders

It is very common for women to experience emotional changes, insomnia, loss of appetite and feelings of overwhelm in the days immediately after birth. These symptoms usually resolve by the tenth day after a birth. Women who experience mild symptoms during the first two weeks after birth warrant closer monitoring but medication is not indicated unless symptoms persist *(20)*. When symptoms persist for more than two weeks after birth evaluation for postnatal depression can determine whether treatment is warranted.

Postnatal depression affects between 6.5% and 20% of new mothers *(21)* and can have serious and long-term negative impacts on their wellbeing, their infants' growth and their breastfeeding outcomes *(22)*. Depressed mothers are more likely to find breastfeeding difficult, are more likely to believe they are not making enough milk for their babies and they tend to attribute these difficulties to their psychological or emotional traits *(11)*. Postnatal anxiety has negative impacts on intention to breastfeed, exclusive breastfeeding and total breastfeeding duration *(23)*. Routine screening with a tool such as the 10-item Edinburgh Postnatal Depression Scale can be used to detect postpartum depression *(24)*.

Postnatal depression affects between 6.5% and 20% of new mothers and can have serious and long-term negative impacts on their wellbeing, their infants' growth and their breastfeeding outcomes

Treating mood disorders is important for the health of both mothers and babies and effective treatments are available. Sertraline and paroxetine are the medications of choice for postnatal depression in breastfeeding women. Only small amounts of these medications transfer into breast milk and adverse effects in infants are rare and not usually serious. Mothers treated with a selective serotonin reuptake inhibitor (SSRI), tricyclic antidepressant, or serotonin and norepinephrine reuptake inhibitor (SNRI) during pregnancy and whose symptoms are well controlled should continue treatment during breastfeeding *(25)*.

Postnatal psychosis is a psychiatric emergency characterized by paranoia, hallucinations, delusions, and suicidal ideation, with the potential risk of suicide and/or infanticide. Often a mother and child have to be separated, which makes breastfeeding difficult, and medications contraindicated in breastfeeding may be necessary. Abrupt withdrawal of breastfeeding can lead to mastitis. Mothers who are separated from their babies for treatment need support to express their breast milk (see **Section 8.5.1**) to prevent mastitis and enable the mother to resume breastfeeding after the psychosis has resolved. If the mother is not taking medication that is contraindicated in breastfeeding, her milk can be given to the baby's caregivers for feeding to the baby.

Mothers experiencing postnatal mood disorders need support and treatment from a mental health professional and more frequent breastfeeding counselling to prevent and address breastfeeding challenges early *(26)*. Many of these mothers and their families find community support invaluable. Connecting these families with available support services may improve outcomes.

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SECTION 8

Routine guidance for breastfeeding mothers

Key messages

- The quality of a woman's diet has little impact on the quality or quantity of her breast milk.
- Mothers of multiples need extra support to breastfeed.
- Breastfeeding can continue through pregnancy and beyond.
- Mothers should be encouraged to express breast milk for their babies when separated for work.
- Some crying is normal; excessive crying should be investigated.
- Breastfeeding should not be withheld from infants or young children during illness.
- Non-hormonal and progestin-only contraceptives are safest while breastfeeding.
- It is not usually necessary to stop or interrupt breastfeeding to take medication.
- Minimizing harm from substance use while breastfeeding is possible with support.

Section 8 outline

8.1 Maternal nutrition

Lactation increases a woman's energy and nutrient requirements by 10–20%. The volume and variety of food breastfeeding women eat each day usually increases slightly to meet their needs and protect her nutrient stores.

Well-nourished women with varied diets who eat according to their appetites will usually take enough food to cover any extra needs. Breastfeeding women should be encouraged not to avoid any foods normally included in the local diet. A breastfeeding mother's diet should include animal-sourced foods such as meat, fish, cheese and milk and vegetables, nuts, seeds and pulses. This will help them feel well and strong while meeting the physical and emotional demands of caring for a new baby *(1)*.

The quality and volume of breast milk a woman produces is not significantly affected by her diet. Even moderately malnourished usually continue to produce good quality breast milk.

The quality and volume of breast milk a woman produces is not significantly affected by diet. Although some micronutrient deficiencies impact micronutrient content in breastmilk, even moderately malnourished women usually continue to produce good quality breast milk.

However, a diet that is poor in quantity or quality may affect a mother's energy and ability to care for her infant or child. In practice, a lactating mother uses about 500 extra kilocalories per day. This is roughly equivalent to one small meal. Women with nutrient-poor diets may not have laid down sufficient nutrient stores in pregnancy. These women may need to eat an extra meal with a variety of foods each day while breastfeeding to meet their needs and protect their nutritional status *(1)*.

Breastfeeding women who eat little or no animal-sourced foods and their babies may become deficient in Vitamin B12 (riboflavin) and require supplementation to become replete *(2, 3)*.

Where household resources are scarce, breast milk is likely to be the most complete and safest food for the baby.

Where household resources are scarce, breast milk is likely to be the most complete and safest food for the baby.

8.2 Multiples

Twins are the most common multiple birth and around 12 in every 1000 births are twin births *(4)*. Around 50% of multiple pregnancies end in preterm or low birth weight births. Considerations for establishing breastfeeding for low-birth-weight infants can be found in **Section 4.5**.

Early skin-to-skin contact helps babies to achieve good attachment at the breast. Exclusive breastfeeding from birth, or as soon as the mother is able to respond, is recommended.

Mothers may require assistance to find the best way to hold two babies to suckle simultaneously. One or both babies can be held in the underarm position for feeding. Support for the babies with pillows or folded clothes is often helpful. The babies should be offered both breasts at every breastfeed and alternate the breast they feed from first.

The demand and supply mechanism of lactation ensures that early, effective suckling stimulates adequate breast milk production for two or more infants. This knowledge can build the mother's confidence that she can make enough milk for her children. Breastfeeding more than one infant is time-consuming. Encouraging relatives to help with other household duties can make it easier for mothers to breastfeed responsively *(5)*.

8.3 Subsequent pregnancy

It is not uncommon for a breastfeeding woman to become pregnant. It is not necessary for a woman who becomes pregnant to stop breastfeeding. Many women continue breastfeeding through pregnancy and breastfeed a new baby alongside an older sibling. There is no evidence that breastfeeding increases the risk of adverse pregnancy outcomes unless there is a history of miscarriage or preterm labour.

It is common for the volume of breast milk a woman produces to decrease sharply during early pregnancy and revert to colostrum during the third trimester. If a woman becomes pregnant while her infant is exclusively breastfed, her infant should be monitored closely to ensure their milk intake is adequate. Assessing breast milk intake is discussed in **Section 7.4**. Older babies and infants may eat more complementary foods if their breast milk intake is reduced. Colostrum has laxative properties. When older infants and young children consume colostrum, their stools may become loose or runny. This is not harmful and should not be confused with diarrhoeal illness.

Pregnancy hormones can increase nipple sensitivity and make breastfeeding uncomfortable, especially during the first trimester. Careful attention to positioning and latch usually alleviates this discomfort.

When a woman gives birth while breastfeeding a new baby, copious milk production (lactogenesis II) may occur sooner than it usually does. Lactogenesis II is described in detail in **Section 2**. The new baby should breastfeed frequently to stimulate lactogenesis II occurs. Some mothers continue breastfeeding through pregnancy and alongside a new baby. Breastfeeding a baby's older sibling may help relieve very full or engorged breasts. The new baby and the older sibling may breastfeed simultaneously, one at each breast, or one at a time. If the children are breastfed one at a time, the mother should be advised not to breastfeed the new baby first at every feed after the first week. This is because breast milk tends to be higher in fat towards the end of a breast feed *(6)*.

8.4 Separation

The most common reason for mothers being separated from their babies for part of the day is because they are employed outside the home.

Mothers should be encouraged to make a plan for feeding their babies after they return to work. It is common for working mothers to breastfeed their babies frequently at home and express breast milk at work (see **Section 8.5.1**) so that someone else can give it to the baby while they are separated. Older infants may prefer to eat complementary foods while they are separated from their

To establish or maintain supply while separated, encourage the mother to express her breast milk – by hand or with a breast pump – as often as the baby would ordinarily feed.

mothers and breastfeed frequently at home. If a mother is separated from her baby frequently, the baby may wake to breastfeed more frequently during the night.

The mother can be encouraged to:

- breastfeed her baby whenever she is at home, such as at night and weekends
- sleep safely with her baby, so that she can breastfeed at night and early in the morning
- express milk, by hand or by pump, before she leaves for work
- express milk, by hand or by pump, while she is at work.

8.5 Expressing and storing breast milk

If a mother does not express breast milk while she is separated from the baby, milk production will decrease and she may be more likely to develop mastitis (see **Section 7.11** for information on managing mastitis).

A mother and her baby may be separated and unable to breastfeed for a variety of reasons. To establish or maintain supply while separated, encourage the mother to express her breast milk – by hand or with a breast pump – as often as the baby would ordinarily feed. If facilities are available, she can store her milk by freezing it. Babies can be fed expressed breast milk from a cup during the separation. Older babies and young children may eat complementary foods during the day and catch up on missed breastfeeds when they are reunited with their mothers.

If prolonged separation cannot be avoided, the mother and baby may require additional skilled lactation support to start or resume breastfeeding when they are reunited.

When livelihood activities or employment necessitates regular periods of separation, mothers can be encouraged to:

- expect the baby to breastfeed more frequently at night and on non-work days;
- practice safe co-sleeping so that the baby can breastfeed at night without disrupting their sleep;
- express milk before work;
- express milk at work; and
- leave EBM for a carer to feed the baby while she is at work.

If a mother does not express breastmilk while separated from the baby, milk production will decrease and she will be at increased risk for mastitis.

8.5.1 Expressing breastmilk by hand

Hand-expressing is a useful skill because it requires almost no equipment and can be done whenever a baby is unable to breastfeed and a mother needs to remove milk from her breasts. All mothers should be taught to express breast milk by hand. Hand expressing can be demonstrated using a breast model or on your own body. Attempting to demonstrate on a mother's own breast can cause injury. **Box 8.1** describes how to express breast milk by hand. **Fig. 8.1** shows a mother expressing her milk.

Box 8.1 Hand expressing

How to express breast milk by hand

1. Prepare a clean dry wide mouthed container for the expressed milk. If expressing colostrum in the first one or two days, collect it in a 2 or 5 mL syringe as it comes from the nipple. This avoids wasting the milk, which can happen with a small volume of milk in a large container.
2. Wash your hands thoroughly with soap and water.
3. She needs to wash her breasts only once a day. Frequent washing, especially with soap, dries the sensitive skin of the areola, increasing the risk of fissures.
4. Sit or stand comfortably and hold the container near the breast.
5. Place thumb above the nipple and areola and first finger or first two fingers below the nipple and areola, opposite the thumb. She supports the breast with her other fingers (see **Fig. 8.1**).
6. Squeeze the breast between thumb and fingers while pressing slightly inwards towards the chest wall to compress the milk ducts. Some mothers can feel the ducts beneath the areola. They are like pods or peanuts.
7. Press and release, press and release. This should not hurt – if it hurts, the technique is wrong. At first no milk may come, but after pressing a few times, milk starts to drip out or flow in streams.
8. Change the position of the fingers, working around the areola so that milk is expressed all segments of the breast.
9. Avoid rubbing or sliding her fingers along the skin. The movement of the fingers should be more like rolling. Avoid squeezing or pulling the nipple itself.
10. Express one breast for at least 3–5 minutes until the flow slows and then express the other side.
11. Repeat the process two or three time times. Stop expressing when the milk no longer flows.
12. Explain that to express breast milk adequately takes 20–30 minutes, especially in the first few days when only a little milk may be produced. It is important not to try to express in a shorter time.

Fig. 8.1 A mother expressing her breast milk

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Hand Expressing Technique

Wash hands before expressing. **Gently** massage the breast in a circular motion, from the outer aspect of the breast in toward the nipple

Place your thumb and forefinger flat on your breast, **approximately 3 cm back from the nipple,** keeping the thumb in line with the forefinger, as if an imaginary line is running through them

Position a clean/decontaminated container resting on the breast, under the nipple, to collect the milk

Collection by oral dispenser is useful for smaller volumes of milk (colostrum)

Press the thumb and forefinger **towards the chest**...

...then compress the thumb and forefinger directly together and hold for 2-3 seconds before releasing. Do not pull your nipple or roll your fingers forward. This compression **should not hurt** – if it is tender, reassess the position of your fingers

Once the flow slows, move your fingers around slightly so you compress a different area (refer back to step 2)

Ensure that all milk containers are labelled with your/your baby's name, time and date of the expression.

To establish your milk supply when baby isn't feeding from your breast, it is necessary to express **at least 8 times a day**.

more milk removed = more milk made

Clinical Multimedia_230418

For more information

- Queensland Health booklet "Child Health Information Your guide to the first 12 months"
- Queensland Health Breastfeeding website: https://www.qld.gov.au/health/children/babies/breastfeeding
- The Australian Breastfeeding Association's Helpline 1800 mum 2 mum (1800 686 268) or https://www.breastfeeding.asn.au/

Partnering with Consumers - National Standard 2. (2.4) Consumers and/or carers provided feedback on this publication

Developed by Women's and Newborn Services, Royal Brisbane and Women's Hospital.

Version No: 4.1 Effective: 11/2023 Review: 11/2026

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Source: State of Queensland (Queensland Health); Metro North Health; Women's Newborn Services. Baby feeding cues (signs) term; version 6.1. Brisbane; Australia (2024). Licensed (https://creativecommons.org/licenses/by-nc-nd/4.0/). Used with permission.

8.5.2 Storing breastmilk

Evidence-based guidelines on the appropriate storage and handling of expressed breast milk include detailed guidance on proper containers, hygiene, temperatures, and storage duration *(7,8)*. If possible, expressed breast milk should be refrigerated immediately and taken home in an insulated container. If that is not possible, breast milk can be kept safely at room temperature (ranging from 27° C to 32° C) for up to four hours. After that, it may have to be discarded. Mothers need to understand that their breasts will make more milk to meet the baby's needs if they continue to breast feed responsively.

8.6 Crying baby

It is normal for babies to cry. Crying is the only way babies can communicate needs such as hunger or comfort. Babies who cry excessively or are difficult to comfort should be examined to rule out pain from illness or injury.

Some babies cry more than others without a specific diagnosis. Often crying occurs at certain times of day, typically the evening. The baby may pull up their legs as if in pain. They want to feed but can be difficult to comfort. Parents need support and reassurance that crying behaviour is likely to improve or resolve over time, usually after the first three months of age.

Babies who cry when lying down and/or vomit large volumes after feeding may be experiencing gastro-oesophageal reflux disease (GORD). Many babies experience a physiological reflux where they regurgitate without crying and gain weight as expected. Gastro-oesophageal reflux disease occurs when reflux episodes are associated with complications such as esophagitis or poor weight gain. Holding the baby upright may help relieve discomfort or regurgitation associated with physiological reflux. Medication may be indicated for infants with gastro-oesophageal reflux disease *(6).*

8.7 Child illness

Mothers should be encouraged to breastfeed sick infants and young children as often as they are willing while they receive other treatment *(9)*. Breast milk is an ideal food during illness, especially for infants younger than six months old. Children older than six months of age may refuse complementary food but may want to breastfeed much more frequently than usual.

If a baby has a blocked nose, the mother can be taught how to use drops of breast milk or prepacked saline to clear the baby's nose. She can give shorter more frequent breastfeeds, allowing the baby time to pause and breathe through the mouth until the nose clears.

If a baby is not able to breastfeed adequately, but can take oral or enteral feeds, the mother should be supported to express her breast milk (see **Section 8.5**). She should express as often as the baby would usually feed and at least eight times in 24 hours to maintain her milk supply. The mother can feed expressed breast milk to the baby by cup and let the baby suckle whenever they want to. If the baby cannot suckle effectively expressed breast milk may be given by nasogastric tube.

If a baby is admitted to a hospital, the mother should be supported to remain with the baby so that she can practice responsive breastfeeding.

If a baby is not able to take any oral or enteral feeds the mother should be supported to continue expressing to maintain her milk supply (see **Section 8.5**). Her expressed breast milk can be stored safely and given to the baby as soon as they start enteral feeds. She may be able to freeze unused milk for later use or when discharged home. If the hospital has milk-banking facilities, the milk may be donated for another child. She can resume breastfeeding as the baby recovers.

If breast milk production decreases during an infant's illness, the mother needs support to rebuild her milk supply. She may need support to relactate and to feed her infant using supplementary suckling to stimulate breast-milk production (see **Section 6.3.4**). With skilled lactation support, mothers can resume exclusive breastfeeding within one to two weeks.

Young children and infants from six months of age may prefer breastfeeding to complementary foods and breastfeed more often than usual while they are ill. More frequent breastfeeding increases fluid intake and supports the baby's immune response. Withholding breastfeeding during illness is not recommended. Mothers should be encouraged and supported to practice responsive breastfeeding.

Families should be encouraged and supported to continue offering these children complementary foods that are softer in consistency and in smaller, more frequent portions than when the child is well. During recovery, caregivers can offer extra food as the child's appetite increases.

8.8 Family planning

Family planning, including non-hormonal and hormonal options should be discussed with all families during prenatal and postnatal care *(10–13)*. Effective counselling includes discussion of efficacy, time since delivery, maternal medical comorbidities, and breastfeeding plans. The Academy of Breastfeeding Medicine has outlined general principles for counselling breastfeeding women about contraceptive use and birth spacing. These are summarized in **Table 8.1** *(13).*

Table 8.1 General principles for counselling women on contraceptive selection and birth spacing in breastfeeding

Issues	Considerations
1. Breastfeeding patterns, status, and plans	• Consider both short- and long-term breastfeeding intent as well as well birth spacing plans. There is the potential for hormonal methods to have an impact depending on when they are started. • Mothers may plan to exclusively breastfeed; some may do so to use LAM, others may use LAM because they are already fully breastfeeding. LAM users should be counseled to have another method in hand for when menses return or breastfeeding patterns change. Effectiveness of LAM in exclusively breastmilk pumping mothers may not be equivalent to direct breastfeeding. • Many women who intend to breastfeed exclusively are not able to achieve their goals.
2. Child's age/time postpartum	• Many methods should not be introduced until breastfeeding is well established (i.e., at 4–6 weeks), as there may be potential for hormonal methods to directly impact lactogenesis and/or to impact the infant.
3. Maternal age and future childbearing plans	• Choices depend on desire to space births or desire to limit family size. Globally recommended interpregnancy intervals are at least 18 months to 2 + years for maternal health, depending on the setting, and about 3–5 years for child health outcomes.
4. Previous contraceptive experience	• Discussion of previous contraceptive experience, including compliance, satisfaction, side effects, and social issues, is essential. These issues can influence compliance and satisfaction, particularly as they pertain to prior lactation experiences.
5. Partners/interactions	• Partner's experiences and opinions may impact compliance, particularly for barrier methods, LAM, and natural family planning. • The woman's social and behavioral considerations, such as number of partners and sexual activity, should be explored. A woman's history of unplanned pregnancy and short interpregnancy interval should be reviewed and discussed.
6. Previous lactation experience/ medical conditions	• Prior insufficient milk supply or inadequate infant growth • Prior breastfeeding experience did NOT meet goals (either exclusivity or duration), AND supply was a potential reason. • Physical examination suggestive of insufficient glandular tissue • Prior breast surgery • Medical conditions potentially adversely affecting supply (polycystic ovary syndrome, infertility, obesity) • Multiple gestation • Preterm infant(s)

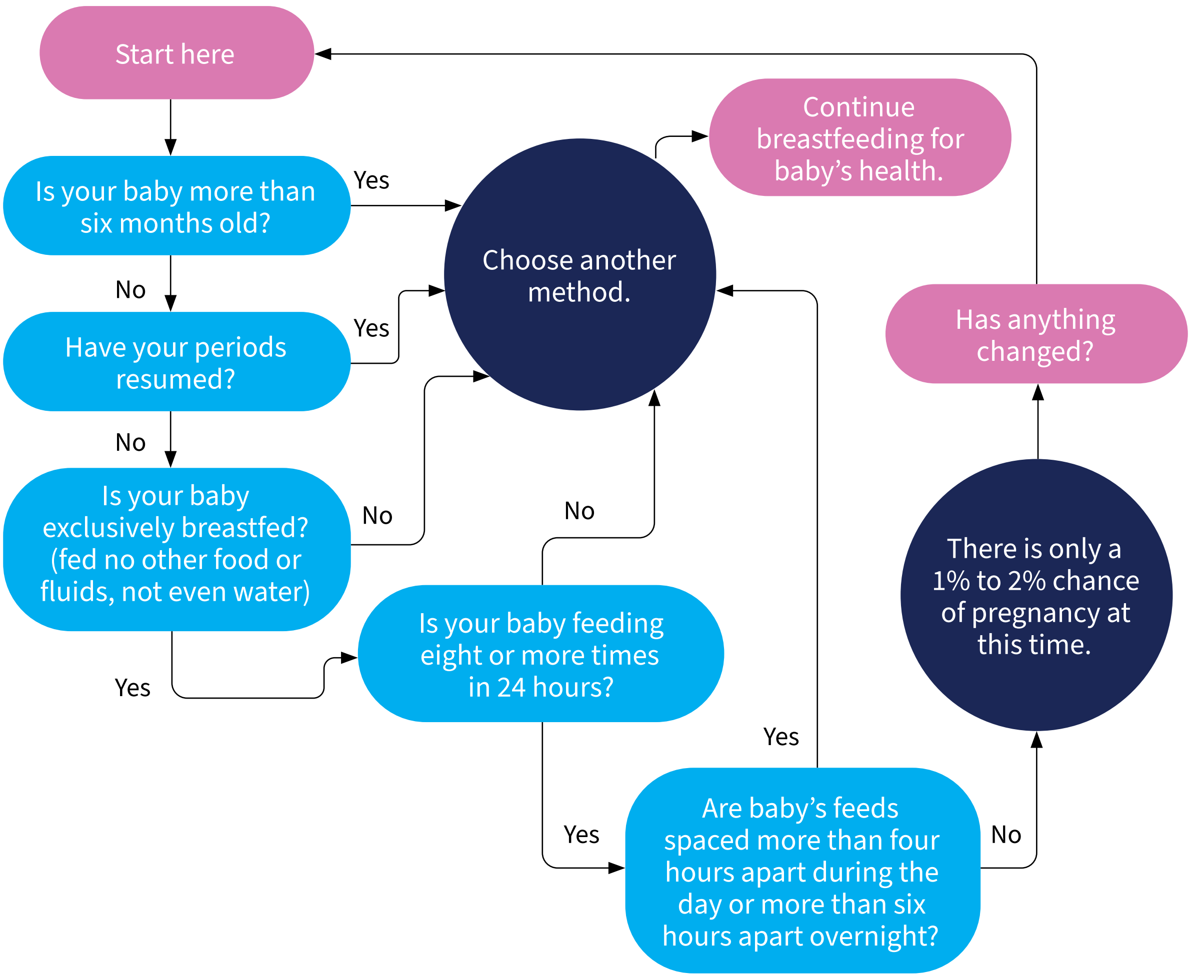
LAM: lactational amenorrhea method.

8.8.1 Lactational amenorrhea method

The lactational amenorrhea method (LAM) describes the contraceptive effect of breastfeeding. Frequent breastfeeding, including breastfeeding at night, can help to delay a subsequent pregnancy. Suckling reduces the release of other pituitary hormones, including gonadotrophin releasing hormone (GnRH), follicle stimulating hormone (FSH), and luteinising hormone (LH). Blocking these hormones suppresses ovulation and menstruation.

The LAM method protects a woman from pregnancy if: her baby is no more than six months of age, is exclusively breastfed (fed no foods or fluids, not even water), is breastfeeding eight or more times in 24hrs, breastfeeds are no more than four hours apart during the day and no more than six hours apart overnight, and her menstrual cycle has not resumed *(1)*. **Fig. 8.2** *(14)* below illustrates a decision tree for assessing whether LAM method is an effective choice.

Fig. 8.2 The lactational amenorrhea method of contraception



Source: Adapted from Labbok M. Breastfeeding, fertility, and family planning. Glob Libr Women's Med 2008.

8.8.2 Natural family planning (symptom-based monitoring)

Symptom-based monitoring, or natural family planning, may be started when normal secretions resume, which typically occurs after six months postpartum, which is usually later than for those who are not breastfeeding *(13).*

8.8.3 Copper intrauterine devices (IUDs)

Copper IUDs are safe and effective while breastfeeding and can be inserted in the immediate 48 hours postpartum. If 48 hours have passed since delivery and a Copper IUD is desired, it is recommended to wait four weeks prior to insertion *(11).*

8.8.4 Condoms, diaphragms and spermicides

Condoms, diaphragms and spermicides do not interfere with lactation. While they are less effective at preventing pregnancy than long-acting contraceptive options, they also protect against sexually transmissible infections. A diaphragm can be fitted from six weeks postpartum *(11).*

8.8.5 Progestin-only contraceptives

Progestin-only contraceptives include progestin-only “mini-pills,” injectables such as Depo-Provera, vaginal rings, implants such as Nexplanon, and progestin-containing intrauterine devices such as Mirena. Progesterone only pills are an effective method of birth control that does not impact milk production and may extend the period of LAM for breastfeeding women.

Progestin-only injectables (including both intermuscular and subcutaneous forms) are safe for mother and baby. There is insufficient evidence to allay concerns that these products could interfere with breastfeeding when given immediately after birth. Therefore, some authorities recommend waiting at least six weeks after birth to initiate treatment with long-acting progestin-containing products *(15, 16)*.

The progestin-releasing vaginal ring is considered safe for breastfeeding women who are at least four weeks postpartum. Breastfeeding women can have a progestin-containing implant inserted at any point after delivery *(16)*.

8.8.6 Levonorgestrel-releasing IUDs

Levonorgestrel-releasing IUDs are safe and effective while breastfeeding and can be inserted in the immediate 48 hours postpartum. However, if 48 hours have passed since delivery and a levonorgestrel-releasing IUD is desired, it is recommended to wait four weeks prior to insertion *(11).*

8.8.7 Oestrogen-containing contraceptives

Oestrogen-containing contraceptives include combined oral contraceptive pills, patches, and rings. Oestrogen-containing methods should be avoided in the initial weeks post-partum due to the increased risk of venous thromboembolism and pulmonary embolism during this period.

Combined oral contraceptives have been demonstrated to decrease breastfeeding duration and increase formula supplementation and should be avoided during breastfeeding. If one of these methods is desired despite this counselling, lower doses of oestrogen may have less of an impact on milk supply. They should be started after breastfeeding is well established and ideally after six months postpartum when complementary foods can be introduced *(11).*

8.9 Medications

Most medications can be taken while breastfeeding without affecting the breastfed infant (see **Table 8.2**) *(17,18).* Many mothers are unnecessarily advised to discontinue breastfeeding or to avoid taking medication to treat their own medical conditions. Some medications can affect a parent's ability to care for an infant safely and this should be discussed with families as required. Information about the safety of medications for the infants of breastfeeding women is available from a number of online and print resources *(17,18)*.

Many mothers are unnecessarily advised to discontinue breastfeeding or to avoid taking medication to treat their own medical conditions.

Breastfeeding should not prevent women from being vaccinated. It is generally safe to vaccinate women while they are breastfeeding *(19, 20)*. Breastfeeding mothers can be safely given inactivated, recombinant, subunit, polysaccharide, conjugate and toxoid vaccines. Breastfeeding is not a reason to deny a vaccination with a live-attenuated vaccine to a person who is otherwise recommended to receive it. For example, yellow fever vaccination is recommended if vaccination is indicated for breastfeeding women in areas where yellow fever is endemic, during outbreaks, and when travel cannot be avoided or postponed *(21)*.

Breastfeeding should not prevent women from receiving medical care. Conversely, infants should not be denied breastfeeding or forced to stop breastfeeding because their mothers need medical care.

Topical iodine or iodophors (e.g. povidone-iodine) when used in large amounts, especially on open wounds or mucous membranes, can result in thyroid suppression or electrolyte abnormalities in the breastfed infant. Alternative antimicrobials should be considered for breastfeeding women *(18).*

Sedating psychotherapeutic drugs, anti-epileptic drugs, and opioids or combinations may cause side effects such as drowsiness and respiratory depression in infants *(18).*

Breastfeeding is contraindicated during treatment with cytotoxic chemotherapeutic agents. Breastfeeding should be withheld during treatment. If a woman plans to breastfeed her baby after she has completed treatment, she should be supported to maintain lactation by expressing and discarding as frequently as her baby would breastfeed until the medication is no longer present in her body. The Memorial Sloane Kettering Cancer Centre advises that mothers can usually, but not always, continue breastfeeding while receiving external beam radiation therapy *(22)*. Radiopharmaceuticals should be avoided if possible. If these medications cannot be avoided, the woman should be supported to maintain her milk supply by expressing and discarding the milk her baby would normally consume until the medication is no longer present in her body.

Health care professionals should be equipped to support shared decision making with breastfeeding women.

Table 8.2 Major classes of medications and breastfeeding

Major classes of medications and breastfeeding	
Breastfeeding contraindicated	
	Anticancer drugs (antimetabolites) Radioactive substances (stop breastfeeding temporarily)
Breastfeeding generally possible	
Monitor baby for drowsiness	Psychotherapeutic drugs and anticonvulsants
Monitor baby for jaundice	Chloramphenicol, tetracyclines, metronidazole, quinolone antibiotics (e.g. ciprofloxacin) Sulfonamides, dapsone, cotrimoxazole (sulfamethoxazole-trimpethoprim), fansidar (sulfadoxine-pyrimetamine)
Breastfeeding generally possible	
May inhibit lactation – use alternative medicine	Oestrogen-containing contraceptives and ovulatory stimulants Progestogen-containing contraceptives in first 6 weeks after delivery Thiazide diuretics Dopamine receptor agonists: bromocriptine, cabergoline Pseudoephedrine
Safe in usual dosage	Most commonly used drugs: Analgesics and antipyretics: short course of paracetamol, acetylsalicylic acid, ibuprofen, occasional doses of morphine and pethidine Anti-infective drugs Antibacterials: ampicilline, amoxicilline, cloxacilline, other penicillins, and erythhromycin Antituberculars, anti-leprotics Antiprotozoals: antimalarials (except mefloquine, fansidar), antihelmintics Antifungals Others: bronchodilators (e.g. salbutamol), corticosteroids, antihistamines, antiacids, drugs for diabetes, most antihypertensives, digoxin, nutritional supplements of iodine, iron and vitamins

Source: Hale TW. Hale's Medications and Mothers' Milk: A Manual of Lactation Pharmacology. 19th ed. Springer; 2023.

8.10 Substance use

The risks of exposure to substances in breast milk must be weighed against the risks – short-term and long-term – of withholding breastfeeding. It may be difficult or dangerous to care for an infant while under the influence of substances with psychoactive properties; women who use these substances should be given information and support about how to minimize the impact of their substance use on their breastfed infants and referred to treatment for substance use disorder as appropriate *(23)*.

8.10.1 Alcohol

Alcohol is known to inhibit the milk ejection reflex, which may result in decreased milk transfer. Breast milk alcohol levels closely parallel blood alcohol levels. There is no consensus around the long-term effects of regular maternal alcohol use during lactation on infant health *(24, 25)*.

8.10.2 Caffeine

Mothers with very high caffeine intakes can be jittery and have poor sleep patterns. Caffeine does appear in breast milk rapidly after maternal ingestion but is believed to be safe with a limit of 300 to 500 mg daily *(26)*.

8.10.3 Cannabis

There are no clear guidelines on the safety of cannabis use while breastfeeding. Data on long-term developmental impacts on infants exposed to cannabis through breast milk are lacking. Breastfeeding mothers who are using cannabis products should be supported to cease or decrease cannabis use *(23, 27)*.

8.10.4 Nicotine

Women who use nicotine-containing products should be encouraged to breastfeed and supported to quit. Breastfeeding protects infants from risks associated with environmental exposure to tobacco smoke and withholding breastfeeding increases these risks. Mothers should be discouraged from smoking for half an hour before breastfeeding. Household members who smoke should avoid smoking indoors, wash their hands and change their clothing before they handle a baby *(28, 29)*.

Women who use nicotine-containing products should be encouraged to breastfeed and supported to quit

8.10.5 Opioids

Breastfeeding mothers with opioid use disorder need active support to initiate methadone or buprenorphine therapy. Breastfeeding mothers on opioid replacement therapy at stable doses should be encouraged to breastfeed, regardless of their medication dose *(30)*.

For infants of mothers with opioid use disorder, breastfeeding may reduce the severity of Neonatal Abstinence Syndrome. Methadone and buprenorphine have been detected at low levels in human breast milk *(31)*. There is mixed evidence on the long-term impact of infant exposure to these medications via breast milk, and this effect is difficult to isolate from that of prenatal exposure and environmental risk factors.

The amounts of buprenorphine in milk may not be sufficient to prevent neonatal withdrawal, and treatment of infant may be required. When the mother weans from the breast, her baby will require monitoring for any possible withdrawal symptoms *(23, 32)*.

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